

Quant Awards

The Quantitative Finance Competition

2024 winners

- 1st prize: Dynamic Economic Tracking Portfolios - Maurits van Altvorst - Erasmus University Rotterdam (Rotterdam, Netherlands)
- 2nd prize: Temperature derivatives pricing based on temperature stochastic modelling - Anne Coll Franck - Universidad Pontificia Comillas (Madrid, Spain)
- 3rd prize: Polynomial Ornstein Uhlenbeck Vol Models- Xuyang LIN - École polytechnique (Paris, France)

Background information

More than a century after the seminal work of Louis Bachelier, the quantitative approach to financial markets has become omnipresent.

Today, many investment firms specialize in researching, developing, and implementing systematic trading strategies, increasingly incorporating data science and AI, while other active managers are adding quantitative approaches to their offerings.

Individual clients, too, can now delegate portfolio management to robo-advisors.

These examples illustrate a shifting landscape, where we believe quantitative portfolio management will continue to grow in importance due to the discipline of scientific methods and the full automation of the investment process.

The QuantAwards competition is open to university students and interns in quantitative finance regardless of specialty. Applications are free. No CFA candidacy or membership is required.

This competition offers students a unique opportunity to showcase their creativity and their understanding of this highly timely subject and to benefit from enhanced visibility in a competitive landscape.

The QuantAwards are organized and run by volunteers from eight CFA Societies, the local associations of CFA charterholders, affiliated with CFA Institute :

CFA Society Belgium, CFA Society Italy, CFA Society France, CFA Society Netherlands, CFA Society Ireland, CFA Society Norway, CFA Society Istanbul and CFA Society Spain

Dynamic Economic Tracking Portfolios: Leveraging Random and Local Linear Forests for Macroeconomic Forecasts

Abstract

This study advances the construction of economic tracking portfolios (ETPs) by integrating random forests (RFs) and local linear forests (LLFs) to capture complex, non-linear relationships between asset returns and macroeconomic factors. We develop RF and LLF-based ETPs to track inflation, consumption growth, and industrial production growth over short (1-month) and long-term (1-year) horizons. Our findings demonstrate that machine learning-based approaches consistently outperform linear ETPs. For the 1-year horizon, LLF ETPs achieve statistically significant improvements in tracking inflation and consumption growth, with Mincer-Zarnowitz R^2 values increasing from 4.4% to 6.9% for inflation and from 3.1% to 7.4% for consumption growth. Shapley value analysis reveals that relationships between assets and macroeconomic factors vary with economic conditions, challenging static factor decompositions. Kernel principal component analysis of LLF kernels identifies distinct economic regimes, offering a new approach to understanding changing economic relationships. While machine learning-based ETPs show improved performance, they exhibit higher turnover and sensitivity to uninformative assets, presenting practical implementation challenges. These advancements provide improved tools for economic forecasting, risk management, and deeper insights into the dynamic relationships between financial markets and macroeconomic conditions.

1 Introduction

The complex relationships between financial markets and macroeconomic variables have long been a subject of study in economics. Understanding how asset returns reflect and predict changes in key economic indicators is crucial for investors, policymakers, and economists alike. Economic tracking portfolios (ETPs), introduced by Breeden, Gibbons and Litzenberger (1989) and expanded by Lamont (2001), have proven to be powerful tools in capturing these market-economy dynamics. These portfolios are designed to track specific macroeconomic variables, offering insights into market expectations about future economic conditions.

However, the current framework for constructing ETPs relies on linear regression models, which may be overly simplistic given the complex nature of modern economies. The global economy is characterized by non-linear interactions, feedback loops, and regime shifts. Economic relationships that hold during periods of stability might change or reverse during crises. The limitations of linear models in capturing these effects can lead to suboptimal portfolio construction and less accurate economic forecasts.

Given these challenges, we propose to enhance the construction of ETPs with the use of machine learning techniques, specifically random forests (RFs) and local linear forests (LLFs). These methods are capable of capturing non-linear relationships and interactions between variables, potentially enhancing the performance of ETPs.

This paper makes several key contributions to the existing literature. First, it introduces two new approaches for constructing ETPs that can capture non-linear relationships and interaction effects between macroeconomic variables and asset returns. Second, by employing Shapley values, we provide insights into how different economic variables contribute to tracking portfolio allocations across different economic regimes. Third, we use kernel principal component analysis on the LLF kernel to identify distinct economic regimes based on the similarity of historical economic environments.

We develop RF and LLF-based ETPs to track inflation, consumption growth, and industrial production growth over short (1-month) and long-term (1-year) horizons. Our findings demonstrate that machine learning-based approaches consistently outperform linear ETPs. For the 1-year horizon, LLF ETPs achieve statistically significant improvements in tracking inflation and consumption growth.

These advancements provide improved tools for economic forecasting, risk management, and deeper insights into the dynamic relationships between financial markets and macroeconomic conditions. By integrating machine learning techniques into the construction of ETPs, we aim to develop more accurate tools for economic forecasting and risk management, potentially leading to more effective monetary policy decisions, improved investment strategies, and more efficient capital allocation.

2 Data and Methodology

2.1 Data

Our study examines ETPs comprising five asset classes: equities, high-yield instruments, credits, bonds, and commodities, using monthly returns from August 1983 to January 2023. This aligns with the dataset used by Bettencourt, Tetereva and Petukhina (2024). Asset returns are sourced from various indices including the Fama-French library, ICE BofA US Corporate Total Return Index, and S&P GSCI Commodity Total Return Index. We track three key macroeconomic factors: inflation, consumption growth, and industrial production growth, obtained from the FRED database. Control variables include year-over-year changes in term spread, credit spread, average S&P 500 dividend and earnings yields, and the macroeconomic factors themselves. To construct zero-cost portfolios, we subtract the risk-free rate from asset returns. All variables were confirmed to be stationary using the Augmented Dickey-Fuller test in Appendix F.

2.2 Linear Economic Tracking Portfolios

We build on the framework introduced by Lamont (2001), which focuses on tracking ‘news’ or unexpected changes in macroeconomic factors. This approach aligns with the rational expectations hypothesis (Muth, 1961), avoiding potential arbitrage opportunities from trading against expected macro factors.

Lamont’s key insight is that asset returns contain information about future discount rates and cash flows, making them predictive of economic indicators. The framework is built on:

$$\Delta \mathbb{E}[y_{t+h} | \mathcal{F}_t] = \mathbb{E}[y_{t+h} | \mathcal{F}_t] - \mathbb{E}[y_{t+h} | \mathcal{F}_{t-1}] = w' \tilde{R}_t + \eta_t, \quad (1)$$

where $y_{t+h} \in \mathbb{R}$ is a macroeconomic variable such as consumption or inflation at time $t + h$ and $\mathcal{F}_t$ is the information set known at the end of timestamp t . $R_t \in \mathbb{R}^K$ denotes the vector of asset returns from the end of timestamp $t - 1$ to the end of timestamp t , and $w \in \mathbb{R}^K$ denotes the desired weights of the tracking portfolio. h corresponds to the desired forecasting horizon for the macroeconomic variable. For instance, when forecasting one-year ahead inflation, we would let $h = 12$ and let y_{t+12} denote the year-over-year inflation from timestamp t to $t + 12$.

By controlling for expected returns using covariates Z_{t-1} , Lamont arrives at the following regression equation for determining the ETP’s weights w :

$$y_{t+h} = w' R_t + c Z_{t-1} + \epsilon_{t,t+h}. \quad (2)$$

This equation is estimated using least-squares, as $\epsilon_{t,t+h}$ is independent of R_t and Z_{t-1} . The resulting estimate provides the portfolio weights w .

2.3 Random Forest ETPs

We extend Lamont’s framework using Random Forests (RFs) to capture non-linear relationships. Our RF ETPs adapt the asset allocation forest framework of Bettencourt et al. (2024), focusing on minimising tracking error.

For each tree, we determine a leaf $L_b(Z_{t-1})$ based on macroeconomic variables Z_{t-1} and compute a target portfolio $\hat{w}_b(Z_{t-1})$ that minimizes tracking error:

$$\hat{w}_b(Z_{t-1}) = \underset{w \in \mathbb{R}^K}{\operatorname{argmin}} \left[\lambda \|w\|_2^2 + \frac{1}{L_b(Z_{t-1})} \sum_{s \in L_b(Z_{t-1})} (w' R_s - y_{s+h})^2 \right],$$

where R_s denotes asset returns at time s , y_{s+h} is the future target factor, and λ is a regularisation parameter determined through 5-fold cross-validation within each leaf.

Our splitting criterion differs from the traditional CART methods (Breiman, Friedman, Olshen & Stone, 1984), focusing on minimising the sum of tracking errors across child nodes instead of impurity:

$$(j^*, \tau^*) = \underset{j \in \{1, \dots, J\}, \tau \in \mathbb{R}}{\operatorname{argmin}} \left[\min_{w_{\text{left}} \in \mathbb{R}^K} \left(\sum_{s: Z_{t-1,j} \leq \tau} \left[(w'_{\text{left}} R_s - y_{s+h})^2 + \lambda_{\text{left}} \|w_{\text{left}}\|_2^2 \right] \right) \right. \\ \left. + \min_{w_{\text{right}} \in \mathbb{R}^K} \left(\sum_{s: Z_{t-1,j} > \tau} \left[(w'_{\text{right}} R_s - y_{s+h})^2 + \lambda_{\text{right}} \|w_{\text{right}}\|_2^2 \right] \right) \right],$$

where j^* corresponds to the optimal splitting feature, τ^* corresponds to the optimal splitting threshold, and $\lambda_{\text{left}}, \lambda_{\text{right}} \in \mathbb{R}_{\geq 0}$ are L_2 regularisation parameters found using 5-fold cross-validation within the leaves. This approach takes the regularisation effect of the Ridge regressions in the leaves into account while constructing the splits and ensures that the resulting splits focus on minimising tracking errors, regardless of the current macroeconomic environment.

Similar to ordinary random forests, we perform recursive subpartitioning of the data where we iteratively split the data into increasingly smaller partitions or leaves until we reach leaf sizes that are smaller than a

predefined threshold. This process allows the trees to capture complex interactions and non-linear relationships in the data. We also make use of bagging, as introduced by Breiman (1996), which involves generating multiple subsets of the training data making use of random sampling with replacement. This enhances model performance by reducing variance. Each tree in the forest is trained on a different sample, ensuring that the trees are diverse which improves the model's overall robustness and out-of-sample performance.

To obtain the overall posterior portfolio weights from the RF ETP, we take the average of the predicted portfolios from all underlying trees:

$$\hat{w}(Z_{t-1}) = \frac{1}{B} \sum_{b=1}^B \hat{w}_b(Z_{t-1}). \quad (3)$$

Appendix C illustrates an example of an asset allocation tree using credit spread and term spread as macroeconomic indicators.

2.4 Local Linear Forest ETPs

Local Linear Forest (LLF) ETPs combine the strengths of random forests with local linear regression. To understand LLFs, we first introduce the kernel interpretation of RFs in the univariate case, then extend it to the multivariate scenario for portfolio construction.

In the univariate prediction scenario, random forests can be reinterpreted as adaptive kernel methods (Athey et al., 2019; Meinshausen & Ridgeway, 2006). Consider a dataset of size S with observations $(Z_1, Y_1), \dots, (Z_S, Y_S)$, where the goal is to predict $\mu(z) = \mathbb{E}[Y | Z = z]$. The RF prediction can be expressed as:

$$\hat{\mu}(z) = \frac{1}{B} \sum_{b=1}^B \sum_{s=1}^S Y_s \cdot \frac{\mathbb{1}\{Z_s \in L_b(z)\}}{|L_b(z)|} = \sum_{s=1}^S Y_s \cdot \frac{1}{B} \sum_{b=1}^B \frac{\mathbb{1}\{Z_s \in L_b(z)\}}{|L_b(z)|} = \sum_{s=1}^S \alpha_s(z) Y_s. \quad (4)$$

The LLF extends this concept by not simply averaging the weighted observations, but by fitting weighted ridge regressions for each prediction point using the kernel weights $\alpha_s(z)$.

While this example illustrates the kernel interpretation for a univariate prediction, our LLF ETP extends this concept to predict entire portfolios. Specifically, instead of predicting a single value $Y \in \mathbb{R}$, we predict a vector of portfolio weights $w \in \mathbb{R}^K$. The LLF ETP combines the kernel interpretation with the regression equation by Lamont (2001). However, instead of taking the average portfolio weight as done with RF ETPs, we fit a weighted ridge regression on all historical data for each prediction. The weights for this regression are provided by $\alpha_s(Z_t)$, the kernel weights derived from the RF structure, where t represents the current timestamp and s represents the timestamp of a historical data point. The prediction formula for the LLF ETP is:

$$\begin{bmatrix} \hat{c}(Z_t) \\ \hat{w}(Z_t) \end{bmatrix} = \left(\begin{bmatrix} Z^{\text{train}'} \\ R^{\text{train}'} \end{bmatrix} A(Z_t) \begin{bmatrix} Z^{\text{train}} & R^{\text{train}} \end{bmatrix} + \lambda I \right)^{-1} \begin{bmatrix} Z^{\text{train}'} \\ R^{\text{train}'} \end{bmatrix} A(Z_t) Y^{\text{train}}, \quad (5)$$

where $R^{\text{train}} \in \mathbb{R}^{S \times K}$ is the matrix of zero-cost asset returns for the S historical train observations, $Z^{\text{train}} \in \mathbb{R}^{S \times J}$ is the matrix of lagged covariates for all historical observations, and $Y^{\text{train}} \in \mathbb{R}^S$ are the corresponding future target factors y_{s+h} for all historical observations. $\lambda \in \mathbb{R}_{\geq 0}$ corresponds to the strength of the L_2 -penalty and is found using 5-fold cross-validation for each separate predicted value. $A(Z_t) \in \mathbb{R}^{S \times S}$ is a diagonal matrix where $A(Z_t)_{ss} = \alpha_s(Z_t) \forall s \in \{1, \dots, S\}$ and $A(Z_t)_{su} = 0 \forall s \neq u$.

$\hat{c}(Z_t) \in \mathbb{R}^J$ and $\hat{w}(Z_t) \in \mathbb{R}^K$ correspond to the final obtained parameters in Lamont (2001)'s regression equation. However, unlike in Lamont's original regression equation, these parameters are dependent on current macroeconomic conditions via the adaptive LLF kernel weighting. The final ETP weights are $\hat{w}(Z_t)$. This approach allows for a flexible, data-driven method to capture both global non-linear structures and local linear trends, potentially improving the model's ability to adapt to different economic regimes.

2.5 Model Training and Evaluation

We use an expanding window approach, retraining models annually with an initial 20-year training set. Performance is evaluated using the Mincer-Zarnowitz regression:

$$y_{t+h} = \alpha + \beta \cdot r_t + \epsilon,$$

where y_{t+h} is the future target variable and r_t are portfolio returns. The primary metric is the R^2 of this regression.

2.6 Interpretation Techniques

To interpret the complex relationships captured by RF and LLF ETPs, we employ Shapley values, derived from cooperative game theory (Shapley, 1953) and widely used in machine learning to interpret model predictions (Lundberg & Lee, 2017; Štrumbelj & Kononenko, 2014). Let w_{t+1} be the portfolio weights constructed at the end of timestamp t , determined by covariates $Z_t \in \mathbb{R}^J$. We define a portfolio metric $M(w_t)$ as a function of the weights. For example, the relative allocation between commodities and bonds:

$$M(w_t) = \frac{w_{t,\text{comm}} - w_{t,\text{bonds}}}{\|w_t\|_1}, \quad (6)$$

where $\|w_t\|_1 = \sum_{k=1}^K |w_{t,k}|$ is the L_1 -norm of the weights. The Shapley value for covariate $j \in \{1, \dots, J\}$ is given by

$$\phi_j = \sum_{Q \subseteq N \setminus \{j\}} \frac{|Q|!(|N| - |Q| - 1)!}{|N|!} [v(Q \cup \{j\}) - v(Q)], \quad (7)$$

where $N = \{1, 2, \dots, J\}$ is the set of all covariates, Q is a subset of N excluding j , and $v(Q)$ is the value function representing the expected portfolio metric when only covariates in Q are known:

$$v(Q) = \mathbb{E}[M(w_t) \mid Z_{t-1,Q}]. \quad (8)$$

This framework quantifies the impact of each macroeconomic covariate on portfolio allocation decisions and allows us to investigate how these relationships evolve dynamically.

Additionally, we use Kernel Principal Component Analysis (KPCA) on the LLF kernel to identify economic regimes. KPCA applies PCA within the high-dimensional feature space implicitly defined by the kernel function $k(z, z')$. Let $\psi : \mathbb{R}^J \rightarrow \mathcal{H}$ be a mapping function projecting data point Z_t into a Hilbert space. The Moore-Aronszajn theorem states that for any symmetric positive definite function $k : \mathbb{R}^J \times \mathbb{R}^J \rightarrow \mathbb{R}$, there exists a unique Hilbert space $\mathcal{H}$ and mapping function ψ such that:

$$k(z, z') := \langle \psi(z), \psi(z') \rangle_{\mathcal{H}}.$$

KPCA is performed by solving the eigenvector equation:

$$S\gamma v = (K - \mathbf{1}_S K - K \mathbf{1}_S + \mathbf{1}_S K \mathbf{1}_S)v,$$

where S is the number of data points, γ and v are eigenvalue-eigenvector pairs, $K \in \mathbb{R}^{S \times S}$ is the kernel matrix, and $\mathbf{1}_S$ is the S -by- S matrix with each element $1/S$.

The resulting principal components represent different economic regimes or factors influencing the relationship between asset returns and macroeconomic variables. Visualizing the data projected onto the first two principal components allows us to identify clusters of time points with similar economic characteristics.

3 Results

3.1 Tracking Performance

Machine learning-based ETPs consistently outperform traditional linear ETPs across various macroeconomic factors and time horizons. Table 1 summarizes the performance metrics. More extensive results including turnover metrics can be found in Appendix D.

Table 1: ETP metrics for inflation, consumption, and growth portfolios. August 2003 - January 2023.

		1 month			1 year		
		Lin. ETP	RF ETP	LLF ETP	Lin. ETP	RF ETP	LLF ETP
Infl.	MZ R^2	0.40	0.41	0.34	0.044	0.059	0.069
	MZ slope	1.15	2.86	1.20	1.66	3.98	1.90
Cons.	MZ R^2	0.020	0.021	0.017	0.031	0.048	0.074
	MZ slope	2.09	4.69	1.44	2.26	8.87	4.79
Growth	MZ R^2	0.047	0.13	0.11	0.068	0.057	0.071
	MZ slope	3.86	6.01	2.48	8.89	3.87	2.44

For short-term forecasts (+1 month), RF ETPs demonstrate superior performance. Notably, the RF ETP achieves a Mincer-Zarnowitz (MZ) R^2 of 13% for tracking industrial production growth, compared to just 4.7% for the linear ETP. This substantial improvement highlights the RF model’s ability to capture complex, non-linear relationships between asset returns and macroeconomic variables.

For longer-term forecasts (+1 year), LLF ETPs consistently achieve the highest MZ R^2 values. This could be due to the LLF ETP’s ability to model both global non-linear relationships and local linear trends. The improvement is particularly pronounced for consumption growth tracking, with the LLF ETP reaching an MZ R^2 of 7.4% compared to 3.1% for the linear ETP.

To assess statistical significance, we conducted one-sided Diebold-Mariano tests. For the one-year horizon, the LLF ETP shows statistically significant improvements over the linear ETP for both inflation (p-value = 0.0125) and consumption tracking (p-value = 0.0113).

3.2 Shapley Value Interpretation: One-Year Ahead Inflation LLF ETP

The LLF ETP demonstrates superior performance in predicting one-year inflation, achieving a MZ R^2 of 6.9% (Table 1). We employ Shapley value analysis to gain deeper insights, focusing on the portfolio metric $M(w_t) = \frac{w_{t,comm} - w_{t,credits}}{\|w_t\|_1}$, representing the relative allocation between commodities and credits, adjusted for overall portfolio leverage.

Figure 9 in Appendix K reveals a non-linear relationship between consumption growth and the relative importance of commodities versus credits. As consumption growth increases, the portfolio tends to allocate more towards commodities, with this effect diminishing at around 2.5% year-over-year growth. This aligns with Hamilton (2009)’s description of strong global economic growth making oil prices more responsive to inflationary pressures.

An interaction effect with the S&P 500 earnings yield is also observed. Low earnings yields correspond to more extreme Shapley values, indicating a stronger influence of consumption growth on portfolio allocation. These dynamics can be interpreted in different macroeconomic environments:

1. Economic Stress: During low consumption growth and low earnings yields, the portfolio shifts towards credits. This aligns with the flight-to-quality phenomenon described by Beber et al. (2009).
2. Economic Growth: As the economy expands, commodities are favoured for inflation tracking, consistent with Gorton and Rouwenhorst (2006)’s findings on commodity returns and unexpected inflation.

3. High Earnings Yield Environments: The effect of consumption growth on allocation is dampened, potentially indicating a more stable economic environment.

Figure 10 in Appendix K shows how different covariates influence the commodity-credit allocation over time. Consumption growth and earnings yield consistently show the largest effects, with year-over-year inflation playing a significant role, particularly during high inflation periods.

These findings highlight that the relationships between asset classes and macroeconomic indicators are complex and vary with changing economic conditions. This underscores the importance of dynamically adjusting tracking portfolio weights based on macroeconomic indicators to enhance tracking performance, rather than relying on a fixed decomposition of asset classes into macroeconomic factors as proposed by Lohre et al. (2020).

3.3 LLF Kernel Interpretation

To interpret the complex kernel generated by the LLF and gain insights into the economic regimes captured by the model, we employ Kernel Principal Component Analysis (KPCA) (Schölkopf et al., 1997). This technique allows us to project the high-dimensional feature space onto lower dimensions, enabling visualisation and interpretation of the economic regimes identified by the LLF.

We focus on the inflation tracking portfolio and use KPCA to project the high-dimensional latent Hilbert space generated by the LLF kernel onto two dimensions. This visualisation allows us to identify which timestamps are perceived as similar and thus in the same regime.

Figure 11 in Appendix K illustrates the data points projected onto two dimensions via KPCA. The clustering of data points reveals distinct economic regimes:

1. Economic Crises (top left): The 2008 financial crisis, Covid-19 pandemic, and the 1991 recession are clustered together. This grouping aligns with Reinhart and Rogoff (2009)’s argument that financial crises, despite varied origins, often share similar patterns.
2. Overheated Economy (right side): Data points corresponding to 2006 (pre-2008 crisis) and 1989 (pre-1991 recession) are grouped together. This cluster likely represents periods of economic overheating, characterised by rapid economic growth and excessive credit growth (Kaminsky & Reinhart, 1999).
3. Reaganomics (bottom right): This cluster represents the economic policies of the 1980s, characterised by tax cuts, deregulation, and increased military spending.
4. Stable Growth (bottom): The late 1990s and the period of growth around 2015 are clustered here, likely representing periods of stable, moderate economic growth.

The Reaganomics cluster’s intermediate positioning between stable growth and economic overheating suggests that the LLF model identifies it as sharing characteristics with both. This aligns with the mixed outcomes of the period described by Niskanen (1988), noting significant economic growth alongside increased federal debt and trade deficits.

The clear separation of these clusters demonstrates the LLF model’s ability to identify and differentiate between distinct economic regimes. The clustering patterns provide insights into the model’s understanding of economic dynamics. For instance, the proximity of pre-crisis periods (e.g., 2006, 1989) suggests that the model captures early warning signs of economic instability, aligning with research on leading indicators of financial crises (Kaminsky & Reinhart, 1999).

4 Conclusion

This study advances the construction of economic tracking portfolios (ETPs) by integrating machine learning techniques, specifically random forests (RFs) and local linear forests (LLFs). Our findings demonstrate that

these approaches consistently outperform traditional linear ETPs across various macroeconomic factors and time horizons, offering improved tools for economic forecasting and risk management.

The superior performance of ML-based ETPs, particularly LLFs for long-term forecasts, highlights their ability to capture complex, non-linear relationships between asset returns and macroeconomic variables. For one-year horizon forecasts, LLF ETPs show statistically significant improvements in tracking both inflation and consumption growth. This enhanced performance could lead to more accurate economic projections, potentially improving monetary policy decisions, investment strategies, and capital allocation.

However, the improved performance comes with trade-offs. ML-based ETPs exhibit higher turnover rates, which could lead to increased transaction costs in practical applications. This presents a challenge for portfolio managers and investors who must weigh the benefits of improved tracking against potentially higher implementation costs.

The Shapley value analysis reveals nuanced, time-varying relationships between macroeconomic variables and portfolio allocations. For instance, the relationship between commodities, credits, and inflation is not static but varies with economic conditions. During periods of low consumption growth and economic stress, credits become more correlated with inflation than commodities. This challenges the fixed factor decomposition proposed in previous studies (Lohre et al., 2020) and underscores the importance of dynamic portfolio adjustment based on changing economic conditions.

The LLF kernel interpretation using KPCA provides a novel approach to identifying economic regimes. The clear separation of clusters corresponding to different economic conditions (e.g., crises, stable growth, overheating) demonstrates the model's ability to capture and differentiate complex economic dynamics. This could have broad applications in economic research and policy analysis, offering a data-driven method for regime identification.

While our results are promising, several limitations and areas for future research emerge. The performance of all ETPs, including ML-based ones, may be influenced by the specific economic conditions during our sample period. Future studies could explore the robustness of these results across different economic regimes and longer time horizons. Additionally, the higher turnover rates of ML-based ETPs warrant further investigation into their net performance after accounting for transaction costs.

In conclusion, this study demonstrates the potential of machine learning techniques, particularly RFs and LLFs, in improving the performance of ETPs. By capturing complex relationships between macroeconomic variables, these models offer enhanced tools for economic forecasting and risk management. The insights gained from interpreting these models contribute to our understanding of dynamic economic relationships and regime changes. These advanced modeling techniques may become invaluable for policymakers, investors, and economists alike.

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A Data Sources and Descriptive Statistics

Table 2: Data sources and series identifiers for asset classes used in the ETPs.

Asset Class	Data Source	Series Name
Equity	Fama French library	‘Mkt-RF’ + ‘RF’
Bonds	10-year maturity US ZCB yields	SVENY10
Credits	ICE BofA US Corporate TR	BAMLCC0A0CMTRIV (FRED)
High Yields	Bloomberg US Corporate High Yield Total Return Index	LF98TRUU
Commodities	S&P GSCI Commodity TR	S&P GSCI Commodity TR
Gold	Bloomberg Gold Spot Rate	XAUUSD
Oil	Bloomberg WTI Crude Oil Subindex	BCOMCL
Risk-Free	Fama French library	RF

Table 3: Data sources and series identifiers for macroeconomic factors tracked and other covariates used in the ETPs.

Variable	Data Source	Series Name
<i>Macroeconomic Factors Tracked</i>		
Inflation	FRED	Consumer Price Index for All Urban Consumers: All Items in U.S. City Average (CPIAUCSL)
Consumption Growth	FRED	Real Personal Consumption Expenditures: Non-durable Goods (DNDGRA3M086SBEA) + Services (DSERRA3M086SBEA)
Industrial Production Growth	FRED	Industrial Production Index (INDPRO)
<i>Other Macroeconomic Covariates</i>		
US Dividend Yield	Quandl	MULTPL/SP500_DIV_YIELD_MONTH
US Earnings Yield	Quandl	MULTPL/SP500_EARNINGS_YIELD_MONTH
Credit Spread	FRED	‘GS20’ - ‘GS30’

Table 4: Pearson correlation coefficients between asset returns and future macroeconomic indicators (1-month and 1-year horizons), August 1983 - January 2023. The upper panel presents correlations with raw assets returns, while the lower panel shows correlations with excess returns (asset returns minus risk-free rate).

	Equity	Bonds	Credits	Comm	HY
+1m inflation	0.14	-0.14	0.07	0.64	0.22
+1y inflation	0.10	-0.06	0.01	0.26	0.05
+1m cons. growth	0.23	-0.04	0.21	0.19	0.22
+1y cons. growth	0.18	0.00	0.10	0.02	0.13
+1m IP growth	0.18	-0.14	0.10	0.30	0.21
+1y IP growth	0.22	-0.08	0.08	0.09	0.23
	Equity minus RF	Bonds minus RF	Credits minus RF	Comm minus RF	HY minus RF
+1m inflation	0.13	-0.16	0.04	0.64	0.20
+1y inflation	0.08	-0.09	-0.03	0.25	0.03
+1m cons. growth	0.23	-0.04	0.21	0.19	0.22
+1y cons. growth	0.18	-0.00	0.09	0.01	0.12
+1m IP growth	0.18	-0.14	0.10	0.30	0.20
+1y IP growth	0.22	-0.09	0.08	0.09	0.23

Table 5: Descriptive statistics of the covariates. Time period is August 1983 to January 2023.

	Min	Max	Med	Avg	SD
YoY term spread change	-0.025	0.032	-0.000	-0.000	0.006
YoY credit spread change	-0.009	0.016	-0.000	-0.000	0.002
YoY div. yield change	-0.004	0.006	-0.000	-0.000	0.001
YoY inflation	-0.020	0.090	0.027	0.028	0.016
YoY consumption growth	-0.116	0.207	0.026	0.025	0.019
YoY IP growth	-0.173	0.162	0.026	0.020	0.043
S&P 500 avg. earn. yield	0.008	0.108	0.050	0.051	0.017

Figure 1 presents the correlation matrix of asset returns, including the risk-free rate.

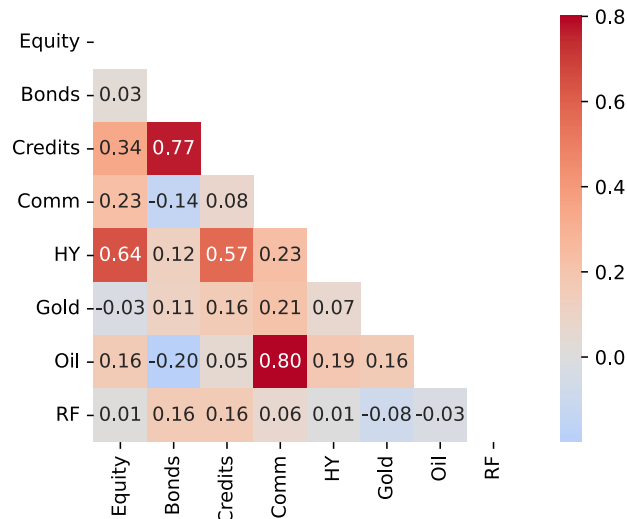


Figure 1: Correlation matrix of asset returns (August 1983 to January 2023)

B Hyperparameter Details

The performance of machine learning models, including our RF and LLF ETPs, can be sensitive to the choice of hyperparameters. Table 6 provides a comprehensive list of the hyperparameters used in our RF and LLF models, along with their descriptions and chosen values. These parameters were used consistently across all models to ensure fair comparison and to avoid potential look-ahead bias that could arise from extensive hyperparameter tuning.

Table 6: Hyperparameters, descriptions, and corresponding values for the RF and LLF ETPs.

Hyperparameter	Description	(Recommended) Value
Number of Trees (B)	Number of trees in the ensemble	200
Honesty Fraction	Fraction of data used for honest estimation	0.5
Minimum Observations per Leaf	Minimum number of samples required to be in a leaf node	36
Maximum Depth of Trees	Maximum depth of the individual trees	3
Inner regression L_2 penalty (λ)	Controls the trade-off between tracking error minimisation and portfolio weight magnitudes	5-fold cross-validation per leaf
Covariate Subset Size	Number of covariates to consider when looking for the best split	$\left\lfloor \sqrt{\frac{\text{number of covariates}}{2}} \right\rfloor = 2$

C Asset Allocation Tree Illustration

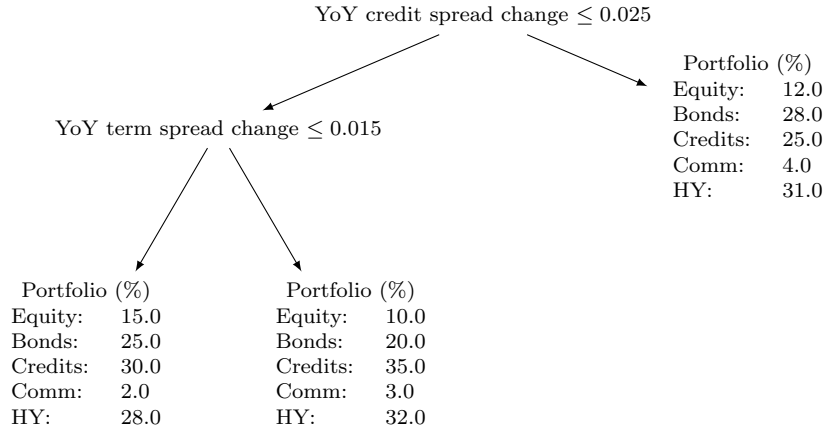


Figure 2: Asset allocation tree based on year-over-year (YoY) changes in credit spread and term spread, illustrating how different economic conditions lead to distinct portfolio allocations.

Figure 2 illustrates an example of an asset allocation tree using credit spread and term spread as macroeconomic indicators. This tree demonstrates how different economic conditions lead to different portfolio allocations.

D Detailed Results Table

Table 7: Detailed ETP metrics for inflation, consumption, and growth portfolios. August 2003 - January 2023.

		1 month			1 year		
		Lin. ETP	RF ETP	LLF ETP	Lin. ETP	RF ETP	LLF ETP
Infl.	Avg. turnover	0.024	0.026	0.065	0.051	0.091	0.195
	Avg. leverage	0.046	0.013	0.031	0.073	0.027	0.055
	MZ R^2	0.40	0.41	0.34	0.044	0.059	0.069
	MZ intercept	0.0020	0.0021	0.0020	0.025	0.025	0.025
	MZ slope	1.15	2.86	1.20	1.66	3.98	1.90
Cons.	Avg. turnover	0.019	0.033	0.062	0.066	0.115	0.152
	Avg. leverage	0.024	0.011	0.019	0.106	0.026	0.054
	MZ R^2	0.020	0.021	0.017	0.031	0.048	0.074
	MZ intercept	0.0014	0.0014	0.0014	0.019	0.019	0.018
	MZ slope	2.09	4.69	1.44	2.26	8.87	4.79
Growth	Avg. turnover	0.030	0.071	0.093	0.038	0.33	0.264
	Avg. leverage	0.041	0.017	0.036	0.074	0.119	0.181
	MZ R^2	0.047	0.13	0.11	0.068	0.057	0.071
	MZ intercept	0.0009	0.0000	0.0000	0.0050	0.0045	0.0043
	MZ slope	3.86	6.01	2.48	8.89	3.87	2.44

E Robustness Check: Hyperparameter Sensitivity

Ideally, annual expanding window hyperparameter optimisation would be conducted through cross-validation, ensuring the hyperparameters are always tuned on the data available at that time. However, due to computation limitations, this approach is impractical. Instead, we chose a fixed set of hyperparameters recommended by prior studies, avoiding any tuning to prevent look-ahead bias (Appendix B).

This makes it important to assess whether our chosen hyperparameters are representative of typical RF and LLF performance. To evaluate this, we conducted robustness checks by varying one hyperparameter at a time from our fixed hyperparameter set. We then fit the portfolio using all data available up to 31 December 2009 and evaluate the performance up to 31 January 2023 without retraining. The average MZ R^2 across all three factors was used to assess performance.

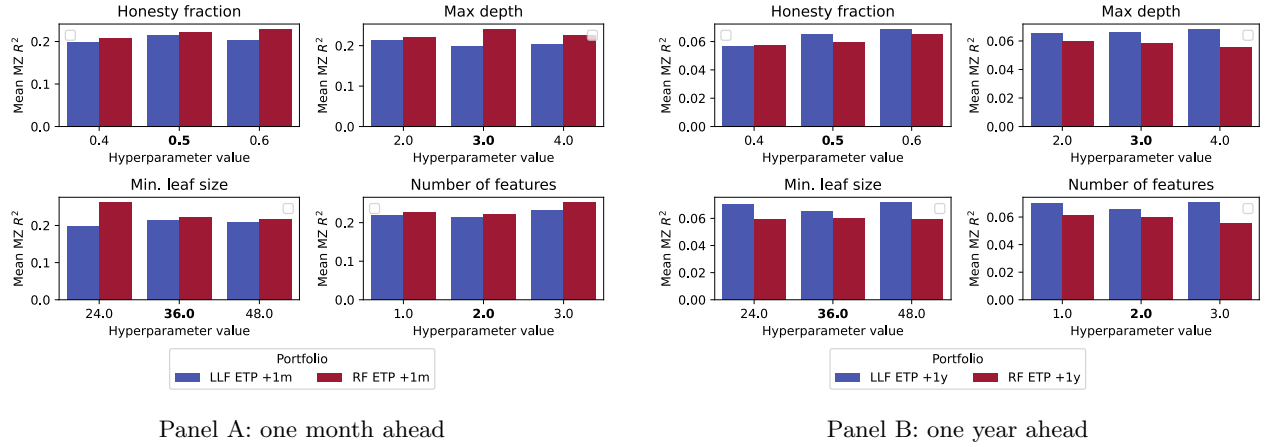


Figure 3: Sensitivity analysis of RF and LLF ETPs to hyperparameter variations. The plots show the average out-of-sample $MZ R^2$ across all macroeconomic factors when varying individual hyperparameters. Panel A: one month ahead, Panel B: one year ahead, both fitted on 31 August 1983 to 31 December 2009, evaluated on 1 January 2010 to 31 January 2023. The original hyperparameter values are highlighted in bold.

Figure 3 demonstrates that for both one-month and one-year horizons, the average $MZ R^2$ is relatively insensitive to hyperparameter variations. For the one-year horizon portfolios, the impact on $MZ R^2$ is minimal, with absolute changes in $MZ R^2$ amounting to only fractions of a per cent. For the one-month horizon portfolios, the variations are slightly more pronounced but still within a few per cent.

For context, linear ETPs typically perform significantly worse. Table 1 reports a 3.1% $MZ R^2$ for the 1-year consumption tracking portfolio and 7.4% for the 1-year LLF ETP. Additionally, the $MZ R^2$ of the 1-month linear growth tracking portfolio is only 4.7% whereas the RF ETP achieves 13%. This indicates that both RF and LLF ETPs consistently outperform their linear counterparts, even with suboptimal hyperparameters.

Finally, the figures suggest that the originally chosen hyperparameters are rarely the optimal ones. This implies that our models' out-of-sample performance can potentially be improved using cross-validation, indicating room for further optimisation.

F Augmented Dickey-Fuller test results

Table 8 presents the test statistics and corresponding p values of the augmented Dickey-Fuller (ADF) test with trend applied to the data. The results indicate that for all covariates, the null hypothesis of non-stationarity is rejected against 5% significance, leading us to conclude that all covariates are stationary. This finding supports the use of these variables in our models.

Table 8: Results of augmented Dickey-Fuller tests with trend for stationarity of macroeconomic covariates, August 1983 - January 2023. Lower p values indicate stronger evidence against the null hypothesis of non-stationarity.

	ADF test statistic	p value
Year-over-year term spread change	-5.81	0.00***
Year-over-year credit spread change	-13.83	0.00***
Year-over-year average dividend yield change	-5.04	0.00***
Year-over-year inflation	-3.61	0.01**
Year-over-year consumption growth	-6.13	0.00***
Year-over-year IP growth	-5.77	0.00***
S&P 500 average earnings yield	-3.49	0.01**

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

G Robustness Check: Covariate Lookbacks

To further validate our findings, we conducted a robustness check using alternative specifications for the macroeconomic covariates. Instead of year-over-year differences, we use first differences (monthly changes) for inflation, consumption, industrial production, dividend yield, credit spread and term spread covariates.

The results in Table 9 largely reinforce our main findings, with tree-based methods outperforming linear ETPs in most cases. However, an anomaly is observed for 1-month consumption forecasts, where the linear ETP shows superior performance.

Table 9: Performance metrics for inflation, consumption, and growth ETPs. August 31, 2003 to January 31, 2023 with yearly retraining for the RF and LLF ETP and monthly retraining for the linear ETP. Bolded values indicate the highest MZ R^2 for each horizon and economic factor.

		1-month			1-year		
		Lin. ETP	RF ETP	LLF ETP	Lin. ETP	RF ETP	LLF ETP
Infl.	Avg. turnover	0.015	0.039	0.083	0.051	0.167	0.272
	Avg. leverage	0.027	0.013	0.030	0.075	0.037	0.061
	MZ R^2	0.22	0.37	0.35	0.043	0.059	0.056
	MZ intercept	0.0022	0.0021	0.0020	0.025	0.025	0.025
	MZ slope	2.13	2.79	1.26	1.67	3.08	1.68
Cons.	Avg. turnover	0.016	0.040	0.097	0.034	0.113	0.219
	Avg. leverage	0.029	0.008	0.014	0.074	0.023	0.052
	MZ R^2	0.092	0.0073	0.0055	0.043	0.066	0.061
	MZ intercept	0.0011	0.0015	0.0016	0.019	0.019	0.019
	MZ slope	4.71	3.08	0.961	3.30	12.1	4.23
Growth	Avg. turnover	0.030	0.091	0.17	0.037	0.467	0.355
	Avg. leverage	0.042	0.017	0.034	0.074	0.150	0.181
	MZ R^2	0.045	0.126	0.068	0.064	0.060	0.065
	MZ intercept	0.0009	0.0002	0.0002	0.0052	0.0045	0.0045
	MZ slope	3.66	5.86	1.62	8.73	3.28	2.41

Further investigation reveals that this anomaly is primarily driven by the extreme economic conditions during the Covid-19 pandemic. When excluding April, May, and June 2020 from the analysis, the performance of RF and LLF ETPs for 1-month consumption forecasts improved significantly:

Table 10: Performance metrics for the 1-month consumption ETP. Data spans August 31, 2003 to January 31, 2023, excluding the period from April 2020 to June 2020 to account for Covid-19 economic disruptions. The highest MZ R^2 is in bold.

		1-month		
		Lin. ETP	RF ETP	LLF ETP
Cons.	Avg. turnover	0.016	0.040	0.086
	Avg. leverage	0.029	0.008	0.016
	MZ R^2	0.035	0.054	0.056
	MZ intercept	0.0015	0.0015	0.0015
	MZ slope	1.31	3.78	1.06

H Diebold-Mariano Test Results

To rigorously assess the significance of the performance improvements by the machine learning-based ETPs, we conduct one-sided Diebold-Mariano (DM) tests. These tests compare the forecasting accuracy of the best-performing machine learning ETP against the linear ETP for each macroeconomic factor and forecast horizon.

Table 11: Diebold-Mariano test results comparing the forecasting accuracy of the best-performing machine learning-based ETP against linear ETPs. The table presents results for 1-month and 1-year forecast horizons across inflation, consumption, and growth factors.

Horizon	Factor	Best Model	Best MZ R^2	Lin. ETP MZ R^2	DM Stat.	p value
1 month	Inflation	RF ETP	0.4089	0.4017	0.1031	0.5412
	Consumpt.	RF ETP	0.0021	0.0199	1.0848	0.8606
	Growth	RF ETP	0.1281	0.0467	-1.1850	0.1183
1 year	Inflation	LLF ETP	0.0685	0.0436	-1.9672	0.0249**
	Consumption	LLF ETP	0.0744	0.0309	-2.0097	0.0225**
	Growth	LLF ETP	0.0705	0.0677	-0.1811	0.4282

Notes: ** indicates significance at the 5% level for a one-sided test.

In this one-sided test, the null hypothesis is that the forecast accuracy of the machine learning-based ETP and the linear ETP are equal. The alternative hypothesis is that the machine learning-based ETP outperforms the linear ETP.

The results in Table 11 demonstrate the statistically significant performance improvements of the LLF ETPs in long-term (1-year) inflation and consumption tracking, with p values below the 5% significance threshold.

I Portfolio Weights Over Time

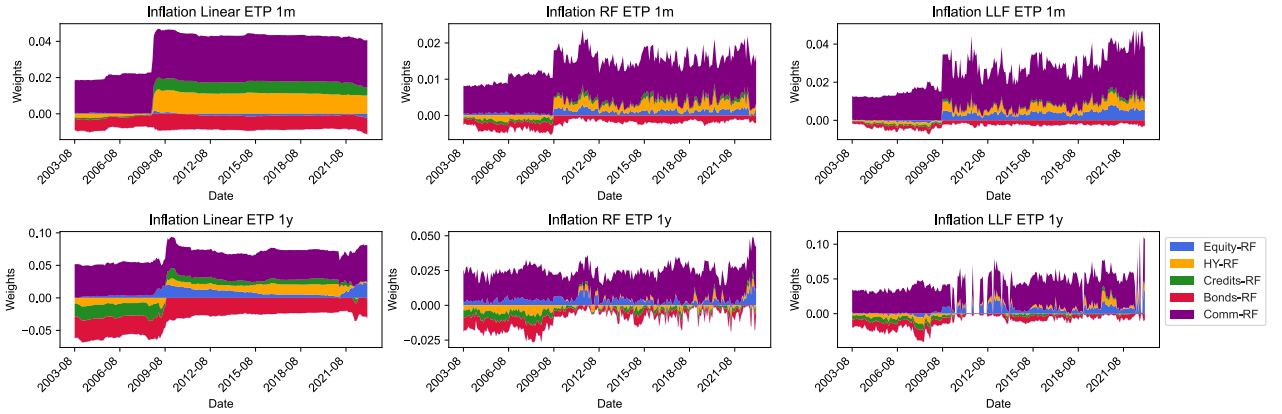


Figure 4: Portfolio weights of inflation tracking portfolios, August 2003 - January 2023 with yearly retraining for the RF and LLF ETPs and monthly retraining for the linear ETP.

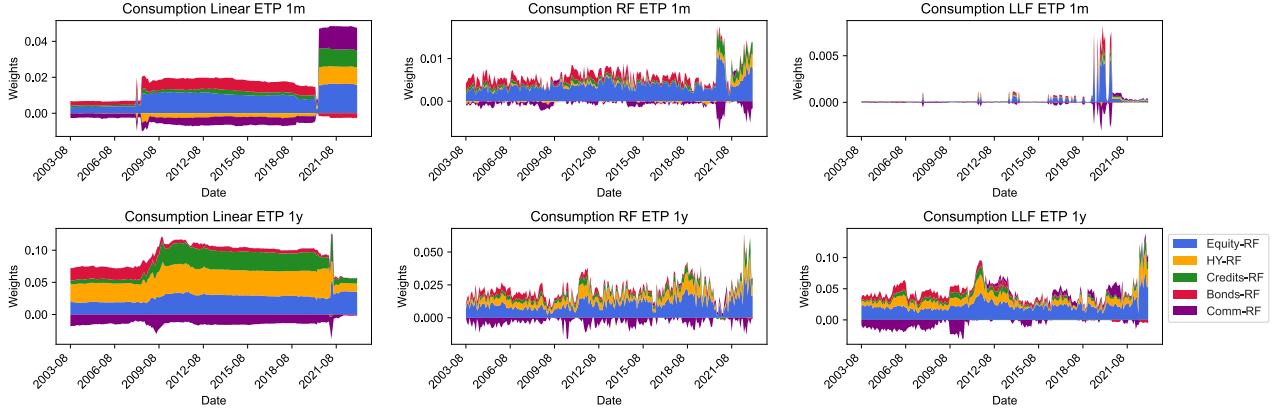


Figure 5: Portfolio weights of consumption tracking portfolios, August 2003 - January 2023 with yearly retraining for the RF and LLF ETPs and monthly retraining for the linear ETP.

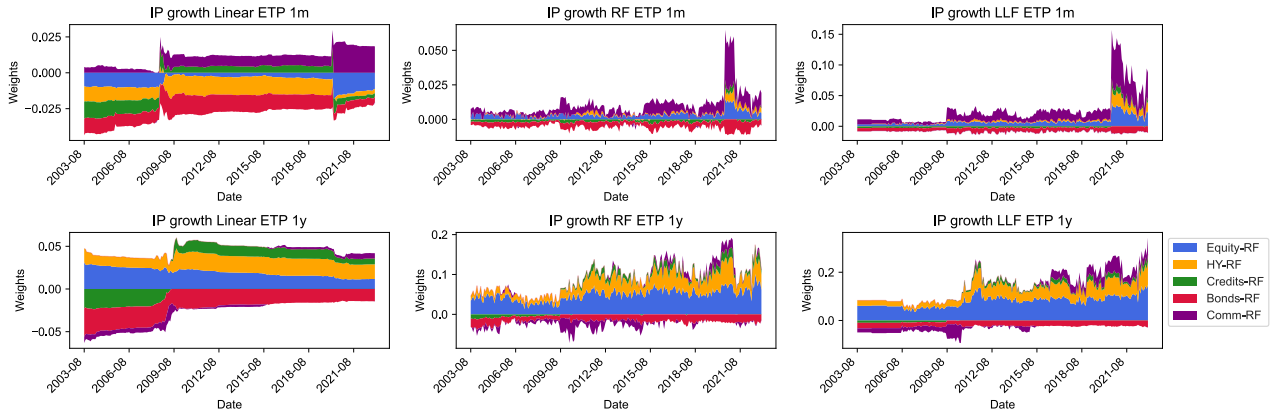


Figure 6: Portfolio weights of IP growth tracking portfolios, August 2003 - January 2023 with yearly retraining for the RF and LLF ETPs and monthly retraining for the linear ETP.

J Commodities, Credits and Inflation during the 2008 Financial Crisis

To provide additional context for the Shapley value analysis in Section 3.2, we examine the changing correlations between inflation, commodities, and credits during the 2008 financial crisis.

Figure 7 shows a significant increase in the correlation between inflation and credits during the 2008 financial crisis that is higher than the increase in correlation between inflation and commodities. This shift in correlations explains the LLF ETP's tendency to favour credits more during economic downturns, as observed in the Shapley value analysis (Figure 9) and the portfolio weights over time (Figure 8).

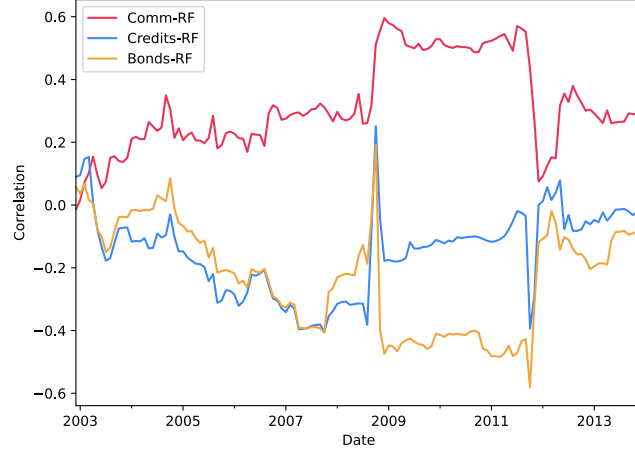


Figure 7: Rolling 3-year correlations of commodities and credits with inflation from January 2001 to January 2014. This period contains significant economic events, including the 2008 financial crisis.

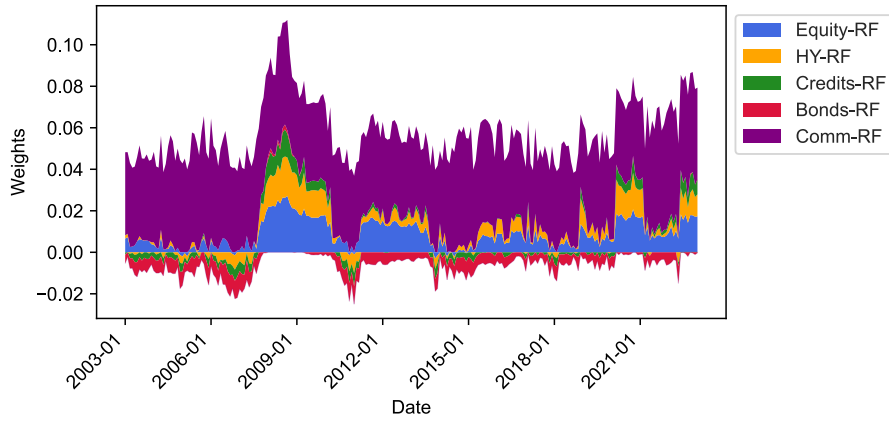


Figure 8: Portfolio weights of LLF ETP from January 2003 to January 2023, model fitted with data until January 2020 which makes the portfolio weights easier to interpret.

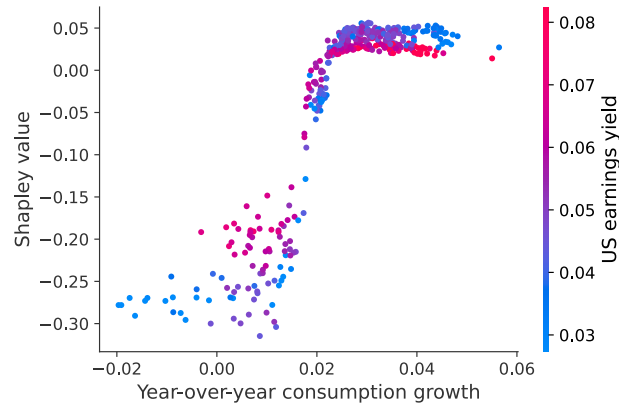


Figure 9: Shapley values with respect to $M(w_t) = \frac{w_{t,comm} - w_{t,credits}}{\|w_t\|_1}$ for the LLF inflation tracking portfolio, as a function of year-over-year consumption growth. Data spans August 1983 to January 2020, colour-coded by the S&P 500 average earnings yield.

K Supplementary Figures

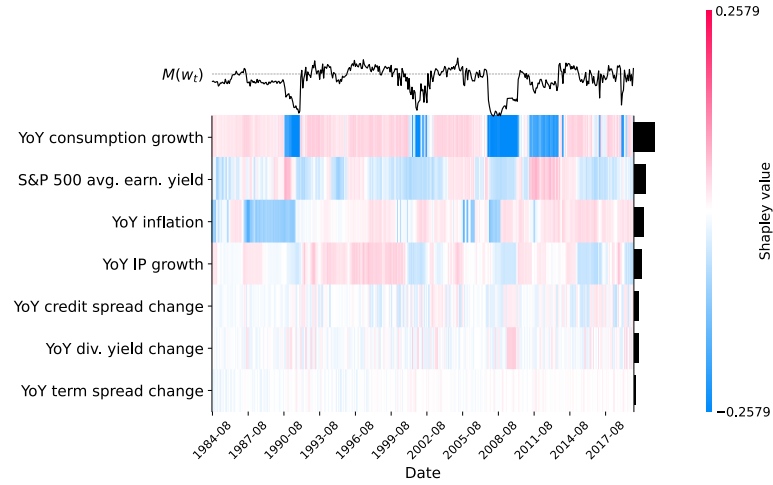


Figure 10: Heatmap of Shapley values with respect to $M(w_t) = \frac{w_{t,comm} - w_{t,credits}}{\|w_t\|_1}$ for the LLF inflation tracking portfolio per covariate over time. Data spans August 1983 to January 2020, colour-coded by Shapley value. Red represents shifts towards commodities and blue towards credits. Black bars show the average absolute Shapley value as a proxy for feature importance.

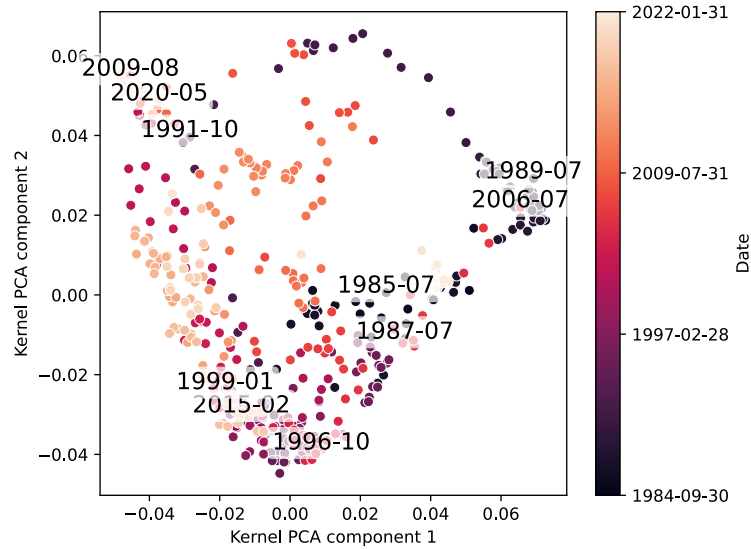


Figure 11: Kernel PCA projection of LLF kernel weights for the 12-month inflation tracking portfolio. Data points from 31 August 1983 to 31 January 2023 are projected onto the first two principal components, revealing distinct clusters corresponding to different economic regimes. Colour represents the date of a data point.

L Programming code

The accompanying code repository contains the following components:

- A Python package called `llaaf` that implements core functionalities, including portfolio construction, backtesting, and evaluation methods.
- Jupyter notebooks in the `notebooks/` directory that execute runs and generate plots and tables used in the report.
- Data assets and intermediate output files stored in the `assets/` and `output/` directories, respectively.

- R code for comparisons with the **grf** package for univariate LLFs and data preparation.

The **llaaf** package is structured into modules for data handling, model implementation, plotting, portfolio management, simulation, optimisation, and interpretation. Notebooks are organised to cover data exploration, portfolio backtesting, result analysis, and robustness checks.

For a comprehensive overview of the repository structure, installation instructions, and detailed descriptions of each component, refer to the **README.md** file included in the repository.

CFA Quant Awards 2024

**Temperature derivatives pricing based on temperature
stochastic modelling**

Abstract

Weather derivatives provide a way to hedge against adverse weather conditions, yet their pricing is complex due to the non-tradable nature of weather variables. Traditional pricing models, such as the Black-Scholes-Merton approach, are not suitable for these derivatives. Instead, indices like the CAT index are used. This paper proposes a new temperature model that accounts for time-dependent mean reversion to enhance the accuracy of temperature derivatives. The model has been tested using daily average temperatures from 37 European weather stations spanning from 1980 to 2019 and incorporates a Kalman filter to probabilistically estimate the speed of mean reversion. Monte Carlo simulation is employed to stochastically estimate the CAT index. The analysis of the CAT index values reveals regional disparities across Europe, with Southern European countries experiencing higher CAT values due to warmer climates, and Eastern and Northern European regions showing lower values.

1 Introduction

Weather derivatives [1] are financial instruments that can be used to hedge against the risk of adverse or unexpected weather conditions. They are based on underlying weather variables such as temperature, rainfall, snowfall, or humidity.

The weather derivatives market is considered incomplete because the underlying weather variables are not storable or tradable. As a result, the traditional Black-Scholes-Merton pricing approach cannot be applied, and the payoffs of weather derivatives must be determined by indices representing average quantities [2], such as the CAT (Cumulative Average Temperature) index for temperature. One approach to simulating weather indices is to simulate the future behaviour of the weather variable.

Ref. [2] states that modelling temperature can lead to more accurate pricing than modelling temperature indices alone. Therefore, the motivation for this work is to propose an accurate temperature model to ensure reliable temperature derivatives. We test our temperature model on the CAT index.

Our temperature model considers the speed of mean reversion to be time-dependent, as in [2]–[4]. By using a Kalman filter to model the speed of mean reversion, we obtain a probabilistic distribution for the speed of mean reversion. Then, we use Monte Carlo simulation to estimate the CAT index stochastically.

The rest of the paper is structured as follows. Section 2 presents two temperature models - one with a time-dependent mean reversion term and the other with a constant mean reversion term - and outlines the methods used to estimate their parameters. Section 3 defines the CAT index model. Section 4 describes the case study used to validate the proposed approach. Section 5 analyses the results from the case study. Finally, Section 6 summarises the key findings of the research.

2 Temperature models

We propose two temperature models: one with a time-dependent mean reversion parameter (Section 2.1) and one with a constant mean reversion parameter (Section 2.2). Sections 2.1.1 and 2.1.2 focus on estimating the trend and seasonality and the volatility for the both models. Section 2.1.3 covers the estimation of the time-dependent mean reversion parameter.

2.1 Temperature model with time-dependent speed of mean reversion

The first model we propose for the temperature T_t is:

$$T_t = S_t + X_{t,1}, \quad (1)$$

$$dX_{t,1} = X_{t,2}X_{t,1}dt + \sigma_t dW_t, \quad (2)$$

$$dX_{t,2} = -\lambda X_{t,2}dt + \sigma_2 dB_t, \quad (3)$$

where $W = (W_t)_{t \geq 0}$, $B = (B_t)_{t \geq 0}$ are two independent Brownian motions, $\lambda \geq 0$, $\sigma_2 > 0$. The functions S_t and σ_t are defined as follows:

$$S_t = a + bt + \alpha_S \cos(2\pi t) + \beta_S \sin(2\pi t), \quad (4)$$

$$\sigma_t^2 = a_\sigma + \alpha_\sigma \cos(2f_\sigma \pi t) + \beta_\sigma \sin(2f_\sigma \pi t), \quad (5)$$

with $a, b, \alpha_S, \beta_S, a_\sigma, \alpha_\sigma, \beta_\sigma \in \mathbb{R}$.

Regarding the main terms of the model:

- T_t is the average daily temperature.
- S_t is a deterministic function that models the trend and seasonality of the average daily temperature. It is estimated in Section 2.1.1.
- σ_t is the daily volatility of temperature variations. It is estimated in Section 2.1.2.
- $X_{t,2}$ is the time-dependent speed of mean reversion. To model $X_{t,2}$, we propose an Ornstein–Uhlenbeck process (3), which has the following solution:

$$X_{t,2} = X_{0,2}e^{-\lambda t} + \sigma_2 \int_0^t e^{-\lambda(t-s)} dB_s \text{ for } t \geq 0.$$

The parameters of $X_{t,2}$ are estimated in Section 2.1.3.

- $X_{t,1}$ represents the detrended and deseasonalized average daily temperature. To model $X_{t,1}$, we propose using an Ornstein–Uhlenbeck process with a time-dependent speed of mean reversion parameter (2). As proposed by [3], the Itô product rule can be used to obtain the following solution to (2):

$$X_{t,1} = X_{0,1}e^{\int_0^t X_{s,2}ds} + e^{\int_0^t X_{s,2}ds} \int_0^t e^{\int_0^s -X_{u,2}du} \sigma(s) dW_s \text{ for } t \geq 0. \quad (6)$$

Since $X_{t,2}$ follows an Ornstein–Uhlenbeck process and σ_t is a continuous, bounded function, we have that

$$E \left[\int_0^t \sigma_s^2 e^{2 \int_0^s -X_{u,2}du} ds \right] < \infty \text{ for every } t \geq 0.$$

Hence, the Itô integral in (6) is well defined.

2.1.1 Estimating trend and seasonality

First, we estimate the parameters of the seasonal and trend function S_t for each weather station using linear regression, as detailed in Algorithm 1.

Fig. 2a displays the average daily temperature (T_t), while Fig. 2b shows the trend and seasonality term (S_t) for a particular weather station.

2.1.2 Estimating the volatility

After estimating the parameters of the function S_t , we detrend and deseasonalise the temperature data, and calculate the residuals as follows:

$$X_{t,1} = T_t - S_t. \quad (7)$$

Fig. 2c shows the time series of term $X_{t,1}$ for a particular weather station.

To estimate the volatility (σ_t) parameters of (5) (f_σ , a_σ , α_σ and β_σ), we undertake a two-step process, as shown in Algorithm 2.

The first step involves estimating the value of f_σ using a periodogram. Fig. 2d shows the periodogram of the term $X_{t,1}^2$ for the illustrative weather station. The highest occurring frequency is $f_\sigma = 1$; that is, the volatility has an annual pattern.

In the second step, we use the definition of the quadratic variation of an Itô process. Ref. [5] also use the quadratic variation to estimate the parameters of the volatility. Let $a = t_0^n < t_1^n < \dots < t_{n-1}^n < t_n^n = b$ be a sequence of partitions of the interval $[a, b]$ such that: $\lim_{n \rightarrow \infty} \max_{i \in \{1, \dots, n\}} \{t_i^n - t_{i-1}^n\} = 0$. The quadratic variation of the process $X_{t,1}$ is defined as:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n (X_{t_i,1} - X_{t_{i-1},1})^2 = \int_a^b a_\sigma + \alpha_\sigma \cos(2f_\sigma \pi s) + \beta_\sigma \sin(2f_\sigma \pi s) ds, \quad (8)$$

where the limit is taken in probability.

We divide the series into N non-overlapping time periods with N_d days each. For each time period $[a_j, b_j]$ for $j = 1, 2, \dots, N$, we compute:

$$\sum_{i=1}^{N_d} (X_{t_i^j,1} - X_{t_{i-1}^j,1})^2 \approx a_\sigma(b_j - a_j) + \frac{\alpha_\sigma}{2f_\sigma \pi} (\sin(2f_\sigma \pi b_j) - \sin(2f_\sigma \pi a_j)) + \frac{\beta_\sigma}{2f_\sigma \pi} (-\cos(2f_\sigma \pi b_j) + \cos(2f_\sigma \pi a_j)), \quad (9)$$

where $a_j = t_0^j < t_1^j < \dots < t_m^j = b_j$.

Then, we estimate the volatility terms a_σ , α_σ and β_σ using linear regression, as shown in Algorithm 2.

The estimated volatility for the temperature of the weather station selected for illustration is shown in Fig. 2e.

2.1.3 Estimating the time-dependent mean reverting parameter

We use a Kalman filter to estimate the value of $X_{t,2}$:

$$y_t = H_t z_t + r_t, \quad (10)$$

$$z_t = F_t z_{t-1} + q_t, \quad (11)$$

where $q_t \sim N(0, Q_t)$, $r_t \sim N(0, R_t)$. y_t is the known state and it depends on z_t , which is the unknown state.

Applying the Euler method to (2) and (3), we obtain:

$$X_{t_{i+1},1} = X_{t_i,1} + (X_{t_{i+1},2} X_{t_i,1}) \Delta + \sigma(t_i) \sqrt{\Delta} Z_{t_{i+1},1}, \quad (12)$$

$$X_{t_{i+1},2} = X_{t_i,2} - \lambda X_{t_i,2} \Delta + \sigma_2 \sqrt{\Delta} Z_{t_{i+1},2}, \quad (13)$$

where Δ represents the time step size, set to 1 day.

By linking the terms in (12) and (13) with those in (10) and (11), we derive:

$$y_t = (X_{t_{i+1},1} - X_{t_i,1}) / \sigma(t_i), \quad (14)$$

$$z_t = X_{t_{i+1},2},$$

$$H_t = X_{t_i,1} \Delta / \sigma(t_i), \quad (15)$$

$$F_t = (1 - \lambda \Delta_i),$$

$$R_t = \Delta,$$

$$Q_t = \sigma_2^2 \Delta.$$

The parameters of the term $X_{t,2}$ (λ , σ_2 and $X_{0,2}$) are estimated with the Kalman filter in an iterative way, using the Expectation-Maximization method (Algorithm 3).

Fig. 2f displays the expected value of the term $X_{t,2}$ and its 95% confidence interval, estimated by the Kalman filter, and the expected value estimated by the Kalman smoother.

2.2 Temperature model with constant speed of mean reversion

Now, we estimate the mean reversion parameter assuming it is constant over time. In this case, the process $X_{t,1}$ satisfies:

$$dX_{t,1} = \beta X_{t,1} dt + \sigma_t dW_{t,1} \quad (16)$$

The parameter β is estimated with Algorithm 6.

3 CAT index

The CAT index for the time interval $[\tau_1, \tau_2]$ is given by the following expression [1]:

$$\text{CAT} = \int_{\tau_1}^{\tau_2} T(s) ds.$$

Our aim is to formulate a mathematical equation to predict the future price of the CAT. The market for weather derivatives is an example of a classic incomplete market and is characterised by low liquidity. In such markets, it's impossible to determine a singular price through the principle of no-arbitrage. Contracts based on temperature are linked to a temperature index, which cannot be easily bought, sold, or stored. To derive a pricing formula, it is necessary to first identify a risk-neutral measure $Q \sim P$, under which all assets become martingales once they are discounted. For weather derivatives, any equivalent measure Q acts as a risk-neutral measure [1]. In this work, we price weather derivatives under the real probability P , as it is done by [6].

The price of the CAT future (FCAT price) is:

$$F_{CAT}(t, \tau_1, \tau_2) = E \left[\int_{\tau_1}^{\tau_2} T(s) ds \middle| \mathcal{F}_t \right] \text{ for } t \leq \tau_1 \leq \tau_2.$$

We propose a simplification of the temperature model proposed in Section 2.1 for pricing the CAT future:

$$\begin{aligned} T_t &= S_t + X_{t,1}, \\ dX_{t,1} &= \kappa X_{t,1} dt + \sigma_t dW_t, \end{aligned}$$

where $\kappa \sim N(\mu_\kappa, \sigma_\kappa^2)$ and it is independent of the Brownian motion W_t . The parameters μ_κ and σ_κ^2 , are calculated from the Kalman filter. The parameter μ_κ is $E[X_{M,2}|x_{0:M,1}]$, that is, the last value of the conditional expected value obtained by the Kalman filter. The parameter σ_κ^2 is $Var[X_{M,2}|x_{0:M,1}]$, that is, the last value of the conditional variance obtained by the Kalman filter. From [1, Proposition 6.2], the price of the CAT future given the value of κ is

$$\begin{aligned} F_{CAT}(t, \tau_1, \tau_2) &= E \left[\int_{\tau_1}^{\tau_2} T(s) ds \middle| \mathcal{F}_t \vee \sigma(\kappa) \right] \\ &= \int_{\tau_1}^{\tau_2} S(s) ds + X_{t,1} \int_{\tau_1}^{\tau_2} e^{\kappa(s-t)} ds \\ &= a(\tau_2 - \tau_1) + \frac{b}{2}(\tau_2^2 - \tau_1^2) \\ &\quad + \frac{\alpha_S}{2\pi} (\sin(2\pi\tau_2) - \sin(2\pi\tau_1)) + \frac{\beta_S}{2\pi} (\cos(2\pi\tau_2) - \cos(2\pi\tau_1)) \\ &\quad + X_{0,1} \frac{e^{-\kappa t}}{\kappa} [e^{\kappa\tau_2} - e^{\kappa\tau_1}]. \end{aligned} \tag{17}$$

We use Monte Carlo and generate 10^5 realisations of the random variable κ . For each realisation, we compute the CAT future price as given in (17). This allows us to generate a distribution of prices rather than a single price.

4 Case study

Our dataset consists of the daily average outdoor temperatures (T) from 37 weather stations located in 37 different European countries, as listed in Table A. The data covers the period from 1st January 1980 to 31st December 2019.

The daily average temperature time series for each weather station is computed as:

$$T = \frac{TMAX + TMIN}{2} \tag{18}$$

The original dataset was sourced from the European Climate Assessment & Dataset project (ECA&D) [7] and contains the daily minimum (TMIN) and maximum (TMAX) outdoor temperatures recorded at these 37 weather stations. We have processed the original data by removing outliers and filling in missing values, following the methodology proposed by [8].

Our case study consists of two analyses:

- In Section 5.1, we study the impact of including a time-dependent or constant mean reversion parameter on the modelling of temperatures across all European stations. This analysis is based on the temperature models described in Sections 2.1 (time-dependent mean reversion parameter) and 2.2 (constant mean reversion parameter).
- In Section 5.2, we analyse the distribution of the FCAT prices for all European stations. This analysis is based on a simplification of the temperature model with the time-dependent constant mean reversion parameter described in Section 2.1 and on the CAT future price model described in Section 3. The FCAT prices are calculated with $t = 0$, $\tau_1 = 0$ (1st January) and $\tau_2 = 0.25$ (1st March). The parameters μ_κ and σ_κ^2 correspond to the distribution of $X_{t,2}$ for 31st December 2019 ($X_{M,2}$).

5 Results

Table 2 presents the estimated parameters a , b , α_S and β_S for the term S_t across all weather stations. Additionally, it includes the volatility parameters for each station (f_σ , a_σ , α_σ and β_σ), as well as the estimated parameters of $X_{t,2}$ (λ , σ_2 and $X_{0,2}$).

Table 3 lists the estimated β parameters for all weather stations and the errors of the two temperature models.

Using the CAT future prices model, we obtain a confidence interval for these prices, allowing for more robust pricing of the CAT future. Table 4 provides the mean, variance and percentiles 2.5% and 97.5% of the prices, as well as these statistics for κ (the distribution of $X_{t,2}$ on the last day of the historical data, estimated via the Kalman filter).

5.1 Analysis of the speed of mean reversion

Table 3 displays the Mean Squared Errors (MSE) for the training (in-sample) data used to adjust the two temperature models, one with the time-dependent mean reversion term and the other with the constant mean reversion term. The errors are consistently slightly lower when considering the mean reversion term to be time-dependent.

However, these results are not conclusive, as the model evaluations were performed on the training data. Future work will involve developing simulation algorithms for both models and comparing their MSE on a test set. This will help determine whether including a constant mean reversion parameter provides any additional benefit.

5.2 Analysis of the distribution of the FCAT prices

Fig. 1 shows the distribution of FCAT prices across the 37 European countries, organised into colour-coded clusters based on their price mean. The error bars represent the variability in FCAT price distributions, defined by the 2.5% and 97.5% percentiles, with longer bars indicating greater variability.

The green group contains Southern European countries such as Malta, Greece, Cyprus, Portugal, Spain, and Italy, which reveal high FCAT prices ranging from 1.6 to 3.0. These countries experience warmer climates with higher average temperatures (refer to the values of a in Table 2), which contribute to the higher FCAT values observed.

The black group includes countries from Eastern and Northern Europe, such as Poland, Sweden, and Russia, with negative FCAT prices ranging from -1.2 to around 0.2. The cooler climates in these regions are associated with lower FCAT values.

The red and blue groups correspond to moderate to slightly positive FCAT prices, ranging from 0.2 to 1.4. This range is associated with milder temperatures compared to the black and green groups. It includes countries across Western, Central, and Eastern Europe, such as France, the United Kingdom, Belgium, Germany, Croatia, and Turkey.

The variability of the FCAT prices is only determined by the term $X_{0,1} \frac{e^{-\kappa t}}{\kappa} [e^{\kappa\tau_2} - e^{\kappa\tau_1}]$ in (17). This term increases with higher values of $X_{0,1}$ and when κ approaches 0. Consequently, countries such as Latvia, Estonia, Finland and Russia demonstrate greater price variability due to their κ values being closer to 0 and their higher $X_{0,1}$ values (see Fig. 3 and Table 2).

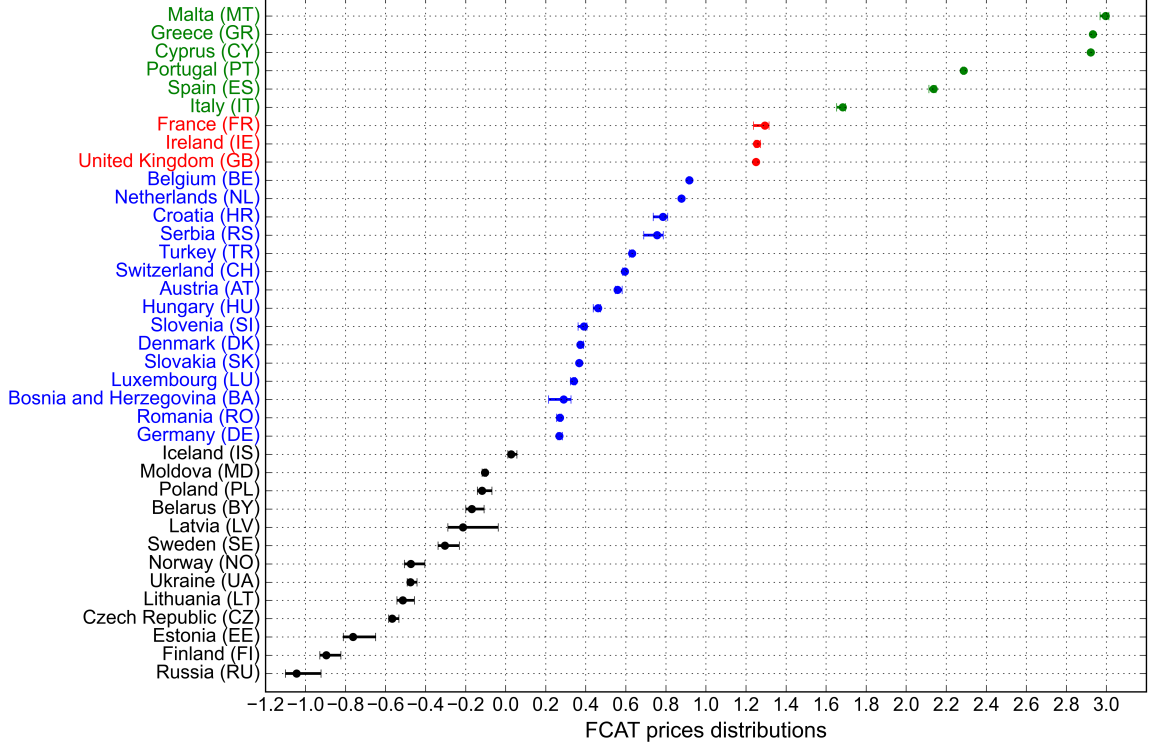


Fig. 1. Distribution of CAT future prices is shown across the 37 weather stations. Each distribution is determined by the mean CAT future price and its 2.5% and 97.5% percentiles, as listed in Table 4. Locations with a mean higher than 1.6 are coloured in green, while those with a mean between 1.2 and 1.4 are coloured in red. Locations with a mean between 0.2 and 1.0 are coloured in blue, and those with a mean lower than 0.2 are coloured in black. The boundaries between the groups were determined visually.

6 Conclusion

This study introduces an approach to modelling temperature for weather derivatives by incorporating a time-dependent mean reversion parameter. By applying a Kalman filter to estimate the speed of mean reversion and combining it with Monte Carlo simulations to determine the FCAT price index, we present a stochastic model for price estimation. Our analysis of the FCAT index across Europe reveals significant regional variations, correlating higher FCAT values with warmer climates in Southern Europe and lower values in the cooler Northern and Eastern regions. Future work will focus on developing simulation algorithms to test the temperature model's performance on out-of-sample data.

References

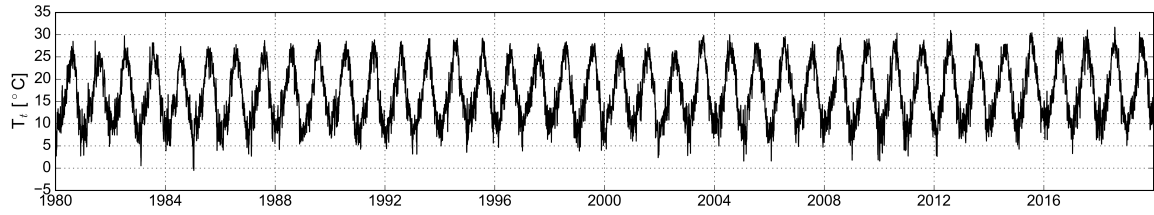
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A Data

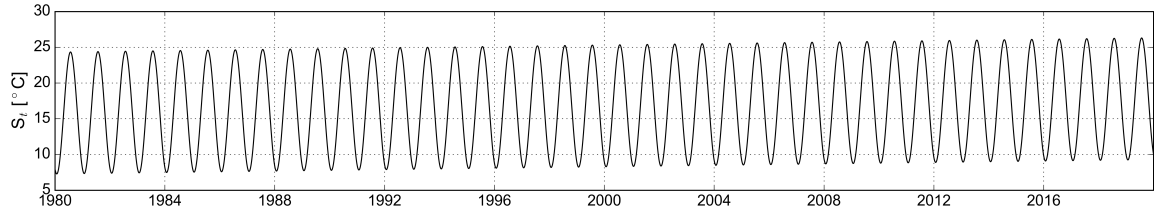
Table 1. Detail of the European weather stations used for each country [8].

Country ID	Country	Station ID	Station	Lat.	Lon.	Hght.
AT	Austria	16	Wien	48.2331	16.35	198
BA	Bosnia and Herzegovina	276	Sarajevo	43.8678	18.4228	630
BE	Belgium	17	Uccle	50.8	4.3664	100
BY	Belarus	653	Brest	52.1167	23.6831	146
CH	Switzerland	240	Genevecontring	46.25	6.1331	420
CY	Cyprus	23	Larnaca	34.8831	33.6331	1
CZ	Czech Republic	510	Milesovka	50.555	13.9331	836
DE	Germany	41	Berlin-Dahlem	52.4639	13.3017	51
DK	Denmark	116	Koebenhavn:Landbohojskolen-1	55.6831	12.5831	40
EE	Estonia	11357	Narva	59.3892	28.1128	28
ES	Spain	230	Madrid-Retiro	40.4117	-3.6781	667
FI	Finland	28	Helsinki:kaisaniemi	60.175	24.9478	4
FR	France	11249	Orly	48.7167	2.3842	89
GB	United Kingdom	1860	Heathrow	51.479	-0.44889	25
GR	Greece	61	Heraklion	35.3331	25.1831	39
HR	Croatia	21	Zagreb-Gric	45.8167	15.9781	156
HU	Hungary	849	Pecs:gongy	46.0067	18.2328	202
IE	Ireland	1718	Dublin:airport	53.4281	-6.2408	71
IS	Iceland	65	Dalatangi	65.2661	-13.5756	9
IT	Italy	174	Brindisi	40.6331	17.9331	15
LT	Lithuania	200	Kaunas	54.8831	23.8831	77
LU	Luxembourg	203	Luxembourg:airport	49.6258	6.2033	376
LV	Latvia	2951	Liye:payamsg	56.55	21.02	4
MD	Moldova	394	Kis:inev	47.02	28.9333	173
MT	Malta	447	Luqa	35.85	14.4831	91
NL	Netherlands	598	De Bilt	51.9606	4.4672	2
NO	Norway	193	Oslo:blindern	59.9428	10.7206	94
PL	Poland	209	Warszawa-Okecie	52.1628	20.9608	107
PT	Portugal	212	Braga:anca	41.8	-6.7331	690
RO	Romania	219	Bucures:ti-Baneasa	44.5167	26.0831	91
RS	Serbia	263	Belgrade (Observatory)	44.8	20.4667	132
RU	Russia	85	St. Petersburg	59.9667	30.3	3
SE	Sweden	10	Stockholm	59.35	18.05	44
SI	Slovenia	228	Ljubljana:bezigrad	46.065	14.5169	299
SK	Slovakia	227	Hurbanovo	47.867	18.1831	115
TR	Turkey	346	Isparta	37.75	30.55	997
UA	Ukraine	252	Kiev	50.4	30.5331	166

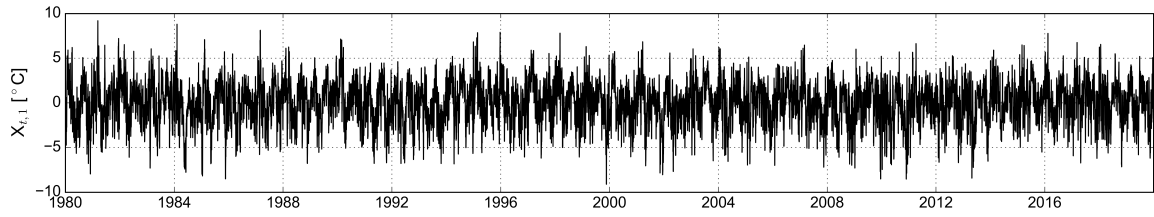
B Additional figures and estimated parameters



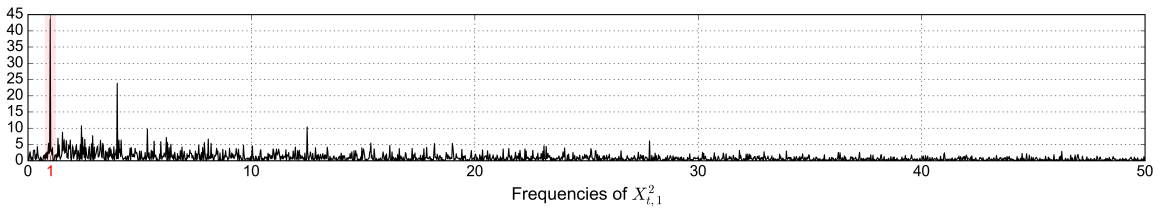
(a) Average daily temperature (T_t).



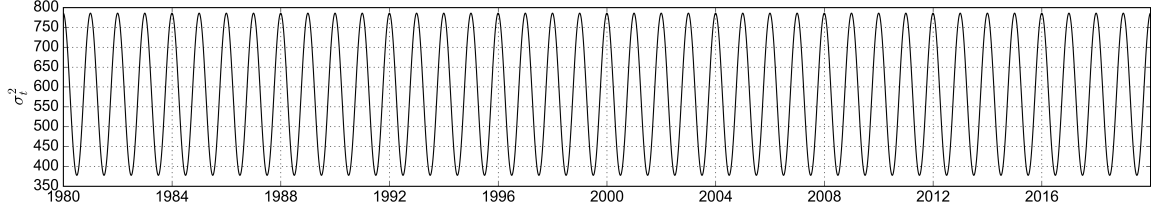
(b) Trend and seasonality component (S_t).



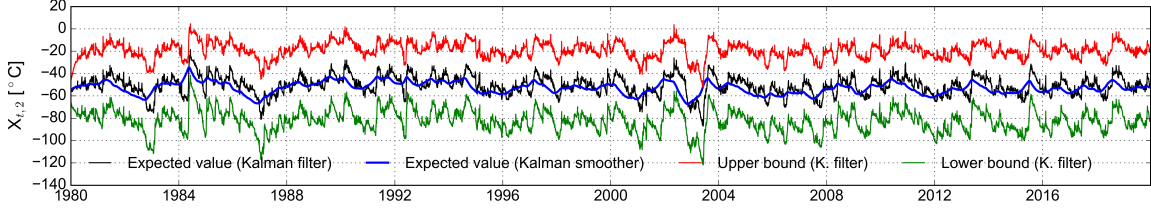
(c) Detrended and deseasonalised average daily temperature ($X_{t,1}$).



(d) Periodogram of $X_{t,1}^2$.



(e) Volatility term (σ_t^2).



(f) Expected value of $X_{t,2}$ and its 95% confidence interval, estimated by the Kalman filter, and expected value estimated by the Kalman smoother.

Fig. 2. Temperature terms for a particular weather station.

Table 2. Estimated parameters of temperature model with time-dependent speed of mean reversion. a , b , α_S and β_S are the parameters of term S_t (applicable to both the time-dependent and constant mean reversion models). f_σ , a_σ , α_σ and β_σ are the parameters of term σ_t^2 (also applicable to both models). λ and σ_2 are the parameters of term $X_{t,2}$. $X_{0,1}$ and $X_{0,2}$ are the values of $X_{t,1}$ and $X_{t,2}$ at $t = 0$.

ID	a	b	α_S	β_S	f_σ	a_σ	α_σ	β_σ	λ	σ_2	$X_{0,2}$	$X_{0,1}$
AT	10.17	5.37e-02	-10.16	-2.54	1.00	1830.26	372.57	163.41	6.29e-02	25.95	-75.71	2.90
BA	9.75	5.46e-02	-9.85	-3.01	1.00	2446.78	674.01	119.95	6.34e-02	25.83	-87.57	-7.24
BE	9.94	4.81e-02	-7.25	-2.55	1.00	1652.51	31.34	139.77	5.03e-02	26.26	-61.16	-0.86
BY	7.73	4.31e-02	-10.92	-2.82	1.00	2335.53	359.69	281.12	6.34e-02	26.67	-63.78	5.97
CH	9.96	4.46e-02	-9.00	-2.81	1.00	1592.88	116.93	67.27	4.80e-02	25.99	-75.60	-1.54
CY	18.69	5.64e-02	-6.85	-4.17	1.00	892.05	9.97	318.29	3.57e-02	26.77	-155.01	0.38
CZ	5.49	6.62e-02	-9.92	-2.63	1.00	2577.98	-510.20	469.72	5.30e-02	26.21	-76.18	4.69
DE	8.65	4.85e-02	-9.53	-2.52	1.00	1450.43	52.37	183.46	8.04e-02	26.13	-67.66	1.50
DK	8.60	4.22e-02	-8.14	-3.22	1.00	1055.05	-21.35	100.46	6.74e-02	26.71	-79.87	2.89
EE	5.27	4.54e-02	-10.03	-3.78	1.00	2080.28	555.48	541.32	6.41e-02	26.85	-72.70	6.88
ES	15.79	5.03e-02	-7.78	-3.46	1.00	581.57	204.11	67.89	1.15e-01	25.97	-55.45	-1.00
FI	5.20	5.22e-02	-10.30	-4.03	1.00	1960.88	1179.79	332.62	6.91e-02	26.76	-67.37	5.19
FR	11.82	4.57e-02	-7.33	-2.85	1.00	721.12	153.28	12.37	1.23e-01	25.75	-47.55	-1.69
GB	10.67	4.09e-02	-6.43	-2.54	1.00	1197.91	26.15	45.02	3.77e-02	26.40	-92.36	1.28
GR	17.91	3.82e-02	-5.96	-3.68	1.00	1218.12	151.86	478.33	2.70e-02	26.67	-143.10	-1.65
HR	11.41	6.87e-02	-10.18	-2.45	1.00	1726.55	291.41	89.73	7.74e-02	25.96	-74.52	-3.68
HU	10.52	4.82e-02	-10.57	-2.84	1.00	1966.88	405.83	118.10	7.10e-02	25.90	-69.90	-2.53
IE	9.46	1.72e-02	-4.95	-2.20	1.00	1549.01	291.77	-70.11	3.19e-02	26.49	-122.02	2.82
IS	3.55	4.19e-02	-3.47	-2.35	1.00	1572.61	830.54	280.61	3.05e-02	26.42	-109.92	7.80
IT	14.58	5.07e-02	-8.56	-3.58	1.00	576.67	268.63	14.65	9.59e-02	26.17	-45.67	-1.68
LT	6.41	5.07e-02	-10.74	-3.12	1.00	2194.40	674.91	316.28	6.40e-02	26.70	-74.74	6.49
LU	8.43	5.75e-02	-8.46	-2.51	1.00	1870.39	-73.95	133.50	6.26e-02	25.92	-68.53	-1.52
LV	6.82	4.24e-02	-9.00	-3.91	1.00	1949.30	408.08	478.91	6.41e-02	27.16	-78.76	5.82
MD	9.44	6.40e-02	-12.09	-3.28	1.00	2356.85	717.39	83.04	6.50e-02	26.20	-85.82	-1.47
MT	18.77	2.79e-02	-6.16	-4.20	1.33	890.77	3.00	70.95	3.18e-02	26.88	-120.79	-4.75
NL	9.54	3.99e-02	-6.75	-2.75	1.00	1670.71	103.99	118.53	4.64e-02	26.61	-73.78	0.27
NO	5.99	5.31e-02	-9.93	-2.99	1.00	1708.79	907.72	93.89	6.33e-02	26.64	-66.12	5.47
PL	7.66	5.94e-02	-10.42	-2.78	1.00	2220.28	432.46	229.43	6.56e-02	26.45	-83.02	4.74
PT	15.25	3.00e-02	-6.53	-2.98	1.00	785.21	-101.11	33.28	7.80e-02	26.01	-55.44	-0.80
RO	10.58	4.00e-02	-11.67	-3.15	1.00	1949.88	836.42	228.77	6.38e-02	25.96	-88.02	-0.91
RS	11.88	6.67e-02	-10.49	-2.93	1.00	2425.04	232.34	160.98	7.20e-02	26.02	-78.21	-5.01
RU	5.13	5.26e-02	-11.79	-3.70	1.00	2257.22	967.75	793.46	8.13e-02	26.40	-63.99	7.60
SE	6.51	4.74e-02	-9.31	-3.41	1.00	1434.42	697.16	154.69	7.40e-02	26.34	-63.11	6.33
SI	9.96	6.70e-02	-10.22	-2.75	1.00	1597.48	371.56	11.37	6.72e-02	25.91	-69.41	-2.32
SK	9.97	6.02e-02	-10.72	-2.64	1.00	2107.71	275.12	103.38	5.83e-02	25.98	-84.09	-0.21
TR	11.31	5.62e-02	-9.66	-4.03	1.00	1398.81	740.66	348.32	6.62e-02	26.10	-116.20	-1.82
UA	7.67	6.48e-02	-12.40	-2.95	1.00	2247.80	579.63	236.50	7.22e-02	26.24	-72.96	3.24

Table 3. Constant mean reversion parameter (β) and Mean Squared Error (MSE) for the forecasts generated by the temperature model with the time-dependent mean reversion parameter (Kalman) and the one with the constant mean reversion parameter (Constant).

ID	β	MSE_{Kalman}	$MSE_{Constant}$
AT	-68.18	2.47e-03	2.49e-03
BA	-72.24	2.45e-03	2.47e-03
BE	-70.34	2.45e-03	2.47e-03
BY	-68.20	2.45e-03	2.48e-03
CH	-76.15	2.43e-03	2.45e-03
CY	-105.46	2.32e-03	2.34e-03
CZ	-74.21	2.44e-03	2.46e-03
DE	-60.14	2.49e-03	2.51e-03
DK	-62.55	2.47e-03	2.51e-03
EE	-71.57	2.45e-03	2.47e-03
ES	-51.60	2.52e-03	2.54e-03
FI	-67.43	2.52e-03	2.55e-03
FR	-48.53	2.53e-03	2.56e-03
GB	-78.49	2.42e-03	2.44e-03
GR	-115.78	2.28e-03	2.30e-03
HR	-64.25	2.48e-03	2.50e-03
HU	-65.06	2.47e-03	2.49e-03
IE	-110.02	2.31e-03	2.33e-03
IS	-95.21	2.33e-03	2.36e-03
IT	-54.06	2.52e-03	2.55e-03
LT	-69.02	2.46e-03	2.48e-03
LU	-69.41	2.46e-03	2.48e-03
LV	-75.10	2.43e-03	2.46e-03
MD	-71.45	2.43e-03	2.46e-03
MT	-110.24	2.30e-03	2.33e-03
NL	-77.54	2.42e-03	2.44e-03
NO	-70.08	2.46e-03	2.49e-03
PL	-70.27	2.45e-03	2.47e-03
PT	-57.48	2.50e-03	2.53e-03
RO	-74.80	2.45e-03	2.47e-03
RS	-67.58	2.46e-03	2.49e-03
RU	-62.36	2.51e-03	2.54e-03
SE	-62.99	2.50e-03	2.53e-03
SI	-67.73	2.47e-03	2.49e-03
SK	-73.59	2.44e-03	2.46e-03
TR	-73.13	2.46e-03	2.48e-03
UA	-64.95	2.47e-03	2.49e-03

Table 4. CAT future prices. μ_κ , σ_κ and the percentiles 2.5% and 97% of κ are the parameters of the Gaussian distribution of $X_{t,2}$ on the last day of the historical data, estimated with the Kalman filter. The mean (μ_{FCAT}), the variance (σ_{FCAT}) and the percentiles 2.5% and 97.5% of the FCAT price are calculated from 10^5 realisations of the FCAT price.

ID	μ_κ	σ_κ^2	P2.5 of κ	P97.5 of κ	μ_{FCAT}	σ_{FCAT}^2	P2.5 of FCAT	P97.5 of FCAT
AT	-85.33	330.41	-120.96	-49.70	0.56	8.64e-05	0.55	0.58
BA	-75.02	308.35	-109.43	-40.60	0.29	1.09e-03	0.21	0.33
BE	-93.62	324.49	-128.92	-58.31	0.92	4.52e-06	0.91	0.92
BY	-73.94	285.22	-107.04	-40.84	-0.17	1.38e-03	-0.20	-0.11
CH	-92.32	353.07	-129.15	-55.49	0.59	3.69e-05	0.58	0.60
CY	-134.11	379.88	-172.31	-95.91	2.92	2.10e-07	2.92	2.92
CZ	-87.98	326.88	-123.41	-52.54	-0.57	1.91e-04	-0.58	-0.53
DE	-76.13	317.51	-111.06	-41.21	0.27	4.37e-05	0.26	0.28
DK	-99.46	323.77	-134.72	-64.19	0.37	3.83e-05	0.36	0.39
EE	-64.44	315.63	-99.26	-29.62	-0.76	3.21e-03	-0.81	-0.65
ES	-52.30	255.78	-83.64	-20.95	2.14	4.10e-02	2.11	2.15
FI	-67.79	306.75	-102.11	-33.46	-0.90	1.37e-03	-0.93	-0.82
FR	-48.36	257.14	-79.79	-16.93	1.29	6.24e-03	1.24	1.31
GB	-124.98	334.75	-160.84	-89.12	1.25	2.76e-06	1.25	1.25
GR	-133.70	435.17	-174.58	-92.81	2.93	4.58e-06	2.93	2.94
HR	-67.21	292.58	-100.74	-33.69	0.79	6.18e-04	0.74	0.81
HU	-74.53	289.40	-107.88	-41.19	0.46	1.20e-04	0.44	0.47
IE	-99.92	390.76	-138.67	-61.18	1.25	4.67e-05	1.25	1.27
IS	-124.90	434.03	-165.74	-84.07	0.03	1.41e-04	0.01	0.06
IT	-61.02	298.46	-94.88	-27.16	1.68	5.55e-04	1.65	1.70
LT	-79.13	306.09	-113.42	-44.84	-0.51	5.81e-04	-0.54	-0.46
LU	-72.79	308.68	-107.23	-38.36	0.34	5.68e-05	0.32	0.35
LV	-56.76	380.99	-95.02	-18.51	-0.21	1.71e+00	-0.29	-0.04
MD	-74.34	289.70	-107.70	-40.99	-0.10	3.89e-05	-0.12	-0.10
MT	-102.87	436.96	-143.84	-61.89	3.00	1.36e-04	2.97	3.01
NL	-93.74	360.09	-130.94	-56.55	0.88	5.15e-07	0.88	0.88

NO	-71.30	328.49	-106.82	-35.77	-0.47	4.02e-03	-0.51	-0.40
PL	-75.60	311.06	-110.16	-41.03	-0.12	4.24e-04	-0.14	-0.07
PT	-82.34	286.59	-115.52	-49.16	2.29	6.20e-06	2.28	2.29
RO	-61.71	308.39	-96.13	-27.29	0.27	1.55e-03	0.25	0.28
RS	-67.41	290.62	-100.82	-33.99	0.76	1.12e-03	0.69	0.79
RU	-60.82	264.45	-92.69	-28.95	-1.04	7.94e-03	-1.10	-0.92
SE	-72.56	301.33	-106.58	-38.54	-0.30	1.13e-03	-0.34	-0.23
SI	-68.70	298.25	-102.55	-34.85	0.39	1.78e-04	0.36	0.41
SK	-82.03	294.32	-115.66	-48.41	0.37	4.69e-07	0.37	0.37
TR	-96.98	417.56	-137.03	-56.93	0.63	2.62e-05	0.62	0.64
UA	-72.43	270.63	-104.67	-40.18	-0.48	2.04e-04	-0.49	-0.44

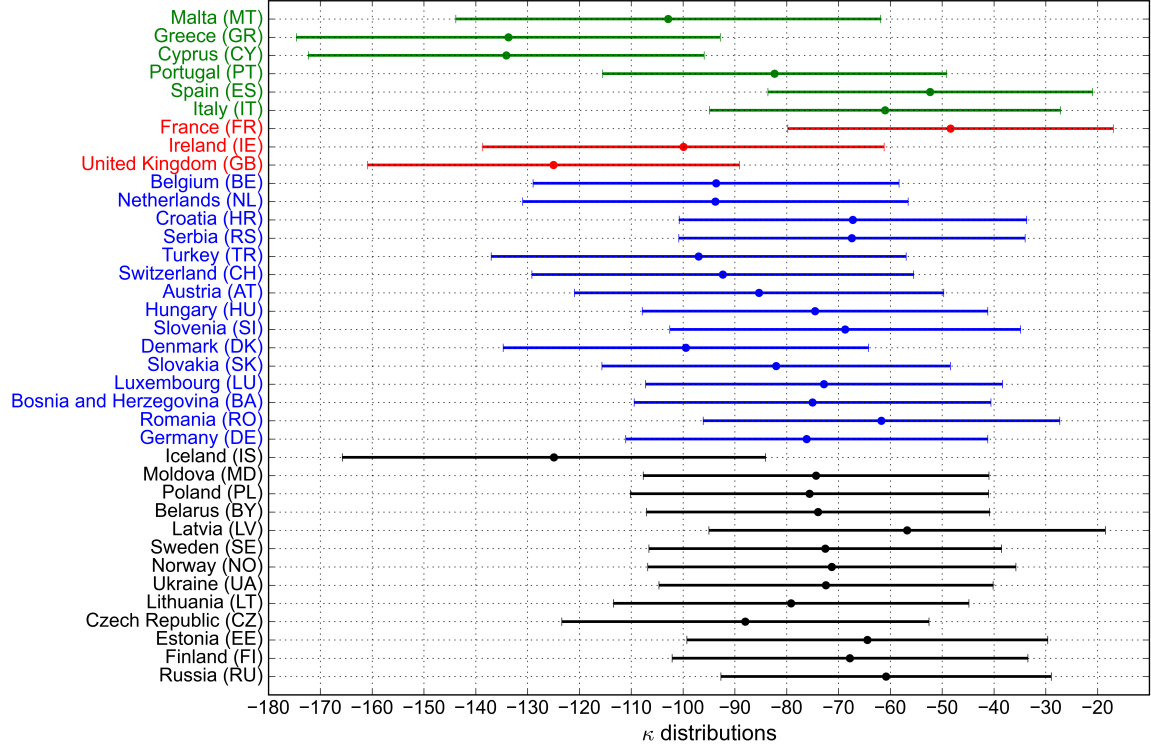


Fig. 3. Distribution of κ is shown across the 37 weather stations. Each distribution is determined by the mean of κ and its 2.5% and 97.5% percentiles, as listed in Table 4. Locations are coloured according to the criteria outlined in Fig. 1.

C Algorithms

Algorithm 1: Estimating the trend and seasonality.

Input: Original T_t time series.
Output: Parameters a, b, α_S and β_S for term S_t .

- 1 Each known term (t , $\cos(2\pi t)$ and $\sin(2\pi t)$) in (4) is considered an input variable (X_1 , X_2 and X_3 , respectively) in a multivariate linear regression model: $T = a + bX_1 + \alpha_S X_2 + \beta_S X_3$.
- 2 **Step 1: Compute input and output variables of the linear regression model:**
- 3 Compute input variable X_1 : t .
- 4 Compute input variable X_2 : $\cos(2\pi t)$.
- 5 Compute input variable X_3 : $\sin(2\pi t)$.
- 6 Compute output variable T : T_t .
- 7 **Step 2: Perform linear regression model:**
- 8 Compute the linear regression parameters a , b , α_S and β_S .
- 9 **return** Parameters a, b, α_S and β_S .

Algorithm 2: Estimating the volatility.

Input: Estimated time series $X_{t,1}$, M (number of days of the temperature time series).
Output: Parameters f_σ , a_σ , α_σ and β_σ for term σ_t .

- 1 **Step 1: Estimate frequency f_σ :**
- 2 Estimate frequency f_σ of $X_{t,1}^2$ with a periodogram.
- 3 Each known term in (9) is considered an input variable in a multivariate linear regression model:
 $Y = a_\sigma X_1 + \alpha_\sigma X_2 + \beta_\sigma X_3$.
- 4 **Step 2: Compute input variables of the linear regression model:**
- 5 Set $N_d = 100$, number of days of each non-overlapping time period.
- 6 Compute $N_t = \frac{M}{N_d}$, number of non-overlapping time periods.
- 7 **for** $j \in \{1, \dots, N_t\}$ **do**
- 8 Compute start period index: $a = (j - 1) \cdot N_d$, and end period index: $b = j \cdot N_d$.
- 9 Compute output variable Y : $\sum_{i=a}^{b-1} (X_{t_{i+1},1} - X_{t_i,1})^2$.
- 10 Compute input variable X_1 : $b - a$.
- 11 Compute input variable X_2 : $\frac{1}{2f_\sigma \pi} \left(\sin(2f_\sigma \pi b) - \sin(2f_\sigma \pi a) \right)$.
- 12 Compute input variable X_3 : $\frac{1}{2f_\sigma \pi} \left(-\cos(2f_\sigma \pi b) + \cos(2f_\sigma \pi a) \right)$.
- 13 **Step 3: Perform linear regression model:**
- 14 Compute the linear regression parameters a_σ , α_σ and β_σ .
- 15 **return** Parameters f_σ , a_σ , α_σ and β_σ .

Algorithm 3: Estimating the speed of mean reversion (time-dependent).

Input: $X_{t,1}$, M , Δ , σ_t and tol .
Output: Parameters of the hidden process $X_{t,2}$: λ , σ_2 and $X_{0,2}$.

1 Step 1: Generate the variables to pass to the Kalman Filter:

2 Compute H_t according to (15): $H_{t_i} = \Delta \frac{X_{t_i,1}}{\sigma_{t_i}}$.

3 Compute y_t according to (14): $y_{t_i} = \frac{X_{t_{i+1},1} - X_{t_i,1}}{\sigma_{t_i}}$.

4 Estimate the initial values of F_0 and μ_0 .

5 Set $R = \Delta$.

6 Set $Q_0 = 2$.

7 Set $\sigma_0 = 2$.

8 Step 2: EM algorithm:

9 Based on [9, Section 29.8.1].

10 Set $tol = 0.01$ and $N_{steps} = 500$.

11 Initialize $value_{tol} = -\infty$, $n_{steps} = 0$, $loglik_0 = -\infty$.

12 **while** $value_{tol} \geq tol$ **and** $n_{steps} \leq N_{steps}$ **do**

13 **Expectation step:**

14 Perform Algorithm 4 (Kalman filter):

- Input: y_t , H_t , F_0 , Q_0 , μ_0 , σ_0 , R , M .
- Output: $\mu_{Kalman,t}$, $\sigma_{Kalman,t}$, K_{Kalman} , $y_{Est,t}$, $loglik_1$.

 Perform Algorithm 5 (Kalman smoother):

- Input: $\mu_{Kalman,t}$, $\sigma_{Kalman,t}$, F_0 , Q_0 , M .
- Output: $\mu_{Smooth,t}$, $\sigma_{Smooth,t}$, $K_{Smooth,t}$.

15 **Maximization step:**

16 Compute P_t and $P_{t,tm1}$.

17 Compute $F_1, Q_1, \mu_1, \sigma_1$.

18 $value_{tol} = \frac{1}{5} \left((Q_1 - Q_0)^2 + (F_1 - F_0)^2 + (\mu_1 - \mu_0)^2 + (\sigma_1 - \sigma_0)^2 + (loglik_1 - loglik_0)^2 \right)$.

19 $n_{steps} = n_{steps} + 1$.

20 Update $F_0 = F_1$, $Q_0 = Q_1$, $\sigma_0 = \sigma_1$, $\mu_0 = \mu_1$ and $loglik_0 = loglik_1$.

21 $\sigma_2 = \sqrt{\frac{Q_0}{\Delta}}$.

22 $\lambda = \frac{1 - F_0}{\Delta}$.

23 $X_{0,2} = \mu_0$.

24 return λ , σ_2 and $X_{0,2}$.

Algorithm 4: Kalman filter. Based on [9, Section 8.2.2].

Input: $y_t, H_t, F_0, Q_0, \mu_0, \Sigma_0, R, M$.
Output: $\mu_{Kalman,t}, \Sigma_{Kalman,t}, K_{Kalman}, y_{Kalman,t}, loglik_1$.

- 1 Initialize $\mu_{Kalman,t} = \emptyset, \Sigma_{Kalman,t} = \emptyset, K_{Kalman,t} = \emptyset, y_{Est,t} = \emptyset, loglik_1 = 0$.
- 2 Add μ_0 to the end of $\mu_{Kalman,t}$.
- 3 Add Σ_0 to the end of $\Sigma_{Kalman,t}$.
- 4 **for** $j \in \{1, \dots, M\}$ **do**
- 5 **Step 1: Predict step:**
- 6 $\mu_1 = F_0 \cdot \mu_0$.
- 7 $\Sigma_1 = F_0 \cdot \Sigma_0 \cdot F_0 + Q_0$.
- 8 **Step 2: Update step:**
- 9 $S = H_{t_j} \cdot \Sigma_1 \cdot H_{t_j} + R$.
- 10 $K = \Sigma_1 \cdot H_{t_j} \cdot \frac{1}{S}$.
- 11 $\mu_0 = \mu_1 + K(y_{t_j} - H_{t_j} \cdot \mu_1)$.
- 12 $\Sigma_0 = (1 - K \cdot H_{t_j}) \cdot \Sigma_1$.
- 13 $y_{Est} = H_{t_j} \cdot \mu_0$.
- 14 Add μ_0 to the end of $\mu_{Kalman,t}$.
- 15 Add Σ_0 to the end of $\Sigma_{Kalman,t}$.
- 16 Add K to the end of $K_{Kalman,t}$.
- 17 Add y_{Est} to the end of $y_{Est,t}$.
- 18 $loglik_1 = loglik_1 + pdf(y_{t_j})$ where pdf is the probability density function of $N \sim (y_{Est}, S)$.
- 19 **return** $\mu_{Kalman,t}, \Sigma_{Kalman,t}, K_{Kalman,t}, y_{Est,t}, loglik_0$.

Algorithm 5: Kalman smoother. Based on [9, Section 8.2.3].

Input: $\mu_{Kalman,t}, \Sigma_{Kalman,t}, F_0, Q_0, M$.
Output: $\mu_{Smooth,t}, \Sigma_{Smooth,t}, K_{Smooth,t}$.

- 1 Initialize $\mu_{Smooth,t} = \emptyset, \Sigma_{Smooth,t} = \emptyset, K_{Smooth,t} = \emptyset$.
- 2 Add the last element of $\mu_{Kalman,t}$ to the end of $\mu_{Smooth,t}$.
- 3 Add the last element of $\Sigma_{Kalman,t}$ to the end of $\Sigma_{Smooth,t}$.
- 4 **for** $j \in \{M-1, \dots, 0\}$ **do**
- 5 $\mu_0 = F_0 \cdot \mu_{Kalman,t_j}$.
- 6 $\Sigma_0 = F_0 \cdot \Sigma_{Kalman,t_j} \cdot F_0 + Q_0$.
- 7 $K = \Sigma_{Kalman,t_j} \cdot F_0 \cdot \frac{1}{\Sigma_0}$.
- 8 $\mu_s = \mu_{Kalman,t_j} + K \cdot (\mu_{Smooth,t_{j+1}} - \mu_0)$.
- 9 $\Sigma_s = \Sigma_{Kalman,t_j} + K \cdot (\Sigma_{Smooth,t_{j+1}} - \Sigma_0) \cdot K$.
- 10 Add μ_s to the end of $\mu_{Smooth,t}$.
- 11 Add Σ_s to the end of $\Sigma_{Smooth,t}$.
- 12 Add K to the end of $K_{Smooth,t}$.
- 13 $\mu_{Smooth,t} = reverse(\mu_{Smooth,t})$.
- 14 $\Sigma_{Smooth,t} = reverse(\Sigma_{Smooth,t})$.
- 15 $K_{Smooth,t} = reverse(K_{Smooth,t})$.
- 16 **return** $\mu_{Smooth,t}, \Sigma_{Smooth,t}, K_{Smooth,t}$.

Algorithm 6: Estimating the speed of mean reversion (constant).

Input: y_t, H_t .
Output: β .

- 1 **Step 1: Compute input variables of linear regression model:**
- 2 Compute input variable X : H_t .
- 3 Compute output variable Y : y_t .
- 4 **Step 2: Perform linear regression model:**
- 5 Compute the linear regression parameter β of the model $Y = \beta X$, equivalent to (10).
- 6 **return** β .

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**Polynomial Ornstein-Uhlenbeck
volatility models for SPX & VIX
pricing**

Abstract

The pricing of VIX derivatives, SPX derivatives, and the SPX-VIX joint calibration problem has been extensively studied in the field of mathematical finance. Despite this extensive research, challenges remain in developing models that are both theoretically sound and computationally efficient. In this paper, we present a broad class of Polynomial Ornstein-Uhlenbeck volatility models, which include the well-known Stein-Stein, Schöbel-Zhu, one-factor Bergomi, and the recently introduced Quintic OU models. On the one hand, these models provide an explicit formula for VIX derivatives, which is straightforward to calculate. On the other hand, SPX derivatives can be calculated more efficiently than with the Monte Carlo method by applying Fourier techniques to the characteristic functional of these models.

From a theoretical perspective, we establish the connection between the expansion of the characteristic functional and the solution of an infinite-dimensional Riccati equation.

From a practical perspective, we demonstrate the efficiency and accuracy of this scheme for pricing SPX derivatives using Fourier and Laplace inversions, with specific examples of the Quintic OU and one-factor Bergomi models and their SPX-VIX joint calibration to real market data.

Keywords: Stochastic volatility, Derivative pricing, Fourier methods, Riccati equations, SPX-VIX calibration

1 Introduction

Fourier inversion techniques hold a pivotal role in stochastic volatility modeling, particularly in the context of pricing the SPX and VIX derivatives (Andersen and Andreasen [8], Carr and Madan [10], Eberlein, Glau, and Papapantoleon [15], Fang and Oosterlee [18], Lewis [25], Lipton [26]). They offer the dual advantage of significantly reducing computational time while maintaining a remarkable degree of accuracy, especially when compared to standard Monte Carlo methods. When it comes to model calibration, the applicability of Fourier methods is paramount, as thousands of derivatives across various maturities and strikes need to be evaluated simultaneously in real time.

Despite their numerical advantage, Fourier techniques have traditionally been confined to specific continuous stochastic volatility models where the characteristic function of the log-price is known in (semi)-closed form. These models usually exhibit Markovian affine structures in the sense of Duffie, Filipović, and Schachermayer [13] such as the renowned Heston [23], Stein-Stein [30] and Schöbel-Zhu [29] models, together with some of their non-Markovian Volterra counterparts (Abi Jaber [1, 2], Abi Jaber, Larsson, and Pulido [5], Cuchiero and Teichmann [11], El Euch and Rosenbaum [17], Gatheral and Keller-Ressel [21]). The key ingredient in all these models is to compute the characteristic function of the log-price by solving a specific system of deterministic Riccati equations.

Numerically solving these equations poses challenges due to their stiffness and complexity. This forms our primary motivation: to establish theoretical results within a specific framework for a larger class of Riccati equations related to integrated quantities of power series of Ornstein-Uhlenbeck processes and to develop more suitable numerical schemes.

Our work demonstrate that Fourier techniques can be effectively extended to a broad class of flexible models previously considered infeasible as they fall beyond the conventional class of affine diffusions, including the celebrated one-factor Bergomi model (Bergomi [9], Dupire [14]) that has been shown to fit well to the SPX smiles and the recently introduced Quintic Ornstein-Uhlenbeck (OU) model of Abi Jaber, Illand, and Li [7], which has demonstrated remarkable capabilities in fitting jointly the SPX-VIX volatility surface for maturities between one week to three months supported by extensive empirical studies on more than 10 years of data in Abi Jaber and Li [4], Abi Jaber, Illand, and Li [6].

Our work are introduced in this paper as follows:

- In Section 2, we introduce the Polynomial Ornstein-Uhlenbeck volatility models, which include the well-known Stein-Stein, Schöbel-Zhu, one-factor Bergomi, and the recently introduced Quintic OU models.
- In Section 3, we introduce the tractability of VIX derivatives within our models by providing an explicit expression for the VIX index, which enables fast VIX derivatives pricing.
- In Section 4, we introduce the characteristic functional and its connection to a system of Riccati equations. We also demonstrate how to apply Fourier inversion techniques to the characteristic functional for fast SPX derivatives pricing, along with a numerical illustration.
- In Section 5, we present the SPX-VIX joint calibration to real market data and provide a numerical illustration.

More details about the theoretical results and additional numerical illustrations are given in the appendix.

2 Polynomial Ornstein-Uhlenbeck volatility models

At first, we introduce the OU(Ornstein-Uhlenbeck) volatility models for a stock price S with the stochastic volatility process σ expressed as a power series of an OU process X :

$$\begin{aligned} dS_t &= S_t \sigma_t dB_t, \quad S_0 > 0, \\ \sigma_t &= g_0(t)p(X_t), \quad p(x) = \sum_{k=0}^{\infty} p_k x^k, \\ dX_t &= (a + bX_t)dt + c dW_t, \quad X_0 \in \mathbb{R}, \end{aligned} \tag{2.1}$$

with $a, b, p_k \in \mathbb{R}$, $c \neq 0$, $B = \rho W + \sqrt{1 - \rho^2} W^\perp$ and $\rho \in [-1, 1]$. Here $(W, W^\perp)$ is a two-dimensional Brownian motion on a risk-neutral filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$ satisfying the usual conditions. The deterministic bounded input curve $g_0 : [0, T] \rightarrow \mathbb{R}_+$ allows the model to match certain term structures of volatility, e.g. the forward variance curve since we have for $g_0(t) := \sqrt{\xi_0(t)/\mathbb{E}[p^2(X_t)]}$:

$$\mathbb{E} \left[\int_0^t \sigma_s^2 ds \right] = \int_0^t \xi_0(s) ds, \quad t \geq 0.$$

We suppose in this paper that p is either a polynomial or an exponential function. These two specifications already provide several interesting models used in practice:

- Stein-Stein [30] and Schöbel-Zhu [29] model: $p(x) = x$;
- Bergomi model [9, 14]: $p(x) = \exp(x)$;
- Quintic OU model [7]: $p(x) = p_0 + p_1 x + p_3 x^3 + p_5 x^5$, $p_0, p_1, p_3, p_5 \geq 0$.

3 Pricing of VIX derivatives

One of the main advantage of our model is its **tractability** of VIX derivatives, i.e. we have an explicit representation of VIX^2 by X_t .

Recall the formula of VIX^2 (see for example [7]),

$$\text{VIX}_T^2 = \frac{100^2}{\Delta} \int_0^\Delta \xi_T(t) dt, \quad \Delta = 30 \text{ days}, \quad \xi_T(t) := \mathbb{E} [\sigma_{T+t}^2 \mid X_T].$$

Proposition 3.1 (Explicit expression for the VIX).

$$\text{VIX}_T^2 = \frac{100^2}{\Delta} \sum_{i=0}^{\infty} \beta_i(T) X_T^i,$$

with

$$p^2(x) = \sum_{k=0}^{\infty} q_k x^k \text{ being the Taylor expansion of } p^2, \quad \binom{k}{i} \text{ the binomial coefficient, and}$$

$$\beta_i(T) = \sum_{k=i}^{\infty} q_k \binom{k}{i} \int_0^{\Delta} g_0^2(T+t) e^{kbt} \mathbb{E}[Y_t^{k-i}] dt,$$

where

$$Y_t = a \int_0^t e^{-bu} du + c \int_0^t e^{-bu} dW_u$$

is Gaussian with explicit moments.

Proof. Since X is an OU process, by its Markovian property, $X_{T+t} = e^{bt}(X_T + \tilde{Y}_t)$, where $\tilde{Y}_t$ has the same distribution as Y_t and is independent of $\mathcal{F}_T$.

Thus

$$\xi_T(t) := \mathbb{E}[\sigma_{T+t}^2 | X_T] = g_0^2(T+t) \mathbb{E}[p^2(X_{T+t}) | X_T] = g_0^2(T+t) \sum_{k=0}^{\infty} q_k e^{bkt} \sum_{i=0}^k \binom{k}{i} X_T^i \mathbb{E}[Y_t^{k-i}].$$

Then we deduce by taking integration and rearranging the sum. □

Pricing VIX derivatives with payoff function Φ and maturity T is thus immediate by integrating directly against the standard Gaussian density:

$$\begin{aligned} \mathbb{E}[\Phi(\text{VIX}_T)] &= \mathbb{E} \left[\Phi \left(\sqrt{\frac{100^2}{\Delta} \sum_{i=0}^{\infty} \beta_i(T) X_T^i} \right) \right] \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \Phi \left(\sqrt{\frac{100^2}{\Delta} \sum_{i=0}^{\infty} \beta_i(T) (\sigma_{X_T} x + m_{X_T})^i} \right) e^{-x^2/2} dx \end{aligned}$$

with $m_{X_T}^2$ the expectation of X_T , $\sigma_{X_T}^2$ the variance of X_T . This integral can be computed efficiently using numerical techniques which leads to fast VIX derivative pricing.

For example, to price VIX futures prices, set $\Phi(\nu) = \nu$ and to price VIX vanilla call price, set $\Phi(\nu) = (\nu - K)^+$.

Notice that, for the Quintic OU model, where $p(x) = p_0 + p_1 x + p_3 x^3 + p_5 x^5$, $p_0, p_1, p_3, p_5 \geq 0$, or the truncation of one-factor Bergomi model, the sum above becomes finite terms.

4 Pricing of SPX derivatives

The SPX index can be defined directly as S_t in our model. And thus the SPX derivatives are defined as $\mathbb{E}[\Phi(S_T)]$, for some payoff functions Φ . Like before, for example, to price SPX futures prices, set $\Phi(\nu) = \nu$ and to price SPX vanilla call price, set $\Phi(\nu) = (\nu - K)^+$.

This requires us to calculate the distribution of S_t , which can be calculated using the Monte Carlo method, since we have an explicit distribution for the OU process X_t .

However, the Fourier and Laplace inversions combined with the characteristic functional of S_t give a more efficient and accurate way to do the SPX derivatives pricing for our model.

4.1 Characteristic functional

In this subsection, we introduce the characteristic functional, which can be viewed also as the joint Fourier-Laplace transform of the log-price and integrated variance

$$\mathbb{E} \left[\exp \left(\int_t^T g_1(T-s) d \log S_s + \int_t^T g_2(T-s) \sigma_s^2 ds \right) \mid \mathcal{F}_t \right], \quad t \leq T.$$

We assume that g_1 and g_2 are regular enough to cut through the details. For more details about their conditions, readers can refer to Appendix C.

It follows from the Markov property of X_t that the above conditional expectation can be reduced to computing the deterministic measurable function $F : [0, T] \times \mathbb{R} \rightarrow \mathbb{C}$ given by

$$F(t, x) := \mathbb{E} \left[\exp \left(\int_t^T g_1(T-s) d \log S_s + \int_t^T g_2(T-s) \sigma_s^2 ds \right) \mid X_t = x \right]. \quad (4.1)$$

To calculate $F(t, x)$ efficiently, our aim is to build the expansion for $F(t, x)$ for its more efficient calculation:

$$F(t, x) = \exp \left(\sum_{k \geq 0} \psi_k(T-t) x^k \right), \quad t \leq T. \quad (4.2)$$

As we can see in the Theorem D.1, ψ_k is supposed to be a solution of a system of Ricatti equations.

$$\begin{aligned} \psi'_k(t) = & \left(g_2(t) + \frac{g_1(t)}{2} (g_1(t) - 1) \right) g_0^2(T-t) (p * p)_k \\ & + bk\psi_k(t) + a(k+1)\psi_{k+1}(t) + \frac{c^2(k+2)(k+1)}{2} \psi_{k+2}(t) \\ & + \frac{c^2}{2} (\tilde{\psi}(t) * \tilde{\psi}(t))_k + \rho g_1(t) g_0(T-t) c (p * \tilde{\psi}(t))_k, \quad \tilde{\psi}_k(t) := (k+1)\psi_{k+1}(t), \end{aligned} \quad (4.3)$$

$$\psi_k(0) = 0, \quad (4.4)$$

where $(p * q)_k := \sum_{l=0}^k p_l q_{k-l}$.

We can prove at first that (4.3)-(4.4) are necessary to (4.2).

Proposition 4.1. *If (4.2) holds, then (4.3)-(4.4) hold.*

Proof. We define that $U_t = \sum_{k \geq 0} \psi_k(T-t) X_t^k + \int_0^t g_1(T-s) d \log S_s + \int_0^t g_2(T-s) \sigma_s^2 ds$.

By Itô's formula:

$$\begin{aligned} dU_t = & g_1(T-t) d \log S_t + g_2(T-t) \sigma_t^2 dt - \sum_k \psi'_k(T-t) X_t^k dt \\ & + \sum_k (k+1) \psi_k(T-t) X_t^k dX_t + \sum_k \frac{1}{2} (k+2)(k+1) \psi_k(T-t) X_t^k d\langle X \rangle_t. \end{aligned}$$

Notice that by (4.2),

$$\exp(U_t) = \mathbb{E} \left[\exp \left(\int_0^T g_1(T-s) d \log S_s + \int_0^T g_2(T-s) \sigma_s^2 ds \right) \mid \mathcal{F}_t \right], \quad t \leq T.$$

Thus $\exp(U_t)$ is a martingale, and thus with zero drift (i.e. dt part of $d \exp(U_t)$ is zero a.s.)

Then we can deduce (4.3) from zero drift of $\frac{d \exp(U_t)}{\exp(U_t)} = dU_t + \frac{1}{2} \langle U \rangle_t$. And (4.4) is directly from $F(T, x) = 1$.

□

The efficient numerical method to solve the stiff equations (4.4)-(4.3) is introduced in the Appendix A.

For the theoretical result, in brief, we give a principle to verify (4.2), build the existence of solution of (4.4)-(4.3), and a discretized version of (4.2). The formal statements of these results can be found in Appendix D. And the proofs of them are given in Appendix E-G.

4.2 Fourier inversion to fast SPX derivatives pricing

In this subsection, we give a concrete example about how to price the call option of SPX by the characteristic functional $F(t, x)$ and Fourier inversion techniques. We use the expression (4.1) with $g_2(t) = 0$ and $g_1(t) = iu$, for $u \in \mathbb{R}$.

There are several Fourier inversion techniques available in the literature to price European-style Vanilla call and put options. Here we adapt the pricing formula suggested in [25] which involves only one integral to evaluate:

$$C_t(S_t, K, T) := \mathbb{E} [(S_T - K)^+ | \mathcal{F}_t] = S_t - \frac{K}{\pi} \int_0^\infty \Re \left[e^{(iu + \frac{1}{2})k_t} \varphi \left(u - \frac{i}{2} \right) \right] \frac{du}{u^2 + \frac{1}{4}},$$

where $C_t(S_t, K, T)$ denotes the European call option price with strike K and maturity $T - t$, $k_t := \log(K/S_t)$ is the log-moneyness and $\varphi(u) := F(t, x)$ the Fourier-Laplace transform of $\log(S_T/S_t)$ by fixing $g_1 \equiv iu$ and $g_2 \equiv 0$. We use the representation of $F(t, x)$ in (4.2) and compute the improper integral numerically via the Gauss-Laguerre quadrature, which has been demonstrated to be efficient, see [3, 24]. In addition, by posing $t = 0$, we obtain the vanilla call option $C_0(K, T) = \mathbb{E}[(S_T - K)^+]$.

4.3 Numerical illustration for Quintic model

In this section, we give the numerical illustration by choosing Quintic OU models in 2 and $g_0(t) = \sqrt{\frac{\xi_0(t)}{\mathbb{E}[p(X_t)^2]}}$, $a = 0$, $b = \alpha \varepsilon^{-1}$ and $c = \varepsilon^\alpha$. This model has shown to produce remarkable joint fits to both SPX and VIX implied volatility surfaces [6, 7].

Since X is an OU process which can be simulated exactly, one could be tempted to use Monte Carlo to estimate the SPX derivatives with appropriate control variates. However, the calibrated values of ε are usually very small and α is negative, pushing the model effectively into a fast regime with large mean reversion of order $\alpha \varepsilon^{-1}$ and large vol of vol ε^α . It is known the standard Euler-scheme for pricing SPX derivatives can reproduce large estimation bias for longer maturities T due to the highly erratic paths of σ , requiring finer step size or other asymptotic approximation techniques [19, 20]. Pricing via Fourier methods hence presents an attractive alternative given the

increased efficiency and accuracy. To demonstrate, we choose the following parameters: $\rho = -0.65$, $\alpha = -0.6$, $(p_0, p_1, p_3, p_5) = (0.01, 1, 0.214, 0.227)$, $\xi_0(t) = 0.025$, $\varepsilon = 1/52$ which are typical values one can expect from calibrating the model to SPX and VIX smiles from [7]. Figure 1 shows the convergence of SPX implied volatility of different maturities as the truncation level M increases:

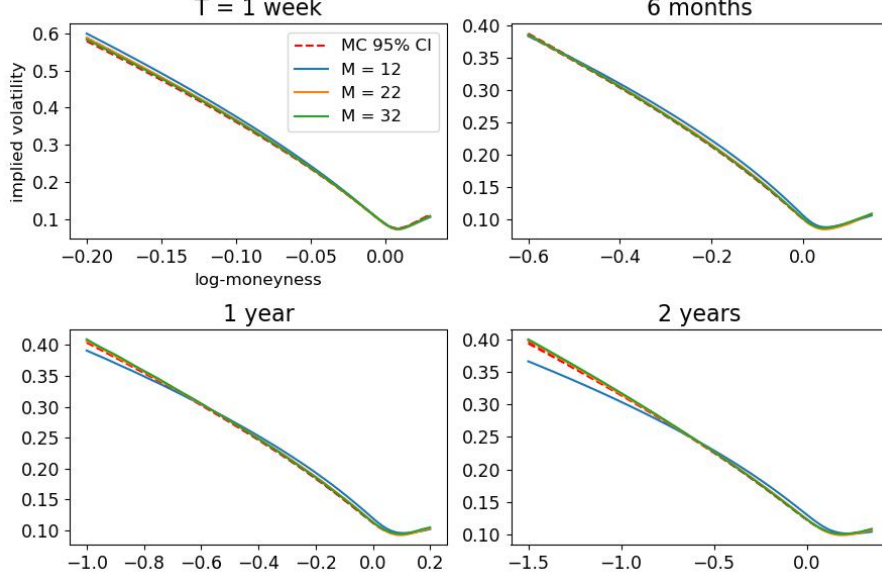


Figure 1: SPX implied volatility of different maturities in the quintic OU model, computed with our algorithm with different level M . Dotted red lines are Monte-Carlo 95% interval computed with 500,000 simulations and $n = 10,000$ number of steps per maturity slice.

The numerical illustration of the same method for the one-factor Bergomi model is given in Appendix B.1.

5 Model calibration to market data via Fourier

In this subsection, we introduce how to calibrate the model based on the method introduced in the precedent sections.

To calibrate our model, we solve the following optimisation problem involving sum of root mean squared error (RMSE):

$$\min_{\Theta} \left\{ c_1 \sqrt{\sum_{i,j} (\sigma_{\text{spx}}^{\Theta}(T_i, K_j) - \sigma_{\text{spx}}^{\text{mkt}}(T_i, K_j))^2} + c_2 \sqrt{\sum_{i,j} (\sigma_{\text{vix}}^{\Theta}(T_i, K_j) - \sigma_{\text{vix}}^{\text{mkt}}(T_i, K_j))^2} + c_3 \sqrt{\sum_i (F_{\text{vix}}^{\Theta}(T_i) - F_{\text{vix}}^{\text{mkt}}(T_i))^2} \right\}. \quad (5.1)$$

Here, $F_{\text{vix}}^{\text{mkt}}(T_i)$ is the market VIX futures price maturing at T_i . $\sigma_{\text{spx}}^{\text{mkt}}(T_i, K_j)$, $\sigma_{\text{vix}}^{\text{mkt}}(T_i, K_j)$ represent market SPX-VIX implied volatility with maturity T_i and strike K_j , calculated by call options and futures from the real market. $\sigma_{\text{spx}}^{\Theta}(T_i, K_j)$, $\sigma_{\text{vix}}^{\Theta}(T_i, K_j)$ and $F_{\text{vix}}^{\Theta}(T_i)$ represent the same in-

struments, but coming from our model. The coefficients c_1 , c_2 , and c_3 are some positive numbers used to assign different weights to the errors in SPX-VIX implied volatility and VIX futures price.

In a nutshell, the calibration of a model involves minimizing the mean square error between prices coming from the model vs. that from market data.

We demonstrate the capability of the quintic OU model as introduced in Section 4.3 to jointly calibrate two slices of maturities of SPX smiles together with one slice of maturity of the VIX smile, with calibrated parameters $\rho = -0.6763$, $\alpha = -0.6821$, $(p_0, p_1, p_3, p_5) = (0.0202, 1.3332, 0.0578, 0.0071)$ and fixed $\varepsilon = 1/52$, with $c_1 = 1, c_2 = 0.1, c_3 = 0.5$, with $\xi_0(t)$ coming directly from market data:

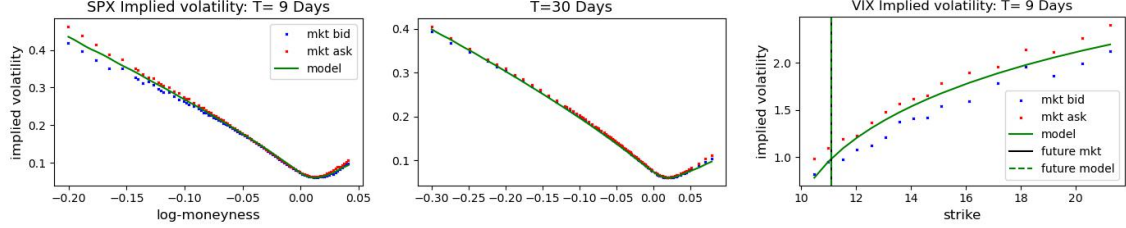


Figure 2: Quintic OU model (green lines) jointly calibrated to the SPX and VIX smiles (bid/ask in blue/red) on 23 October 2017 via Fourier using the Nelder-Mead optimization algorithm. The truncation level of the Riccati solver is set at $M = 32$.

The numerical illustration for the same method of calibration of one-factor Bergomi Model is given in Appendix B.2.

6 Conclusion

In this paper, we have successfully demonstrated the efficacy of pricing VIX, SPX derivatives, and the SPX-VIX joint calibration within the framework of Ornstein-Uhlenbeck (OU) volatility models, encompassing the Stein-Stein, Schöbel-Zhu, one-factor Bergomi, and the recently introduced Quintic OU models.

Our models provide a more efficient and accurate pricing scheme compared to the Monte Carlo Method. For VIX derivatives, the tractability stems from their explicit representations within our models, facilitating fast pricing. SPX derivatives can also be efficiently priced by applying Fourier and Laplace inversion techniques to the characteristic functional and its relation to an infinite-dimensional Riccati equation.

The integration of SPX and VIX derivatives enables the SPX-VIX joint calibration to real market data, demonstrating significant capabilities in jointly fitting the SPX-VIX volatility surface across maturities.

Additionally, we have established several theoretical results concerning the connection between the characteristic functional and the infinite-dimensional Riccati equation, including the equivalent verification condition, the existence of solutions to Riccati equations, and a corresponding approximation method.

Future research could explore the broad class of polynomial Ornstein-Uhlenbeck (OU) volatility models to identify one with superior numerical performance. Furthermore, additional properties of the characteristic functional and the infinite-dimensional Riccati equation are expected to be established using partial differential equation methods.

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A Numerical scheme for high-dimensional Riccati equations

In this section, we show how to solve the infinite dimensional Riccati equations ψ in (4.3)-(4.4) numerically for derivative pricing and hedging. The idea is to solve a truncated version of the ODE by assuming $\psi_k \approx 0, k > M$ for some integer M .

Even the truncated version of the infinite dimensional system of Riccati equations can be extremely difficult to solve. Standard techniques such as the Runge–Kutta methods usually fail to solve the ODE, especially when the parameters b and c in (2.1) are large. For example, see [12, Figures 2 and 3] for other attempts to a similar problem.

We present a customized algorithm that combines variation of constants and the implicit Euler scheme. For this section alone, all matrix and vector indices will start from 0. With some abuse of notation, we re-write the Riccati ODE (4.4) in the following matrix form:

$$\psi'(t) = P(t) + A\psi(t) + Q\psi(t) + \frac{c^2}{2}(\tilde{\psi}(t) * \tilde{\psi}(t)) + l(t)N\psi(t),$$

with $\psi(t)$ denoting the vector $(\psi_0(t), \psi_1(t), \dots)^\top$ and $\psi'(t)$ the vector of element-wise derivative of $\psi(t)$. $P(t)$ is a vector with its k^{th} element being $\left(g_2(t) + \frac{g_1(t)}{2}(g_1(t) - 1)\right)g_0^2(T - t)(p * p)_k$ for $k \in \{0, 1, \dots\}$. A is a diagonal matrix with its diagonal $A_{k,k} = bk$. Q is an upper triangular matrix with $Q_{k,k+1} = a(k+1)$, $Q_{k,k+2} = c^2(k+1)(k+2)/2$ and zero elsewhere. The term $(\tilde{\psi}(t) * \tilde{\psi}(t))$ denotes the discrete convolution of the vectors $\tilde{\psi}(t)$, with its k^{th} element given by $(\tilde{\psi}(t) * \tilde{\psi}(t))_k$. Finally, N is a matrix with $N_{j,k} = kp_{j+1-k}$ where p_m denotes the power series coefficient of p such that $p_m = 0$ whenever $m < 0$, with $l(t)$ collecting the term $c\rho g_1(t)g_0(T - t)$.

Variation of constants gives:

$$\begin{aligned} \psi(t_{i+1}) &= e^{A\Delta t}\psi(t_i) + \int_{t_i}^{t_{i+1}} e^{A(t_{i+1}-s)}P(s)ds + \int_{t_i}^{t_{i+1}} e^{A(t_{i+1}-s)}Q\psi(s)ds \\ &+ \frac{c^2}{2} \int_{t_i}^{t_{i+1}} e^{A(t_{i+1}-s)}(\tilde{\psi}(s) * \tilde{\psi}(s))ds + \int_{t_i}^{t_{i+1}} e^{A(t_{i+1}-s)}l(s)N\psi(s)ds, \end{aligned} \quad (\text{A.1})$$

with $\Delta = t_{i+1} - t_i$. By truncating the vector ψ up to some level M and assuming $(\psi_k)_{k>M} = 0$, we approximate the solution of $\psi(t_{i+1})$ in (A.1) by the following quasi-implicit scheme:

$$\begin{aligned} \psi(t_{i+1}) &\approx e^{A\Delta t}\psi(t_i) + G_\Delta P(t_i) + G_\Delta Q\psi(t_{i+1}) \\ &+ \frac{c^2}{2}G_\Delta(\tilde{\psi}(t_i) * \tilde{\psi}(t_{i+1})) + G_\Delta l(t_i)N\psi(t_{i+1}), \end{aligned}$$

where $G_\Delta = \int_{t_i}^{t_{i+1}} e^{A(t_{i+1}-s)}ds$ is a diagonal matrix with $G_{k,k} = (e^{b(k-1)\Delta} - 1)/(b(k-1))$, $1 < k \leq M+1$ and $G_{1,1} = \Delta$. Similarly, Q and N are now $(M+1) \times (M+1)$ matrices as defined above, with ψ and $P(t_i)$ both vectors of dimension $M+1$.

For the quadratic term $(\tilde{\psi}(t_i) * \tilde{\psi}(t_{i+1}))$, we define

$$R^{(i)}\psi(t_{i+1}) := (\tilde{\psi}(t_i) * \tilde{\psi}(t_{i+1})),$$

where $R^{(i)}$ is a $(M+1) \times (M+1)$ matrix with $R_{j,k}^{(i)} = (j+2-k)k\psi_{j+2-k}(t_i)$, $(\psi_z(t_i))_{z<0} = 0$. We can now solve ψ iteratively by:

$$\psi(t_{i+1}) \approx J^{-1} \left(e^{A\Delta t}\psi(t_i) + G_\Delta P(t_i) \right),$$

where $J = \left(I - G_\Delta \left(\frac{c^2}{2}R^{(i)} + l(t_i)N + Q \right) \right)$.

Matrix Q contains very large coefficients resulting from the term $\left(c^2(k+2)(k+1)/2\right)$ when k is large. This term introduces numerical instability so we capped $(k+2)(k+1)$ to some level k_{max}^2 to ensure the scheme does not blow up.

We first test our algorithm on the term $\mathbb{E}[\exp(-\frac{W_t^4}{4!})]$ considered in [12, Figures 2 and 3], which can be expressed by the solution of a particular case of the Riccati ODE in (4.3)-(4.4) with initial condition $\psi_4(0) = -1/4!$, $(a, b, g_1, g_2) = 0$ and $c = 1$. The reference value can be computed via a numerical integration quadrature with respect to the Gaussian density, allowing us to evaluate our algorithm's performance. Figure 3 illustrate numerical convergence in terms of M , with step size $n = 100$ and $k_{max} = 15$ fixed. We clearly observe the instability of the Runge-Kutta scheme with increasing truncation M , whereas the scheme we propose remains stable with the increase of M .

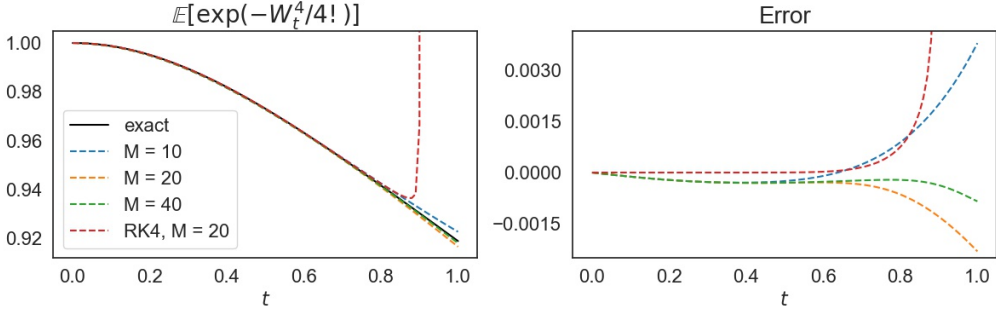


Figure 3: Numerical solution of $\mathbb{E}[\exp(-\frac{W_t^4}{4!})]$ with different values of M using our algorithm vs. the exact solution. For comparison the explicit Runge-Kutta 4 solution with $M = 20$ are added in red.

B Numerical illustration for one-factor Bergomi mode

B.1 SPX implied volatility

In this section adapt the Section 4.3 to the case of one-factor Bergomi model [9, 14], which assumes σ_t to be log-normal:

$$\begin{aligned}\sigma_t &= \sqrt{\xi_0(t)} \exp\left(\frac{1}{2}\eta X_t - \frac{1}{4}\eta^2 \mathbb{E}(X_t^2)\right), \\ X_t &= \varepsilon^\alpha \int_0^t e^{\alpha\varepsilon^{-1}(t-s)} dW_s,\end{aligned}\tag{B.1}$$

with $\varepsilon > 0$ and $\alpha \leq 0$. Similar to quintic OU model before, we have $a = 0, b = \alpha\varepsilon^{-1}$ and $c = \varepsilon^\alpha$. We now approximate the exponential as a truncated sum up to level N :

$$\tilde{\sigma}_t = \sqrt{\frac{\xi_0(t)}{\mathbb{E}[p(X_t)^2]}} p(X_t), \quad p(X_t) = \sum_{k=0}^N p_k X_t^k, \quad p_k = \frac{\eta^k}{2^k k!},$$

where $\tilde{\sigma}_t$ converges to σ_t in (B.1) when sending $N \rightarrow \infty$. We now fix $\rho = -0.7, \varepsilon = 1/52, \alpha = -0.7, \eta = 1.2, \xi_0(t) = 0.025$ for the numerical experiment and set $N = 8$. Figure 4 shows that our numerical scheme quickly converges to Monte-Carlo estimates of the original one-factor Bergomi model:

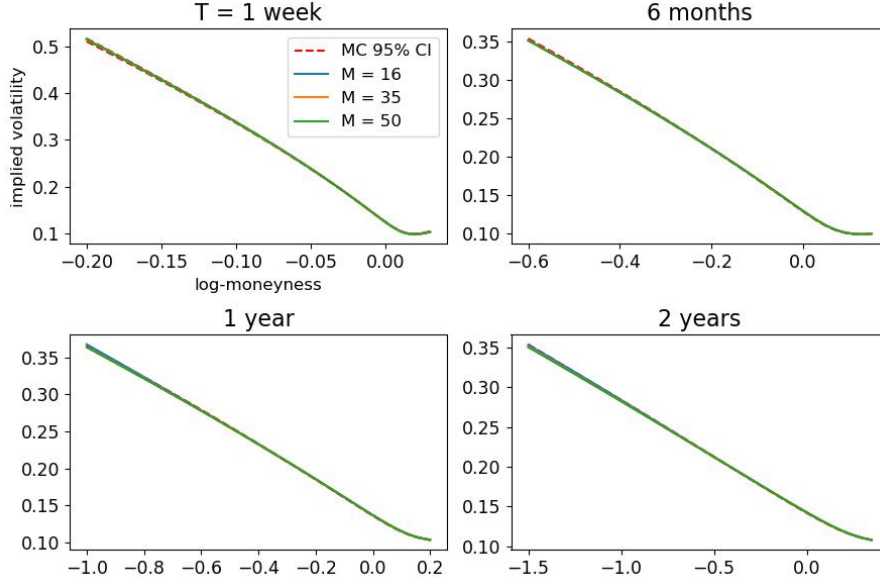


Figure 4: SPX implied volatility of different maturities in the one-factor Bergomi model, computed with our algorithm with different level M . Dotted red lines are Monte-Carlo 95% interval computed with 500,000 simulations and $n = 10,000$ number of steps per maturity slice.

B.2 Model calibration to market data via Fourier

In this section, we illustrate the calibration of real market data by minimizing (5.1) introduced in section 5.

Next, we showcase the calibration results of the one-factor Bergomi model (B.1) on four slices of maturities of SPX smiles between around 1 week to 3 months, with calibrated parameters $\eta = 1.1416$, $\rho = -0.6744$, $\alpha = -0.7377$, $c_1 = 1$, $c_2 = c_3 = 0$, fixed $\varepsilon = 1/52$:

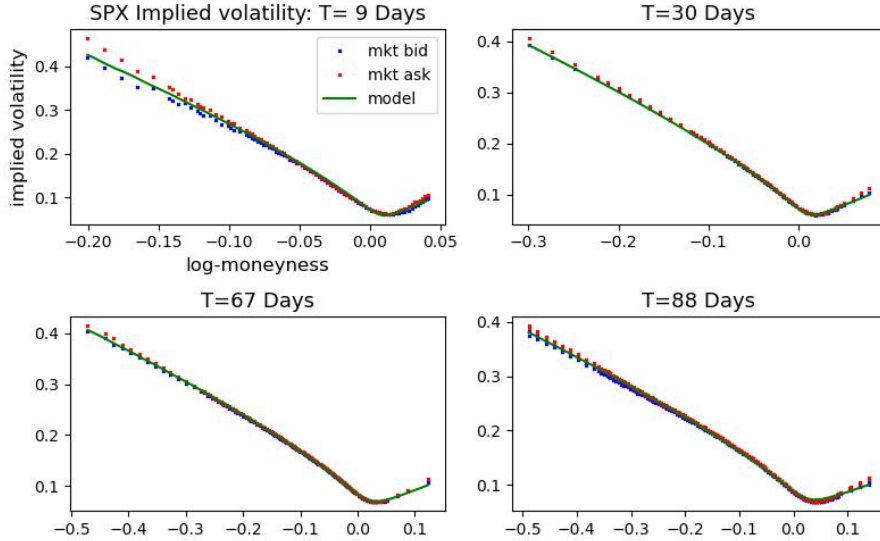


Figure 5: One-factor Bergomi model (green lines) calibrated to the SPX smiles (bid/ask in blue/red) on 23 October 2017 via Fourier using the Nelder-Mead optimization algorithm. The truncation level of the Riccati solver is set at $M = 32$.

The market data of SPX and VIX volatility surface is purchased from the CBOE website <https://datashop.cboe.com/>.

C Precise condition of the models

In this section, to clarify our models, we give the notations and more explicit and general definitions of the functions g_0, g_1, g_2, p mentioned before. This serves also to prove that the functions (as the characteristic functional $F(t, x)$) mentioned before are well-defined.

C.1 Notations

For a power series q , we denote by q_k the coefficients of x^k , i.e. $q(x) = \sum_{k=0}^{\infty} q_k x^k$. We denote by $|q|$ the **absolute power series** of q , i.e. $|q|(x) = \sum_{k=0}^{\infty} |q_k| x^k$. We remark that $|q|_k = |q_k|$. It is well-known that if p and q have an infinite radius of convergence, then $p(x)q(x) = \sum_{k=0}^{\infty} (p * q)_k x^k$ also has an infinite radius of convergence, with $(p * q)_k := \sum_{l=0}^k p_l q_{k-l}$. Unless stated otherwise, we will assume that all power series in this paper have infinite radius of convergence. Moreover, we specify that the term *continuity* means *joint continuity* when applied to a function taking (t, x) as variables. For example, a function $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{C}$ is C^∞ in x if all its space derivatives in x exist and are continuous in (t, x) . We also adopt the following notations for the partial derivatives: $f_t = \frac{\partial f}{\partial t}, f_x = \frac{\partial f}{\partial x}, f_{xx} = \frac{\partial^2 f}{\partial x^2}$ when they exist. We say that a function $f : \mathbb{R} \rightarrow \mathbb{C}$ is (real)-entire on $\mathbb{R}$, if it is equal to a power series q (with infinite radius of convergence), that is for all $x \in \mathbb{R}$, $f(x) = \sum_{k=0}^{\infty} q_k x^k$. We say that it is (complex)-entire when $\mathbb{R}$ is replaced by $\mathbb{C}$. Clearly, any real-entire function admits a complex-entire extension. When there is no-ambiguity, we will simply use the word entire. We say that a function does not vanish if it is non-zero at each point of its domain.

C.2 Conditions over g_0 and p in Section 2

First of all, for g_0 defined in (2.1), we suppose it to be a measure function $[0, T] \rightarrow \mathbb{R}_+$. For function p , we will introduce a general class of functions which includes the polynomial and exponential functions as we mentioned.

We will consider the following class of power series p for which $\int_0^T \mathbb{E}[p^2(X_s)] ds < \infty$.

Definition C.1. A power series $p : x \mapsto \sum_{k=0}^{\infty} p_k x^k$ is said to be **negligible to double factorial** if the power series $\sum_{k=0}^{\infty} (k-1)!! p_k x^k$ has an infinite radius of convergence, i.e.

$$\limsup_{k \rightarrow \infty} (|p_k| (k-1)!!)^{\frac{1}{k}} = 0,$$

with the convention $(-1)!! = 0!! = 1$.

By Definition C.1, a power series p negligible to double factorial also has infinite radius of convergence. This allows us to show that our class of Polynomial Ornstein-Uhlenbeck volatility models (2.1) is well-posed in Proposition C.3. For this we need a simple lemma about the absolute power series which will be useful later.

Lemma C.2.

- (i) for all $x \in \mathbb{R}$, $|p(x)| \leq |p|(|x|)$ and $|p|$ is monotonically increasing on $\mathbb{R}^+$;
- (ii) for all $x, y, z \geq 0$, $|p|(x + y + z) \leq (|p|(3x) + |p|(3y) + |p|(3z))$.

Proof. (i) is obvious; for (ii), it suffices to notice that $(x + y + z)^k \leq \frac{1}{3}((3x)^k + (3y)^k + (3z)^k) \leq ((3x)^k + (3y)^k + (3z)^k)$. $\square$

Proposition C.3. *Let $g_0 : [0, T] \rightarrow \mathbb{R}$ be a measurable and bounded function. Let p be a power series with infinite radius of convergence such that p^2 is negligible to double factorial. Then,*

$$\int_0^T \mathbb{E}[p^2(X_t)] dt < \infty, \quad (\text{C.1})$$

so that the stochastic integral $\int_0^\cdot \sigma_s dB_s$, with $\sigma_s = g_0(s)p(X_s)$, is well-defined and

$$S_t = S_0 \exp \left(-\frac{1}{2} \int_0^t \sigma_s^2 ds + \int_0^t \sigma_s dB_s \right), \quad t \geq 0, \quad (\text{C.2})$$

is the unique strong solution to (2.1).

Proof. Under (C.1), it is straightforward to obtain that the unique strong solution S to (2.1) is given by (C.2). It suffices to prove (C.1). For this, we note that the explicit solution of X is given by

$$X_t = e^{bt} X_0 + a \int_0^t e^{b(t-s)} ds + c \int_0^t e^{b(t-s)} dW_s.$$

We set $w(t) := a \int_0^t e^{b(t-s)} ds$, which is a deterministic continuous function of t , and $\widetilde{W}_t := c \int_0^t e^{b(t-s)} dW_s$, which is a Gaussian random variable $\sim \mathcal{N}(0, c^2 \frac{e^{2bt}-1}{2b})$ for $t \in [0, T]$. Applying Lemma C.2 yields

$$\begin{aligned} \left| \int_0^T |p^2| \left(e^{bs} X_0 + w(s) + \widetilde{W}_s \right) ds \right| &\leq \int_0^T |p^2| \left(e^{bs} |X_0| + |w(s)| + |\widetilde{W}_s| \right) ds \\ &\leq \int_0^T \sum_{k=0}^{\infty} |p^2|_k \left((3e^{bs} |X_0|)^k + |3w(s)|^k + |3\widetilde{W}_s|^k \right) ds, \end{aligned}$$

and since $e^{bs} \leq e^{|b|T}$ with $w(s)$ bounded in $[0, T]$, there exists a constant $C_T \geq 0$ such that

$$\int_0^T \sum_{k=0}^{\infty} |p^2|_k \left((3e^{bs} |X_0|)^k + |3w(s)|^k + |3\widetilde{W}_s|^k \right) ds \leq \int_0^T \sum_{k=0}^{\infty} |p^2|_k \left(|3e^{|b|T} |X_0|)^k + (3C_T)^k + |3\widetilde{W}_s|^k \right) ds.$$

By assumption, p^2 has an infinite radius of convergence, then so does $|p^2|$, therefore

$$\sum_{k=0}^{\infty} |p^2|_k |3e^{|b|T} |X_0||^k < \infty, \quad \sum_{k=0}^{\infty} |p^2|_k (3C_T)^k < \infty,$$

and thus we only need to prove

$$\int_0^T \mathbb{E}[|p^2|(|3\widetilde{W}_t|)] dt < \infty.$$

The random variable $3\widetilde{W}_t$ is Gaussian with bounded variance in $[0, T]$, so there exists a constant $C > 0$ such that $\mathbb{E}[|3\widetilde{W}_t|^k] = \mathbb{E}[|CZ|^k]$ for all $t \in [0, T]$, where Z is a standard normal variable. Given $\mathbb{E}[|Z|^k] \leq (k-1)!!$, hence

$$\mathbb{E}[|p^2|(C|Z|)] \leq \sum_{k=0}^{\infty} |p^2|_k (k-1)!! C^k < \infty,$$

since p^2 is negligible to double factorial. This completes the proof. $\square$

Remark C.4. In Proposition C.3, p^2 needs to be negligible to double factorial. As we will see later in Lemma F.3 below, the square of a power series negligible to double factorial is also negligible to double factorial. Hence p negligible to double factorial is sufficient for Proposition C.3 to hold.

Example C.5. Polynomials of finite degree (e.g. the Stein-Stein and the Quintic OU models) and the exponential function (e.g. the one-factor Bergomi model) are clearly negligible to double factorial. For the exponential function $p(x) = e^{\varepsilon x}$, $\varepsilon > 0$, it suffices to observe that $\lim_{k \rightarrow \infty} (\frac{\varepsilon^k (k-1)!!}{k!})^{\frac{1}{k}} = \lim_{k \rightarrow \infty} (\frac{\varepsilon^k}{k!})^{\frac{1}{k}} = 0$. However, the function $p(x) = e^{\varepsilon x^2}$ is not negligible to double factorial. Indeed, $\limsup_{k \rightarrow \infty} (|p_k| (k-1)!!)^{\frac{1}{k}} = \sqrt{2\varepsilon}$. In particular, if $b > 0$, then the expectation of $p(X_t)$ does not exist for all $t \geq \frac{1}{2c^2\varepsilon}$.

C.3 Conditions over g_1, g_2 in Section 4.1

In the following, if not mentioned particularly, we assume that g_1, g_2 are measurable functions $[0, T] \rightarrow \mathbb{C}$, such that $\Re(g_1) = 0, \Re(g_2) \leq 0$, and p^2 is negligible to double factorial, then the exponential above exists and with modulus of at most 1, so that (4.1) is well-defined.

D Theoretical result of the characteristic functional

We will present three main results related to the computation of the characteristic functional F in (4.1). The first result can be seen as a verification result and uncovers an affine structure in infinite dimension in terms of the powers $(1, X, X^2, X^3, \dots)$ by making a connection with infinite-dimensional Riccati deterministic equations (Theorem D.1 and Corollary D.5). The second result is concerned with the existence of a solution to the infinite-dimensional Riccati deterministic equation (Theorem D.4). The last result provides an approximation procedure for obtaining the characteristic functional (Theorem D.12).

D.1 A verification result

Our first main result uncovers an affine structure in infinite dimension expressed in powers $(1, X, X^2, \dots)$ and makes a connection with the following system of infinite dimensional deterministic Riccati equations:

$$\begin{aligned} \psi'_k(t) = & \left(g_2(t) + \frac{g_1(t)}{2}(g_1(t) - 1) \right) g_0^2(T-t)(p * p)_k \\ & + bk\psi_k(t) + a(k+1)\psi_{k+1}(t) + \frac{c^2(k+2)(k+1)}{2}\psi_{k+2}(t) \\ & + \frac{c^2}{2}(\tilde{\psi}(t) * \tilde{\psi}(t))_k + \rho g_1(t)g_0(T-t)c(p * \tilde{\psi}(t))_k, \quad \tilde{\psi}_k(t) := (k+1)\psi_{k+1}(t), \end{aligned}$$

$$\psi_k(0) = 0.$$

Theorem D.1. Let $g_0 : [0, T] \rightarrow \mathbb{R}$, $g_1, g_2 : [0, T] \rightarrow \mathbb{C}$ be measurable and bounded functions. Let p be a power series with infinite radius of convergence such that p^2 is negligible to double factorial. Assume that there exists a continuously differentiable solution $(\psi_k)_{k \geq 0}$ to the system of infinite dimensional Riccati equations (4.4) such that the power series $\sum_{k \geq 0} \sup_{t \in [0, T]} |\psi_k(t)| x^k$ has an infinite radius of convergence. Define the process

$$U_t = \sum_{k \geq 0} \psi_k(T-t) X_t^k + \int_0^t g_1(T-s) d \log S_s + \int_0^t g_2(T-s) \sigma_s^2 ds.$$

Then the process $M := \exp(U)$ is a local martingale. If in addition M is a true martingale, then the following expression holds for the joint characteristic functional F given in (4.1):

$$F(t, x) = \exp \left(\sum_{k \geq 0} \psi_k(T-t)x^k \right), \quad t \leq T.$$

Proof. The proof is given in Section E. □

Remark D.2. Theorem D.1 is in the spirit of [12, Theorem 5.5] which provides a similar verification result for the characteristic function of (real)-entire functions of solutions to stochastic differential equations with (real)-entire coefficients. Contrary to [12, Theorem 5.5], Theorem D.1 deals with time-integrated quantities.

A possible strategy for obtaining the representation (4.2) would involve verifying the assumptions outlined in Theorem D.1. These assumptions include ensuring the existence of a solution for the system of Riccati equations (4.3)-(4.4), such that the power series $(\psi_k)_{k \geq 0}$ has infinite radius of convergence; together with proving that the local martingale M is a true martingale. The latter can typically be achieved by arguing, for instance, that $\sum_{k \geq 0} \Re(\psi_k(t))x^k \leq 0$ to obtain that M is uniformly bounded by 1 whenever g_1, g_2 are purely imaginary for instance.

In the specific case of the Stein-Stein model, i.e. when p is an affine function, these assumptions are comparatively easier to confirm, as highlighted in the next example.

Example D.3. In the case of the classical Stein-Stein model [30], the volatility process process is defined as:

$$\sigma_t = g_0(t)X_t.$$

This is equivalent to (2.1) by setting $p_1 = 1$ and $p_k = 0$ for $k \neq 1$. Notice the convolution term $(p * p)_k = 1$ for $k = 2$ and zero otherwise. In this case, the infinite dimensional Riccati equations (4.4)-(4.3) reduce to the following:

$$\begin{aligned} \psi'_0(t) &= a\psi_1(t) + c^2\psi_2(t) + \frac{c^2}{2}\psi_1^2(t), \\ \psi'_1(t) &= b\psi_1(t) + 2a\psi_2(t) + 2c^2\psi_1(t)\psi_2(t) + \rho g_1(t)g_0(T-t)c\psi_1(t), \\ \psi'_2(t) &= \left(g_2(t) + \frac{g_1(t)}{2}(g_1(t) - 1) \right) g_0^2(T-t) \\ &\quad + 2b\psi_2(t) + 2c^2\psi_2^2(t) + 2\rho g_1(t)g_0(T-t)c\psi_2(t), \\ \psi_k(0) &= 0, \quad k \in \{0, 1, 2\}, \\ \psi_k &\equiv 0, \quad k \geq 3, \end{aligned}$$

which is a system of standard finite-dimensional Riccati equations whose existence is well-known whenever g_1, g_2 are such that

$$\Re \left(g_2(t) + \frac{g_1(t)}{2}(g_1(t) - 1) \right) \leq 0, \quad t \leq T.$$

Hence the characteristic functional of the classical Stein-Stein model is affine in $(1, X, X^2)$:

$$F(t, x) = \exp \left(\psi_0(T-t) + \psi_1(T-t)x + \psi_2(T-t)x^2 \right), \quad t \leq T,$$

for which ψ_0, ψ_1 and ψ_2 can even be solved explicitly when g_0, g_1 and g_2 are constants, see [27].

Obtaining the representation (4.2) under the Stein-Stein model can be attributed to the finite number of terms, i.e. (ψ_0, ψ_1, ψ_2) of the Riccati equations, involved in the sum. However, when dealing with an infinite sum, a notable challenge arises in proving that the log of the characteristic

functional $\log F$ is entire in the variable x on $\mathbb{R}$. In Section D.2, we show how to generate a solution for the system of infinite-dimensional Riccati equations modulo a non-vanishing condition of $\log F$. In Section D.3, we provide an expression for the characteristic functional using approximation arguments where the approximations are entire functions thanks to the Gaussianity of the process X .

D.2 Existence for the Riccati equations

Our second main result generates a solution to the infinite dimensional system of Riccati equations (4.3)-(4.4) by differentiating the logarithm of the characteristic functional F :

$$\psi_k(t) := \frac{1}{k!} \partial_x^k \log F(T-t, x) \Big|_{x=0}, \quad t \geq 0.$$

This requires the logarithm of a complex-valued function, see Appendix I for its precise definition.

Theorem D.4. *Fix $g_0 : [0, T] \rightarrow \mathbb{R}$, $g_1, g_2 : [0, T] \rightarrow \mathbb{C}$ continuously differentiable such that $\Re(g_1) = 0$ and $\Re(g_2) \leq 0$. Let p be a power series negligible to double factorial. Then, F in (4.1) is well-defined for any $t \leq T$, $x \in \mathbb{R}$ and $(t, x) \mapsto F(t, x)$ is continuous. If in addition F does not vanish on $[0, T] \times \mathbb{R}$, then $\log F$ can be defined as in Definition I.5. Furthermore, $\log F$ is C^∞ in x and C^1 in t . In particular, the family of functions*

$$\psi_k(t) = \frac{1}{k!} \partial_x^k \log F(T-t, x) \Big|_{x=0}, \quad t \geq 0, \quad k \geq 0, \quad (\text{D.1})$$

solves the system of Riccati ODEs (4.3)-(4.4).

Proof. The proof is given in Section F. □

Theorem D.4 establishes the existence of a solution to the Riccati ODEs (4.3)-(4.4) when the coefficients are real as shown in the following corollary.

Corollary D.5. *Fix $g_0 : [0, T] \rightarrow \mathbb{R}$, $g_1 = 0$ and $g_2 : [0, T] \rightarrow \mathbb{R}_-$ continuously differentiable. Let p be a power series negligible to double factorial. Then, $\log F$ is C^∞ in x and C^1 in t and the family of functions $(\psi_k)_{k \geq 0}$ given by (D.1) solves the system of Riccati ODEs (4.3)-(4.4).*

Proof. In this case, F reduces to the Laplace transform of the integrated variance:

$$F(t, x) = \mathbb{E} \left[\exp \left(\int_t^T g_2(T-s) \sigma_s^2 ds \right) \mid X_t = x \right],$$

which shows that $F > 0$ and that it trivially satisfies the non-vanishing condition on $[0, T] \times \mathbb{R}$ in Theorem D.4, so that an application of Theorem D.4 yields the result. □

Remark D.6. *Theorem D.4 is in the spirit of [12, Proposition 6.2], which provides the existence of a solution to the infinite dimensional ODE for the case of the characteristic function of powers of a single Brownian motion modulo similar assumptions. In contrast, Theorem D.4 deals with more involved time-integrated quantities which requires a more delicate analysis.*

Although Theorem D.4 does not establish that $\log F(t, \cdot)$ is entire, its analyticity can be inferred from the properties of solutions to parabolic partial differential equations (PDEs).

Remark D.7. In order to prove Theorem D.4, we establish the following PDE for $f(t, x) = F(t, x) \exp(v(t)q(x))$ in Theorem of F.11:

$$\begin{cases} f_t(t, x) + f_x(t, x)(a + bx) + \frac{1}{2}c^2 f_{xx}(t, x) + f(t, x) \sum_{i=1}^3 u_i(t)p_i(x) = 0, \\ f(T, x) = \exp(v(T)q(x)). \end{cases}$$

where v, u_i are continuous and p_i are analytic functions. Following [16, Theorem 6.2 in Section II.2.], given that functions $u_i(t)$ are continuous in $[0, T]$, $a + bx, p_i(x)$ are bounded, Hölder continuous and analytic in variable x within any open bounded set of $\mathbb{R}$, one would obtain that $f(t, \cdot)$ is analytic. Notice that $\exp(v(t)q(\cdot))$ is also analytic, then so is $F(t, \cdot)$. Since analyticity is a local property, if F does not vanish, then $\log F(t, \cdot)$ is also analytic. Therefore, we can deduce a local representation: for any $t \leq T$, there exists $r_t > 0$ such that

$$F(t, x) = \exp\left(\sum_{k=0}^{\infty} \psi_k(T-t)x^k\right), \quad x \in (-r_t, r_t),$$

where the $(\psi_k)_{k \geq 0}$ are defined by (D.1). In our case, both p_i and q are not only analytic but also entire. Extending the proof of analyticity using PDE techniques to establish that $F(t, \cdot)$ is entire might be possible, and that will lead to a global representation as in (4.2). However, this needs a more delicate analysis at the PDE level, diverging from our probabilistic approach that allowed us to obtain a global representation in terms of approximations by exponentially entire functions, see Theorem D.12 below. Such a question holds independent interest for PDEs in its own right.

D.3 The Fourier-Laplace transform by approximation

Our third main result provides an approximation procedure for the characteristic functional F . The main idea is to approximate the Riemann sum on the sample paths of the Ornstein-Uhlenbeck process X to exploit an underlying Gaussian density in finite dimension.

For this, we need first to get rid of the stochastic integrals and express them in terms of functions of X_T and Lebesgue's integrals on $(X_s)_{s \leq T}$. This is done by combining Itô's Lemma and the Romano-Touzi conditioning trick [28] in the next Lemma.

Lemma D.8. For p negligible to double factorial, $g_0 : [0, T] \rightarrow \mathbb{R}$, $g_1, g_2 : [0, T] \rightarrow \mathbb{C}$ continuously differentiable with $\Re(g_1) = 0, \Re(g_2) \leq 0$, there exists power series $q, p_i, i = 1, 2, 3$ with infinite radius of convergence, and functions $v, u_i, i = 1, 2, 3$ such that:

$$F(t, x) = \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s)p_i(X_s)ds + v(T)q(X_T) - v(t)q(X_t) \right) \mid X_t = x \right].$$

In addition, u_i, v are continuous on $[0, T]$, $\Re(u_1) \leq 0, \Re(u_2), \Re(u_3), \Re(v) = 0$, $p_1 = p^2$ such that p_i, q are all negligible to double factorial.

Proof. The proof and the precise expressions of u_i, v, p_i, q are given in Appendix H. □

We now introduce F_n the discretized version of F .

Definition D.9. For $\mu_n = \frac{T}{n} \sum_{i=0}^{n-1} \delta_{i\frac{T}{n}}$, where δ_t is the Dirac mass at a point t , we define the function

$$F_n(t, x) := \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s)p_i(X_s)\mu_n(ds) + v(T)q(X_T) - v(t)q(X_t) \right) \mid X_t = x \right].$$

Definition D.10. Given an entire function g on $\mathbb{R}$, we define the extension of g to the complex plane by $g_c(z) = \sum_{k=0}^{\infty} \frac{g^{(k)}(0)}{k!} z^k, z \in \mathbb{C}$.

Remark D.11. Given an entire function g on $\mathbb{R}$, the power series $\sum_{k=0}^{\infty} \frac{g^{(k)}(0)}{k!} x^k$ has infinite radius of convergence, so does $\sum_{k=0}^{\infty} \frac{g^{(k)}(0)}{k!} z^k$, thus $g_c(z)$ is well-defined in $\mathbb{C}$, and is entire on $\mathbb{C}$.

Theorem D.12. Suppose that p is negligible to double factorial, $g_0 : [0, T] \rightarrow \mathbb{R}$, $g_1, g_2 : [0, T] \rightarrow \mathbb{C}$ continuously differentiable such that $\Re(g_1) = 0, \Re(g_2) \leq 0$, then $F_n(t, \cdot)$ as specified in Definition D.9 is well-defined and entire in $x \in \mathbb{R}$. If in addition, the extension of $F_n(t, \cdot)$ to the complex plane $(F_n(t, \cdot))_c$ does not vanish for all $n \in \mathbb{N}$ and $t \leq T$, then, $\log F_n$ is entire in $x \in \mathbb{R}$ such that

$$F_n(t, x) = \exp \left(\sum_{k \geq 0} \psi_{n,k}(T-t)x^k \right), \quad t \leq T,$$

and

$$F(t, x) = \lim_{n \rightarrow \infty} F_n(t, x),$$

where for all $n \in \mathbb{N}$, the family of functions $\psi_{n,k}$ is defined by

$$\begin{aligned} \psi_{n,k}(t) &= \phi_{n,k}(t) - \rho g_1(t) g_0(T-t) \frac{p_{k-1}}{ck}, \quad k \geq 1, \\ \psi_{n,0}(t) &= \phi_{n,0}(t), \end{aligned}$$

where the family $\phi_{n,k}$ solves a (step-wise) ‘discretized’ system of Riccati ODEs:

$$\begin{aligned} \phi'_{n,k}(t) &= bk\phi_{n,k}(t) + a(k+1)\phi_{n,k+1}(t) + \frac{c^2(k+2)(k+1)}{2}\phi_{n,k+2}(t) \\ &\quad + \frac{c^2}{2}(\tilde{\phi}_n(t) * \tilde{\phi}_n(t))_k, \quad \tilde{\phi}_{n,k} = (k+1)\phi_{n,k+1}, \quad \frac{nt}{T} \notin \{0, 1, 2, \dots, n\} \\ \phi_{n,k}(t) &= \lim_{s \rightarrow t^-} \phi_{n,k}(s) + \frac{T}{n} \sum_{i=1}^3 u_i(T-t)(p_i)_k, \quad \frac{nt}{T} \in \{1, 2, \dots, n\} \\ \phi_{n,k}(0) &= v(T)q_k \end{aligned}$$

Proof. The proof is given in Section G. □

Remark D.13. One of the features of Theorem D.12 is to obtain that the functions $F_n(t, \cdot)$ are entire in $x \in \mathbb{R}$ with no additional assumptions. Unfortunately, $\log F_n(t, \cdot)$ inherits this property only when the extension of $F_n(t, \cdot)$ to the complex plane $(F_n(t, \cdot))_c$ does not vanish, see Lemma I.3. The requirement of F_n being non-vanishing on the real line is not enough, see Remark I.4. For this reason, we could not obtain a Corollary of Theorem D.12 without the non-vanishing condition to deal with Laplace transforms as we did in Corollary D.5.

E Proof of Theorem D.1

We first introduce a lemma to justify the use of Itô’s formula to the series $\sum_{k \geq 0} \psi_k(T-t)X_t^k$.

Lemma E.1. Under the condition of Theorem D.1, $h(t, x) := \sum_{k \geq 0} \psi_k(T-t)x^k$ is C^2 in x and C^1 in t , with $h_x(t, x) = \sum_{k \geq 0} (k+1)\psi_{k+1}(T-t)x^k$, $h_{xx}(t, x) = \sum_{k \geq 0} (k+2)(k+1)\psi_{k+2}(T-t)x^k$, $h_t(t, x) = \sum_{k \geq 0} -\psi'_k(T-t)x^k$.

Proof. First, $|\psi_k(T-t)x^k| \leq \sup_{t \in [0, T]} |\psi_k(t)| |x|^k$, and $\sum_{k \leq 0} \sup_{t \in [0, T]} |\psi_k(t)| |x|^k < \infty$. So $h(t, x) = \sum_{k \geq 0} \psi_k(T-t)x^k$ is well defined. In addition, if we restrict x in a bounded set of $\mathbb{R}$, then $h^n(t, x) := \sum_{k=0}^n \psi_k(T-t)x^k$ is continuous and converges uniformly to $h(t, x)$ when $n \rightarrow \infty$, so $h(t, x)$ is continuous.

Notice that the domain of convergence for $\sum_{k \leq 0} \sup_{t \in [0, T]} |\psi_k(t)| x^k < \infty$ is $\mathbb{R}$, then so is $\sum_{k \geq 0} \psi_k(T - t)x^k$. So $h_x(t, x), h_{xx}(t, x)$ are well-defined, continuous and have the expression as the statement for the same reason as for $h(t, x)$.

To treat $h_t(t, x)$, we should at first prove that $\sum_k \sup_{t \in [0, T]} |\psi'_k(t)| x^k$ has also an infinite radius of convergence.

Indeed, using the Riccati expression (4.3), we only have to check that we have an infinite radius of convergence for

$$\sum_{k \geq 0} \sup_{t \in [0, T]} |a_{t,k}| x^k, \quad (\text{E.1})$$

where $a_{t,k}$ is among $(p * p)_k, k\psi_k(t), (k+1)\psi_{k+1}, \frac{(k+1)(k+1)}{2}\psi_{k+1}, (\tilde{\psi}(t) * \tilde{\psi}(t))_k, (p * \tilde{\psi}(t))_k$.

By assumption and Cauchy–Hadamard Theorem, we know that $\limsup_{n \rightarrow \infty} (\sup_{t \in [0, T]} |\psi_n(t)|)^{\frac{1}{n}} = 0$ and thus we obtain that when $a_{t,x} = k\psi_k(t), (k+1)\psi_{k+1} = \tilde{\psi}(t)_k, \frac{(k+1)(k+2)}{2}\psi_{k+2}, (\text{E.1})$ has an infinite radius of convergence.

Also, for two power series $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ both with infinite radius of convergence, their product $\sum_{n=0}^{\infty} (a * b)_n x^n$ also has an infinite radius of convergence. And notice that $(p * p)_k, (\tilde{\psi}(t) * \tilde{\psi}(t))_k, (p * \tilde{\psi}(t))_k$ are dominated by $(|p| * |p|)_k, (\sup_{t \in [0, T]} |\tilde{\psi}(t)| * \sup_{t \in [0, T]} |\tilde{\psi}(t)|)_k, (|p| * \sup_{t \in [0, T]} |\tilde{\psi}(t)|)_k$. Thus when $a_{t,k}$ is among $(p * p)_k, (\tilde{\psi}(t) * \tilde{\psi}(t))_k, (p * \tilde{\psi}(t))_k, (\text{E.1})$ has also an infinite radius of convergence. Therefore $\sum_k \sup_{t \in [0, T]} |\psi'_k(t)| x^k$ has an infinite radius of convergence, so that $\sum_{k \geq 0} -\psi'_k(T - t)x^k$ also has infinite radius of convergence and converges uniformly when x is in a bounded subset of $\mathbb{R}$ and also on $t \in [0, T]$. So $\sum_{k \geq 0} -\psi'_k(T - t)x^k$ is well-defined and also continuous.

Next, notice that $h_t^n(t, x) = \sum_{k=0}^n -\psi'_k(T - t)x^k$ and given the the uniform convergence of both series $\sum_{k \geq 0} -\psi'_k(T - t)x^k$ and $\sum_{k \geq 0} \psi_k(T - t)x^k$, we deduce that $h_t(t, x) = (\sum_{k \geq 0} \psi_k(T - t)x^k)_t = \sum_{k \geq 0} -\psi'_k(T - t)x^k$.

□

Proof of Theorem D.1. We first notice that since the power series $\sum_k |\psi_k(t)| x^k$ has an infinite radius of convergence, U_t is well-defined for all $t \leq T$.

With Lemma E.1, we can now apply Itô's formula on the semimartingale M :

$$\begin{aligned} \frac{dM_t}{M_t} &= dU_t + \frac{1}{2} \langle U \rangle_t, \\ dU_t &= g_1(T - t) d \log S_t + g_2(T - t) \sigma_t^2 dt - \sum_k \psi'_k(T - t) X_t^k dt \\ &\quad + \sum_k (k+1) \psi_k(T - t) X_t^k dX_t + \sum_k \frac{1}{2} (k+2)(k+1) \psi_k(T - t) X_t^k d\langle X \rangle_t. \end{aligned} \quad (\text{E.2})$$

Plugging (2.1) into (E.2), we have:

$$\begin{aligned}
dU_t &= \left((g_2(T-t) - \frac{g_1(T-t)}{2}) \sigma_t^2 - \sum_k \psi'_k(T-t) X_t^k \right. \\
&\quad + \sum_k a(k+1) \psi_k(T-t) X_t^k + \sum_k bk \psi_k(T-t) X_t^k + \sum_k c^2 \frac{(k+2)(k+1)}{2} \psi_k(T-t) X_t^k \Big) dt \\
&\quad + \left(\sum_k c(k+1) \psi_k(T-t) X_t^k + \rho g_1(T-t) \sigma_t \right) dW_t + \sqrt{1-\rho^2} g_1(T-t) \sigma_t dW_t^\perp, \\
d\langle U \rangle_t &= \left(\sum_k ck \psi_k(T-t) X_t^k + \rho g_1(T-t) \sigma_t \right)^2 dt + (1-\rho^2) g_1^2(T-t) \sigma_t^2 dt.
\end{aligned}$$

Applying the Cauchy product on the power series $\sum_k p_k X_t^k$ leads to

$$\begin{aligned}
\frac{dM_t}{M_t} &= \left(\sum_k \left[(g_2(T-t) - \frac{g_1(T-t)}{2}) g_0^2(t) (p * p)_k - \psi'_k(T-t) + a(k+1) \psi_{k+1}(T-t) \right. \right. \\
&\quad + bk \psi_k(T-t) + \frac{c^2(k+2)(k+1)}{2} \psi_{k+2}(T-t) \\
&\quad + \frac{c^2}{2} (\tilde{\psi}(T-t) * \tilde{\psi}(T-t))_k + \frac{g_1^2(T-t) g_0^2(t)}{2} (p * p)_k \\
&\quad + \rho g_1(T-t) g_0(t) c (p * \tilde{\psi}(T-t))_k \Big] X_t^k \Big) dt \\
&\quad + \left(\sum_k ck \psi_k(T-t) X_t^k + \rho g_1(T-t) \sigma_t \right) dW_t + \sqrt{1-\rho^2} g_1(T-t) \sigma_t dW_t^\perp.
\end{aligned}$$

M is a local martingale if and only if it has zero drift (i.e. dt part is zero a.s.). This is true for all values of X by assumption from (4.3), so that M is a local martingale. Moreover, if M is a true martingale, we have

$$\begin{aligned}
&\mathbb{E} \left[\exp \left(\int_t^T g_1(T-s) d \log S_s + \int_t^T g_2(T-s) \sigma_s^2 ds \right) \middle| \mathcal{F}_t \right] \\
&= \exp \left(\sum_{k \geq 0} \psi_k(T-t) X_t^k \right), \quad t \leq T.
\end{aligned}$$

which follows from the martingality of M and the assumption of the initial condition of ψ_k from (4.4). This completes the proof. $\square$

F Proof of Theorem D.4

To prove Theorem D.4, we adopt the representation of F as mentioned in Lemma D.8:

$$F(t, x) = \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s) p_i(X_s) ds + v(T) q(X_T) - v(t) q(X_t) \right) \middle| X_t = x \right],$$

where the functions $p_1 = p^2, p_2, p_3, q$ are all negligible to double factorial and (u_1, u_2, u_3, v) are continuous such that $\Re(u_1) \leq 0, \Re(u_2) = \Re(u_3) = \Re(v) = 0$. Consider the following function f defined by:

$$\begin{aligned}
f(t, x) &:= \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s) p_i(X_s) ds + v(T) q(X_T) \right) \middle| X_t = x \right] \\
&= F(t, x) \exp(v(t) q(x)),
\end{aligned} \tag{F.1}$$

for all $t \leq T$ and $x \in \mathbb{R}$. We recall that f is well-defined, since

$$\Re \left(\int_t^T \sum_{i=1}^3 u_i(s) p_i(X_s) ds + v(T) q(X_T) \right) = \int_t^T \Re (u_1(s) p^2(X_s)) ds \leq 0.$$

Our proof is composed of five parts:

- In Subsection F.1, we start by deriving some properties of power series negligible to double factorial which will be used later.
- In Subsection F.2, we prove the regularity in x of f given by (F.1), i.e. f is C^∞ in x such that the partial derivatives $\frac{\partial^k f}{\partial x^k}$ are bounded in a sense in preparation for the Feynman-Kac formula in Subsection F.3.
- In Subsection F.3, we prove that f solves the associated PDE coming from Feynman-Kac formula. In particular, we prove that $\frac{\partial f}{\partial t}$ indeed exists.
- In Subsection F.4, we introduce a system of ODE and obtain a solution by comparing the derivatives of all orders of both sides of Feynman-Kac formula at $x = 0$.
- In Subsection F.5, we prove that the system of ODE in Theorem D.4 has a solution, via a system of ODE introduced in Subsection F.4.

F.1 Properties of power series negligible to double factorial

We collect some properties of power series that are negligible to double factorial as defined per Definition C.1.

Given a power series $p(x) = \sum_{k=0}^{\infty} p_k x^k$, we define the set $\mathcal{A}_D(p)$ as the $\mathbb{R}$ -algebra generated by all higher-order derivatives $\{p, p', p'' \dots\}$ of p :

$$\mathcal{A}_D(p) := \left\{ \sum_{s=0}^l c_s f_{s,1} f_{s,2} \cdots f_{s,m_s} : l \in \mathbb{N}, m_s \in \mathbb{N}, c_s \in \mathbb{R}, f_{s,i} \in \{p, p', p'' \dots\} \text{ for every } i \text{ and } s \right\}.$$

Notice that if $m_s = 0$, then by convention the product $f_{s,1} f_{s,2} \cdots f_{s,m_s} = 1$.

Remark F.1. *This definition comes from the calculations of the successive partial derivatives $F_x, F_{xx}, \dots$. For example, in the simple case when $g_0 = 1, g_1 = 0, g_2 = -1$, by setting $Z = \exp \left(- \int_0^{T-t} p^2(X_s) ds \right)$, we have $F(t, x) = \mathbb{E}[Z | X_0 = x]$ by the Markov property of X . Since we can write $X_s = e^{bs} X_0 + Y_s$, where Y_s does not depend on X_0 , this means formally, $F_x = \mathbb{E}[Z_x | X_0 = x]$, where $Z_x = -Z \int_0^{T-t} 2e^{bs} p p'(X_s) ds$ and $pp' \in \mathcal{A}_D(p)$ appears. Higher order derivatives like Z_{xx}, Z_{xxx} will also formally generate the elements in $\mathcal{A}_D(p)$.*

Now we prove some properties of power series which are negligible to double factorial. For the lemma below, we recall the convention $(-1)!! = (0)!! = 1$. Notice that although the proof is a little cumbersome, the essential idea is the observation that $\ell!!$ behaves like $(\lfloor \frac{\ell}{2} \rfloor)! 2^{\frac{\ell}{2}}$ and $\sum_{k=0}^{\ell} \binom{\ell}{k} = 2^\ell$.

Lemma F.2. *There exists a constant $C > 0$, such that for all $\ell \in \mathbb{N}$, we have that*

$$\sum_{k=0}^{\ell} \frac{1}{(k-1)!! (\ell-k-1)!!} \leq \frac{C \ell^3 2^{\frac{\ell}{2}}}{(\ell-1)!!}.$$

Proof. For $k = 0, \ell$, we have $\frac{1}{(k-1)!!(\ell-k-1)!!} = \frac{1}{(\ell-1)!!}$. So we need only to prove that for a constant $C > 0$,

$$\sum_{k=1}^{\ell-1} \frac{1}{(k-1)!!(\ell-k-1)!!} = \sum_{k=0}^{\ell-2} \frac{1}{k!!(\ell-2-k)!!} \leq \frac{C\ell^3 2^{\frac{\ell}{2}}}{(\ell-1)!!}.$$

Case 1: $\ell = 2\ell_1$ is an even number. If $k = 2k_1$ is an even number, $k!!(\ell-2-k)!! = 2^{\ell_1-1}k_1!(\ell_1-1-k_1)!$. If $k = 2k_1 + 1$ is an odd number, $k!!(\ell-2-k)!! = (2k_1+1)!!(\ell-3-2k_1)!! \geq \frac{(2k_1+1)!!(\ell-1-2k_1)!!}{\ell-1-2k_1} \geq \frac{(2k_1)!!(\ell-2-2k_1)!!}{\ell} = \frac{2^{\ell_1-1}k_1!(\ell_1-1-k_1)!}{\ell}$. Therefore,

$$\begin{aligned} \sum_{k=0}^{\ell-2} \frac{1}{k!!(\ell-2-k)!!} &\leq \sum_{k_1=0}^{\ell_1-1} \frac{\ell+1}{2^{\ell_1-1}k_1!(\ell_1-1-k_1)!} = \frac{\ell+1}{(\ell_1-1)!} = \frac{(\ell+1)2^{\ell_1-1}}{(\ell-2)!!} \\ &\leq \frac{(\ell+1)(\ell-1)2^{\ell_1-1}}{(\ell-1)!!} \leq \frac{\ell^2 2^{\frac{\ell}{2}}}{(\ell-1)!!}. \end{aligned}$$

Case 2: $\ell = 2\ell_1 + 1$ is an odd number, then

$$\sum_{k=0}^{\ell-2} \frac{1}{k!!(\ell-2-k)!!} \leq \sum_{k=0}^{\ell-3} \frac{1}{k!!(\ell-3-k)!!} + \frac{1}{(\ell-2)!!},$$

and since $\ell-3$ is even, as the Case 1, we have, for a constant $C > 0$:

$$\sum_{k=0}^{\ell-3} \frac{1}{k!!(\ell-3-k)!!} + \frac{1}{(\ell-2)!!} \leq \frac{(\ell-1)2^{\frac{\ell-1}{2}}}{(\ell-2)!!} + \frac{1}{(\ell-2)!!} \leq \frac{C\ell^3 2^{\frac{\ell}{2}}}{(\ell-1)!!}.$$

Therefore there exists constant C such that for all ℓ , $\sum_{k=0}^{\ell-2} \frac{1}{k!!(\ell-2-k)!!} \leq \frac{C\ell^3 2^{\frac{\ell}{2}}}{(\ell-1)!!}$. $\square$

Lemma F.3. *The following statements are true.*

- (i) *If p is negligible to double factorial, then so is $q(x) = \int_0^x p(y)dy$.*
- (ii) *The set of power series which are negligible to double factorial is closed under addition, differentiation, scalar multiplication, and multiplication. Thus if p is negligible to double factorial, then for any $q \in \mathcal{A}_D(p)$, q is also negligible to double factorial.*
- (iii) *If $p(x) = \sum p_k x^k$ is negligible to double factorial, then so is $q(x) = \sum p_k c^k x^k$, where c is a constant.*
- (iv) *If p is negligible to double factorial, then so is $|p|$.*

Proof.

- (i) Note that $q(x) = \sum_{k=1}^{\infty} \frac{p_{k-1}}{k} x^k$. It follows that

$$\begin{aligned} \limsup_{k \rightarrow \infty} (|q_k|(k-1)!!)^{\frac{1}{k}} &= \limsup_{k \rightarrow \infty} (|p_{k-1}| \frac{(k-1)!!}{k})^{\frac{1}{k-1}})^{\frac{k-1}{k}} \\ &\leq \limsup_{k \rightarrow \infty} (|p_{k-1}|(k-2)!!)^{\frac{1}{k-1}})^{\frac{k-1}{k}} = 0, \end{aligned}$$

using that $\frac{(k-1)!!}{k} \leq \frac{k!!}{k} = (k-2)!!$.

(ii) Let p, q be negligible to double factorial and note that $q'(x) = \sum_{k=0}^{\infty} (k+1)q_{k+1}x^k$. Then,

$$\limsup_{k \rightarrow \infty} (|q'_k|(k-1)!!)^{\frac{1}{k}} = \limsup_{k \rightarrow \infty} ((|q_{k+1}|(k+1)!!)^{\frac{1}{k+1}})^{\frac{k+1}{k}},$$

then by $(k+1)!! \leq (k+1)k!!$,

$$\limsup_{k \rightarrow \infty} (|q'_k|(k-1)!!)^{\frac{1}{k}} \leq \limsup_{k \rightarrow \infty} ((|q_{k+1}|k!!)^{\frac{1}{k+1}})^{\frac{k+1}{k}} (k+1)^{\frac{1}{k}} = 0.$$

By $|a+b|^{\frac{1}{k}} \leq |a|^{\frac{1}{k}} + |b|^{\frac{1}{k}}$, $\limsup_{k \rightarrow \infty} (|q_k + p_k|(k-1)!!)^{\frac{1}{k}} = 0$, and $\limsup_{k \rightarrow \infty} (|cq_k|(k-1)!!)^{\frac{1}{k}} = 0$ for any constant c .

Next, we will show that pq is again negligible to double factorial. For any $\epsilon > 0$, there exists N , such that for all $k > N$, we have that $(|q_k|(k-1)!!)^{\frac{1}{k}} \leq \epsilon^k$, $(|p_k|(k-1)!!)^{\frac{1}{k}} \leq \epsilon^k$. Therefore there exists constant C_ϵ such that $(|q_k|(k-1)!!)^{\frac{1}{k}} \leq C_\epsilon \epsilon^k$, $(|p_k|(k-1)!!)^{\frac{1}{k}} \leq C_\epsilon \epsilon^k$ for all k . Thus by Lemma F.2,

$$\begin{aligned} |(pq)_l| &= \left| \sum_{k=0}^l p_k q_{l-k} \right| \\ &\leq \sum_{k=0}^l |p_k| |q_{l-k}| \leq \sum_{k=0}^l \frac{C_\epsilon^2 \epsilon^l}{(k-1)!! (l-k-1)!!} \leq \frac{C_\epsilon^2 C(\sqrt{2}\epsilon)^l l^3}{(l-1)!!}, \end{aligned}$$

thus $\limsup_{l \rightarrow \infty} (|(pq)_l|(l-1)!!)^{\frac{1}{l}} \leq \sqrt{2}\epsilon$. Since ϵ can be any arbitrary positive number, we obtain

$$\limsup_{l \rightarrow \infty} (|(pq)_l|(l-1)!!)^{\frac{1}{l}} = 0.$$

Therefore, pq is also negligible to double factorial.

(iii) Notice that

$$\limsup_{k \rightarrow \infty} (|q_k|(k-1)!!)^{\frac{1}{k}} = |c| \limsup_{k \rightarrow \infty} (|p_k|(k-1)!!)^{\frac{1}{k}} = 0.$$

(iv) It is obvious, since the modulus of p has no impact on the $\limsup_{k \rightarrow \infty} (|p_k|(k-1)!!)^{\frac{1}{k}}$.

□

F.2 Regularity in space

In this subsection, we prove the regularity of f in x given by (F.1), i.e. f is C^∞ in x such that the partial derivatives $\frac{\partial^k f}{\partial x^k}$ are bounded in a sense in preparation for the Feynman-Kac formula in Subsection F.3.

Theorem F.4. *The function f given by (F.1) is C^∞ in x , i.e. the partial derivatives $\frac{\partial^k f}{\partial x^k}$ exist and are continuous in (t, x) , for $k \in \mathbb{N}$. Furthermore, for each $k \in \mathbb{N}$, there exists a power series q_k negligible to double factorial and a constant C_k independent of (t, x) , such that*

$$\left| \frac{\partial^k f}{\partial x^k}(t, x) \right| \leq |q_k|(|x|) + C_k, \quad t \in [0, T], \quad x \in \mathbb{R}. \quad (\text{F.2})$$

Before proving Theorem F.4, we first simplify the expression of f in (F.1) to get rid of the conditioning. Thanks to the Markovianity of X , we can write f after a change of variable as:

$$f(t, x) = \mathbb{E}[Z(t, x)], \quad (\text{F.3})$$

with

$$Z(t, x) := \exp \left(\int_0^{T-t} \sum_{i=1}^3 u_i(t+s) p_i \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds \right. \\ \left. + v(T) q(e^{b(T-t)} x + w(T-t) + \widetilde{W}_{T-t}) \right). \quad (\text{F.4})$$

Notice $X_t = e^{bt} x + w(t) + \widetilde{W}_t$ when $X_0 = x$, with $w(t) := a \int_0^t e^{b(t-u)} du$ and $\widetilde{W}_t := c \int_0^t e^{b(t-u)} dW_u$. From now on, we will consider mainly the representation (F.3) for $f(t, x)$.

The main idea for proving Theorem F.4 consists in taking successive derivatives in x inside the expectation above and applying the dominated convergence theorem. To illustrate this, we first calculate formally $\frac{\partial f}{\partial x}$. Notice that the derivative in x for the terms inside the exponential of $Z(t, x)$ is:

$$h(x) := \int_0^{T-t} \sum_{i=1}^3 u_i(t+s) e^{bs} p'_i \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds \\ + v(T) e^{b(T-t)} q' \left(e^{b(T-t)} x + w(T-t) + \widetilde{W}_{T-t} \right), \quad (\text{F.5})$$

so that one expects

$$\frac{\partial f}{\partial x}(t, x) = \mathbb{E} [h(x) Z(t, x)]. \quad (\text{F.6})$$

Notice that $|Z(t, x)| \leq 1$ since the terms inside the exponential of $Z(t, x)$ have a non-positive real part as a result of Lemma D.8. Therefore we only need to bound $h(x + \theta \Delta x)$ for $\theta \in [0, 1]$ and small enough $\Delta x \in \mathbb{R}$. Notice that u_i, v are continuous and thus bounded on $[0, T]$, e^{bs} is also bounded in $[0, T]$, p'_i, q' are negligible to double factorial by the statement (ii) of Lemma F.3. So we only need to bound the expressions of the following form

$$\int_0^t q \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds \quad \text{and} \quad q \left(e^{bt} x + w(t) + \widetilde{W}_t \right), \quad (\text{F.7})$$

where q is any power series negligible to double factorial.

First of all, we introduce a new definition and a lemma to help bound in a certain sense the quantities in (F.7). We also introduce the notation $\widetilde{W}_t^* := \sup_{s \leq t} |\widetilde{W}_s|$, which will be useful for applying Doob's inequality.

Definition F.5. We say that a family of processes $(M_t(x))_{t \leq T, x \in \mathbb{R}}$ is **estimable** if there exists a power series q negligible to double factorial such that for all fixed $t \in [0, T]$ and $x \in \mathbb{R}$,

$$|M_t(x)| \leq |q|(|x|) + |q|(\widetilde{W}_T^*), \quad a.s.$$

We will prove that the expressions in (F.7) are estimable in Proposition F.7 below. For now, let us first prove that $|q|(|x|) + |q|(\widetilde{W}_T^*)$ is integrable via the following lemma.

Lemma F.6. The set of estimable family of processes is closed under addition and multiplication. In addition, for all power series q negligible to double factorial, $\mathbb{E} [|q|(\widetilde{W}_T^*)] < \infty$. Therefore, if $M_t(x)$ is estimable, then there exists a power series q negligible to double factorial and a constant C , such that for all $t \in [0, T]$ and $x \in \mathbb{R}$, $\mathbb{E}[|M_t(x)|] \leq |q|(|x|) + C$.

Proof. For the first part of this lemma, recall that by the statements (ii) and (iv) of Lemma F.3, the property of being negligible to double factorial is closed under addition, multiplication, and taking

absolute power series. So the set of estimable family of process is obviously closed under addition. For the multiplication, it suffices to use the basic inequality $(a+b)(c+d) \leq a^2 + b^2 + c^2 + d^2$.

To show that $\mathbb{E} \left[|q|(\widetilde{W}_T^*) \right] < \infty$, recall $\widetilde{W}_t = e^{bt} \int_0^t e^{-bs} cdW_s$, where $\int_0^t e^{-bs} cdW_s$ is a true martingale. There is a positive constant C_T such that $\frac{1}{C_T} < e^{bt} < C_T$ when $0 \leq t \leq T$. We use $\overline{W}_t$ to denote $\int_0^t e^{-bs} cdW_s$ and $\overline{W}_t^*$ to denote its maximal process $\sup_{s \leq t} |\overline{W}_s|$. By Doob's maximal inequality for $k \geq 2$,

$$\mathbb{E} \left[(\widetilde{W}_T^*)^k \right] \leq C_T^k \mathbb{E} \left[(\overline{W}_T^*)^k \right] \leq C_T^k \left(\frac{k}{k-1} \right)^k \mathbb{E} \left[|\overline{W}_T|^k \right] \leq 4C_T^k \mathbb{E} \left[|\overline{W}_T|^k \right] \leq 4C_T^{2k} \mathbb{E} \left[|\widetilde{W}_T|^k \right].$$

For $k = 1$,

$$\mathbb{E} \left[\widetilde{W}_T^* \right] \leq \frac{1}{2} \left(1 + \mathbb{E} \left[(\widetilde{W}_T^*)^2 \right] \right) \leq 1 + 4C_T^4 \mathbb{E} \left[(\widetilde{W}_T)^2 \right].$$

Notice that $\widetilde{W}_T$ is a centred normal distribution and that the constant C_T can be taken large enough so that the variance of $\widetilde{W}_T$ is smaller than C_T^2 . It follows that $\mathbb{E} \left[|\widetilde{W}_T|^k \right] \leq (C_T^2)^{\frac{k}{2}} (k-1)!!$. By modifying a little the coefficients $|q|_0, |q|_1, |q|_2$ in the power series $|q|$, which does not influence the conclusion, we obtain that $C = 4 \sum_{k=0}^{\infty} |q|_k (k-1)!! C_T^{3k}$ satisfies the requirements and is finite, since q is negligible to double factorial. $\square$

Proposition F.7. *If q is negligible to double factorial, then the expressions in (F.7) are estimable, and*

$$\mathbb{E} \left[\left| \int_0^t q \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds \right| \right] + \mathbb{E} \left[\left| q \left(e^{bt} x + w(t) + \widetilde{W}_t \right) \right| \right] \leq |\tilde{q}|(|x|) + C, \quad t \in [0, T], x \in \mathbb{R},$$

where $\tilde{q}$ is negligible to double factorial and only depends on T, q , and C is a constant.

Proof. Fix $t \in [0, T]$ and $x \in \mathbb{R}$. We first bound $q \left(e^{bt} x + w(t) + \widetilde{W}_t \right)$:

$$\left| q \left(e^{bt} x + w(t) + \widetilde{W}_t \right) \right| \leq |q| \left(e^{bt} |x| + |w(t)| + |\widetilde{W}_t| \right) \leq \sum_{k=0}^{\infty} |q|_k \left((3e^{bt} |x|)^k + |3w(t)|^k + |3\widetilde{W}_t|^k \right)$$

by Lemma C.2. Since e^{bt} and $w(t)$ are both bounded in $[0, T]$, there exists a constant $C_T \geq 0$ such that

$$\begin{aligned} \sum_{k=0}^{\infty} |q|_k \left((3e^{bt} |x|)^k + |3w(t)|^k + |3\widetilde{W}_t|^k \right) &\leq \sum_{k=0}^{\infty} |q|_k \left(|3C_T x|^k + (3C_T)^k + |3\widetilde{W}_t|^k \right) \\ &\leq |q|(|3C_T x|) + |q|(|3C_T|) + |q|(|3\widetilde{W}_t^*|). \end{aligned}$$

Setting $\tilde{q}(x) = |q|(|3(C_T + 1)x|) + |q|(|3C_T|)$, we get $\left| q \left(e^{bt} x + w(t) + \widetilde{W}_t \right) \right| \leq \tilde{q}(|x|) + \tilde{q}(\widetilde{W}_T^*) = |\tilde{q}|(|x|) + |\tilde{q}|(\widetilde{W}_T^*)$. By Lemma F.3, $|q|$ is also negligible to double factorial, and $\tilde{q}$ is still negligible to double factorial. So $q \left(e^{bt} x + w(t) + \widetilde{W}_t \right)$ is estimable. An application of Lemma F.6 yields the bound for the expectation. Furthermore, the term $\int_0^t q \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds$ is clearly bounded by $T|\tilde{q}|(|x|) + T|\tilde{q}|(\widetilde{W}_T^*)$. So we can update $\tilde{q}$ to $\tilde{q}(x) = T \vee 1 |q|(|3(C_T + 1)x|) + T \vee 1 |q|(|3C_T|)$ to obtain the claimed bound and end the proof. $\square$

We have now that $\left| \int_0^t q \left(e^{bs} (x + \theta \Delta x) + w(s) + \widetilde{W}_s \right) ds \right| \leq |\tilde{q}|(|x + \theta \Delta x|) + |\tilde{q}|(\widetilde{W}_T^*) \leq |\tilde{q}|(|x| + 1) + |\tilde{q}|(\widetilde{W}_T^*)$ since we can choose that $|\Delta x| \leq 1$, $\theta \in [0, 1]$ with coefficient of $|\tilde{q}|$ all positive. The dominating function $|\tilde{q}|(|x| + 1) + |\tilde{q}|(\widetilde{W}_T^*)$ is integrable. For the term $q \left(e^{bt} (x + \theta \Delta x) + w(t) + \widetilde{W}_t \right)$, we can build a dominating function in a similar way.

Going back to (F.6), we now have all the ingredients to apply the dominated convergence theorem when $\Delta x \rightarrow 0$ on

$$\frac{f(t, x + \Delta x) - f(t, x)}{\Delta x} = \mathbb{E} \left[\frac{Z(t, x + \Delta x) - Z(t, x)}{\Delta x} \right] = \mathbb{E} [h(x + \theta \Delta x) Z(t, x + \theta \Delta x)],$$

where $\theta \in [0, 1]$. Recall $h(x)$ from (F.5), and that $|Z(t, x)| \leq 1$ and u_i, v are bounded, e^{bs} is bounded in $[0, T]$, p'_i, q' are negligible to double factorial. Therefore, (F.6) holds. Of course, we would like to prove by induction that

$$\frac{\partial^k f}{\partial x^k}(t, x) = \mathbb{E} [H_k(t, x) Z(t, x)]$$

holds for all $k \in \mathbb{N}$ to show that f is C^∞ in x , where

$$H_{k+1}(t, x) = \frac{\partial H_k(t, x)}{\partial x} + H_k(t, x)h(x), \quad H_0(t, x) = 1, \quad H_1(t, x) = h(x). \quad (\text{F.8})$$

To achieve this, we need to prove first that H_k are well-defined. Then, similarly to before, we will bound $H_k(x)$ so that we can apply by induction the dominated convergence theorem to

$$\begin{aligned} & \lim_{\Delta x \rightarrow 0} \frac{\frac{\partial^k f}{\partial x^k}(t, x + \Delta x) - \frac{\partial^k f}{\partial x^k}(t, x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \mathbb{E} \left[\frac{H_k(t, x + \Delta x) Z(t, x + \Delta x) - H_k(t, x) Z(t, x)}{\Delta x} \right] \\ &= \lim_{\Delta x \rightarrow 0} \mathbb{E} [H_{k+1}(t, x + \theta \Delta x) Z(t, x + \theta \Delta x)]. \end{aligned}$$

For $H_2(t, x)$, we need to compute $\frac{\partial h(x)}{\partial x}$, recall that $h(x)$ is just a sum of the Riemann integral of the continuously differentiable function and a differentiable function with respect to x for a fixed ω outside a null-set. Therefore,

$$\begin{aligned} \frac{\partial h(x)}{\partial x} &= \int_0^{T-t} \sum_{i=1}^3 u_i(t+s) e^{2bs} p_i'' \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds \\ &\quad + v(T) e^{2b(T-t)} q'' \left(e^{b(T-t)} x + w(T-t) + \widetilde{W}_{T-t} \right). \end{aligned}$$

Thus H_2 can be obtained explicitly, and similarly for $H_3, H_4, \dots, H_k, \dots$. This procedure will differentiate many times the function $h(x)$, so it is useful to define:

$$\begin{aligned} h_k(x) &:= \int_0^{T-t} \sum_{i=1}^3 u_i(t+s) e^{kbs} p_i^{(k)} \left(e^{bs} x + w(s) + \widetilde{W}_s \right) ds \\ &\quad + v(T) e^{kb(T-t)} q^{(k)} \left(e^{b(T-t)} x + w(T-t) + \widetilde{W}_{T-t} \right). \end{aligned}$$

Definition F.8. We define the set $\mathcal{A}_x(h)$:

$$\mathcal{A}_x(h) := \left\{ \sum_{s=0}^l c_s h_{s,1} h_{s,2} \cdots h_{s,m_s} : l \in \mathbb{N}, m_s \in \mathbb{N}, c_s \in \mathbb{R}, h_{s,i} \in \{h\}_k \text{ for every } i \text{ and } s \right\}.$$

as the $\mathbb{R}$ -algebra generated by higher order derivatives of the function $h(x)$. We call h_k generating elements of $\mathcal{A}_x(h)$.

Notice that if $m_s = 0$, then by convention the product $h_{s,1} h_{s,2} \cdots h_{s,m_s} = 1$.

In the next Lemma F.9, we will characterize H_k defined in (F.8).

Lemma F.9. *For every $k \in \mathbb{N}$, the function H_k in (F.8) is well-defined. In addition, H_k is differentiable in x , continuous in (t, x) and estimable.*

Proof. First, note the following three facts:

- (i) $h(x) = h_1(x)$ is a generating element of $\mathcal{A}_x(h)$,
- (ii) $H_0 = 1 \in \mathcal{A}_x(h)$,
- (iii) All generating elements $h_k(x)$ of $\mathcal{A}_x(h)$ are differentiable in x by applying the Leibniz's rule, and $\frac{\partial h_k(x)}{\partial x} = h_{k+1}(x)$. So if $g \in \mathcal{A}_x(h)$, then $\frac{\partial g}{\partial x}$ exists and $\in \mathcal{A}_x(h)$.

By induction and noticing that $\mathcal{A}_x(h)$ is closed under linear sum and multiplication, H_k are well-defined and in $\mathcal{A}_x(h)$. For fixed $\omega \in \Omega$ outside a null-set, the generating elements h_k are continuous in (t, x) and differentiable in x . This implies that H_k is also continuous in (t, x) and differentiable in x .

We know u_i, v are continuous (and thus bounded), e^{kbs} is bounded in $[0, T]$, $p_i^{(k)}, q^{(k)}$ are negligible to double factorial by the statement 2 of Lemma F.3. By Proposition F.7, generating elements h_k of $\mathcal{A}_x(h)$ are estimable. Since $H_k \in \mathcal{A}_x(h)$ is a linear sum of finite product of generating elements of $\mathcal{A}_x(h)$, by Lemma F.6 H_k is also estimable. $\square$

We are now ready to prove Theorem F.4.

Proof of Theorem F.4. Take H_k as defined in (F.8) which are estimable by Lemma F.9. By applying Lemma F.6, the term $H_k(t, x)Z(t, x)$ is estimable with its expectation bounded by $|q_k|(|x|) + C_k$. Using induction, we first notice that for the case $k = 0$, $\left| \frac{\partial^k f}{\partial x} \right| = |f(t, x)| \leq \mathbb{E}[|Z(t, x)|] \leq 1$ which trivially satisfies the inequality in (F.2). Next, suppose that it is true for up to case k , recall the definition of Z from (F.4) and that $Z_x = hZ$, where h is defined in (F.5). Thus $(H_k Z)_x = H_{k+1} Z$. Since both Z and H_{k+1} are differentiable in x as per F.9, by the Mean value theorem:

$$\frac{1}{\Delta x} \left(H_k(t, x + \Delta x)Z(t, x + \Delta x) - H_k(t, x)Z(t, x) \right) = H_{k+1}(t, x + \theta \Delta x)Z(t, x + \theta \Delta x),$$

where $\theta \in [0, 1]$. Strictly speaking, since $H_k Z$ is a complex-valued functions, we need to apply the Mean value on both the real part and imaginary part separately with different θ . However, this does not change the proof at all, so to simplify the notation and discussion, only θ is used.

By Lemma C.2, Lemma F.9 and the fact that $|Z| \leq 1$, and $H_{k+1}(t, x + \theta \Delta x)$ is dominated by $|q_{k+1}|(|x|+1) + |q_{k+1}|(\widetilde{W}_T^*)$ with bounded expectation by Lemma F.6 if we choose that $|\Delta x| \leq 1$, where q_{k+1} is negligible to double factorial. Applying dominated convergence theorem, for $\Delta x \rightarrow 0$ we have:

$$\frac{\partial^{k+1} f}{\partial x^{k+1}}(t, x) = \mathbb{E} \left[H_{k+1}(t, x)Z(t, x) \right].$$

Therefore, for all $k \in \mathbb{N}$, where q_k negligible to double factorial and a constant C_k , we have that (F.2) holds. Finally, for the continuity of $\frac{\partial^k f}{\partial x^k}(t, x)$, it suffices to notice that before taking the expectation, the random variable $H_k(t, x)Z(t, x)$ is continuous with respect to (t, x) . Then fixing (t_0, x_0) , again by Proposition F.7, Lemma F.9, its expectation is uniformly bounded with $t \leq T$, and $|x| \leq |x_0|+1$ bounded. So again by the dominated convergence theorem and taking the limit at (t_0, x_0) , the continuity holds. $\square$

F.3 Feynman-Kac

In this subsection, we derive the Feynman-Kac formula in Theorem F.11. Since we do not have that f_t exists a priori in our setting, we shall prove its existence and obtain the Feynman-Kac formula at the same time.

First, we introduce a lemma which will be useful later.

Lemma F.10. *Let $z \in \mathbb{C}$ such that $\Re(z) \leq 0$, then $|\exp(z) - 1| \leq 3|z|$.*

Proof. We write $z = x + yi$, $x \leq 0$ and $y \in \mathbb{R}$. Then $|\exp(z) - 1| = |\exp(x + yi) - 1| = |\exp(yi)(\exp(x) - 1) + \exp(yi) - 1| \leq |\exp(x) - 1| + |\cos(y) - 1| + |\sin(y)|$.

If $x = 0$, then $|\exp(x) - 1| = 0$. If $x < 0$, $|\frac{\exp(x)-1}{x}| \leq |\frac{\exp(x)-1}{x}| \leq 1$.

If $y = 0$, then $|\cos(y) - 1| = 0$, $|\sin(y)| = 0$. If $y \neq 0$, $|\frac{\cos(y)-1}{y}| \leq |\frac{\cos(y)-1}{y}| \leq 1$, $|\frac{\sin(y)}{y}| \leq |\frac{\sin(y)}{y}| \leq 1$. $\square$

Theorem F.11. *The function f given by (F.1) and equivalently (F.3) is C^∞ in x , C^1 in t , and solves the following partial differential equation (PDE):*

$$\begin{cases} f_t(t, x) + f_x(t, x)(a + bx) + \frac{1}{2}c^2 f_{xx}(t, x) + f(t, x) \sum_{i=1}^3 u_i(t) p_i(x) = 0, \\ f(T, x) = \exp(v(T)q(x)). \end{cases} \quad (\text{F.9})$$

Proof. By Theorem F.4, f is C^∞ in x . We will compute the infinitesimal generator

$$\lim_{r \rightarrow 0^+} \frac{1}{r} (\mathbb{E}^x [f(t_0, X_r) - f(t_0, x)])$$

in two ways to obtain the PDE and the existence of f_t at the same time. Here $\mathbb{E}^x$ means the conditional expectation with $X_0 = x$.

First way: fix t_0 , we compute the quantity using the classical definition of generator. By Itô's formula:

$$df(t_0, X_t) = f_x dX_t + \frac{1}{2} f_{xx} d\langle X \rangle_t = c f_x dW_t + \left(\frac{1}{2} c^2 f_{xx} + (a + bX_t) f_x \right) dt.$$

Note that f_x, f_{xx} are evaluated at (t_0, X_t) here. By Theorem F.4, statement (ii) of Lemma F.3, $f_x(t, x), (a + bx)f_x(t, x), f_{xx}(t, x), f_x^2(t, x)$ are all bounded by $|q|(|x|) + C$, where q is negligible to double factorial. Combining with Proposition F.7, $\mathbb{E}^x [\int_0^r f_x^2(t_0, X_t) dt]$ is bounded by $|\tilde{q}|(|x|) + C$ for $r \in [0, T]$, where $\tilde{q}$ is negligible to double factorial. Thus $\int_0^r c f_x(t_0, X_t) dW_t$ is in L^2 and is a true martingale for $t \in [0, T]$. Similarly, the Riemann integral $\int_0^r (\frac{1}{2} c^2 f_{xx}(t_0, X_t) + (a + bX_t) f_x(t_0, X_t)) dt$ is estimable with finite expectation. In addition, we have

$$\left| \frac{1}{r} \int_0^r \left(\frac{1}{2} c^2 f_{xx}(t_0, X_t) + (a + bX_t) f_x(t_0, X_t) \right) dt \right| \leq \sup_{s \in [0, r]} \frac{1}{2} c^2 |f_{xx}(t_0, X_s)| + |(a + bX_s) f_x(t_0, X_s)|,$$

where the right hand side is still estimable and dominated by a random variable with finite expectation. By dominated convergence theorem,

$$\begin{aligned} & \lim_{r \rightarrow 0^+} \frac{1}{r} (\mathbb{E}^x [f(t_0, X_r) - f(t_0, x)]) \\ &= \lim_{r \rightarrow 0^+} \frac{1}{r} \mathbb{E}^x \left[\int_0^r \left(\frac{1}{2} c^2 f_{xx}(t_0, X_t) + (a + bX_t) f_x(t_0, X_t) \right) dt \right] \\ &= \frac{1}{2} c^2 f_{xx}(t_0, x) + (a + bx) f_x(t_0, x). \end{aligned} \quad (\text{F.10})$$

Second way: compute the generator $\lim_{r \rightarrow 0^+} \frac{1}{r} (\mathbb{E}^x [f(t_0, X_r) - f(t_0, x)])$ directly by applying the Markov property of X_t . Define $Z_t := \exp(\int_0^t \sum_{i=1}^3 u_i(T-t+s)p_i(X_s) ds)$ and $Y_t = v(T)q(X_t)$, we apply Markov property of X_t to the representation of $f(t, x)$ in (F.1), i.e.

$$f(t, x) = \mathbb{E} \left[\exp \left(\int_0^{T-t} \sum_{i=1}^3 u_i(t+s)p_i(X_s) ds + v(T)q(X_{T-t}) \right) \mid X_0 = x \right],$$

and by Markov property for $r \in [0, t]$,

$$f(t, X_r) = \mathbb{E} \left[\exp \left(\int_r^{T-t+r} \sum_{i=1}^3 u_i(t-r+s)p_i(X_s) ds + v(T)q(X_{T-t+r}) \right) \mid X_r \right].$$

By the tower property of conditional expectation, we have

$$\begin{aligned} \mathbb{E}^x [f(t, X_r)] &= \mathbb{E}^x \left[\mathbb{E} \left[\exp \left(\int_r^{T-t+r} \sum_{i=1}^3 u_i(t-r+s)p_i(X_s) ds + v(T)q(X_{T-t+r}) \right) \mid X_r \right] \right] \\ &= \mathbb{E}^x \left[\exp \left(\int_r^{T-t+r} \sum_{i=1}^3 u_i(t-r+s)p_i(X_s) ds + v(T)q(X_{T-t+r}) \right) \right] \\ &= \mathbb{E}^x \left[Z_{T-t+r} \cdot \exp \left(- \int_0^r \sum_{i=1}^3 u_i(t-r+s)p_i(X_s) ds \right) \exp(Y_{T-t+r}) \right]. \end{aligned}$$

Therefore,

$$\begin{aligned} &\frac{1}{r} (\mathbb{E}^x [f(t_0, X_r) - f(t_0, x)]) \\ &= \frac{1}{r} \mathbb{E}^x \left[Z_{T-t_0+r} \exp \left(- \int_0^r \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) \exp(Y_{T-t_0+r}) - Z_{T-t_0} \exp(Y_{T-t_0}) \right] \\ &= \frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r} - \exp(Y_{T-t_0}) Z_{T-t_0} \right] \\ &\quad + \frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r} \left(\exp \left(- \int_0^r \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) - 1 \right) \right] \\ &= \frac{1}{r} (f(t_0-r, x) - f(t_0, x)) \\ &\quad + \frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r} \left(\exp \left(- \int_0^r \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) - 1 \right) \right]. \end{aligned}$$

Now we want to apply dominated convergence theorem when $r \rightarrow 0^+$ to the term

$$\frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r} \left(\exp \left(- \int_0^r \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) - 1 \right) \right].$$

To dominate the term inside the expectation, notice that at first $\Re(Y_t) = \Re(v(T)q(X_t)) = 0$, thus $|\exp(Y_{T-t_0+r})| = 1$. In addition,

$$\begin{aligned} &\left| \frac{1}{r} Z_{T-t_0+r} \left(\exp \left(- \int_0^r \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) - 1 \right) \right| \\ &= \left| \frac{1}{r} \exp \left(\int_r^{T-t_0+r} \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) \left(1 - \exp \left(\int_0^r \sum_{i=1}^3 u_i(t_0-r+s)p_i(X_s) ds \right) \right) \right|. \end{aligned}$$

Since $\Re(u_1) \leq 0$, $\Re(u_2), \Re(u_3) = 0$, $p_1 = p^2$, therefore

$$\Re\left(\int_r^{T-t_0+r} \sum_{i=1}^3 u_i(t_0 - r + s) p_i(X_s) ds\right) = \int_r^{T-t_0+r} \Re\left(u_1(t_0 - r + s) p_1(X_s)\right) ds \leq 0$$

and thus $\left|\exp\left(\int_r^{T-t_0+r} \sum_{i=1}^3 u_i(t_0 - r + s) p_i(X_s) ds\right)\right| \leq 1$. Therefore we only need to dominate the term

$$\frac{1}{r} \left(\exp\left(\int_0^r \sum_{i=1}^3 u_i(t_0 - r + s) p_i(X_s) ds\right) - 1 \right).$$

Since $\Re\left(\int_0^r \sum_{i=1}^3 u_i(t_0 - r + s) p_i(X_s) ds\right) \leq 0$, by Lemma F.10 we just need to dominate the term

$$\left| \frac{1}{r} \int_0^r \sum_{i=1}^3 u_i(t_0 - r + s) p_i(X_s) ds \right| \leq \sum_{i=1}^3 \|u_i\|_\infty \sup_{s \in [0, r]} |p_i(X_s)|.$$

Notice $|u_i(s)|$ are bounded, p_i are negligible to double factorial so $|p_i(X_s)|, s \in [0, T]$ is dominated when conditioned $X_0 = x$ by Proposition F.7. Thus by the dominated convergence theorem :

$$\begin{aligned} & \lim_{r \rightarrow 0^+} \frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r} \left(\exp\left(-\int_0^r \sum_{i=1}^3 u_i(t_0 - r + s) p_i(X_s) ds\right) - 1 \right) \right] \\ &= - \sum_{i=1}^3 u_i(t_0) p_i(x) f(t_0, x) \end{aligned}$$

which is finite. Equating with (F.10), the term $\lim_{r \rightarrow 0^+} \frac{1}{r} (f(t_0 - r, x) - f(t_0, x))$ is thus well-defined. Of course, one can also replace t_0 by $t_0 + r$ and perform similar computations as per the above two methods on the quantity

$$\lim_{r \rightarrow 0^+} \frac{1}{r} (\mathbb{E}^x [f(t_0 + r, X_r) - f(t_0 + r, x)])$$

to obtain the existence of f_t and also the PDE in (F.9). A subtle point to note is that for the first method, we do not apply Itô's formula to $f(t_0 + r, X_r)$ for the variable r , which requires the existence of the partial derivative f_t a priori. Instead, we obtain the existence of f_t by applying Itô's formula to X_r in $f(t_0 + u, X_r)$ and evaluate at $u = r$:

$$\begin{aligned} & \frac{1}{r} \left(\mathbb{E}^x [f(t_0 + r, X_r) - f(t_0 + r, x)] \right) \\ &= \frac{1}{r} \mathbb{E}^x \left[\int_0^r \left(\frac{1}{2} c^2 f_{xx}(t_0 + r, X_t) + (a + bX_t) f_x(t_0 + r, X_t) \right) dt \right], \end{aligned}$$

and taking the limit for $r \rightarrow 0^+$ as above. The continuity of f_t is directly obtained from the continuity of the other terms in this equation, with boundary condition at T comes from the continuity of f and its definition. $\square$

F.4 Infinite dimensional ODE

In this subsection, we obtain a solution of the system of ODE by comparing the derivatives of both sides of the PDE (F.9) when $x = 0$.

Since f is continuous by Theorem F.4, in addition if f does not vanish, we can define $\log f$ such that $\exp(\log f) = f$ and $\log f(T, 0) = v(T)q(0)$, see Lemma I.1. Note $\log f$ is also continuous in (t, x) , and $\exp(\log f)(T, x) = f(T, x) = \exp(v(T)q(x))$ with $\log f(T, x) = v(T)q(x)$.

Theorem F.12. *If f does not vanish, then*

$$\phi_k(t) := \frac{1}{k!} \partial_x^k \log f(T-t, x) \Big|_{x=0}, \quad t \geq 0$$

solves the system of ODE

$$\begin{aligned} \phi'_k(t) &= \sum_{i=1}^3 u_i(T-t)(p_i)_k \\ &\quad + bk\phi_k(t) + a(k+1)\phi_{k+1}(t) + \frac{c^2(k+2)(k+1)}{2}\phi_{k+2}(t) \\ &\quad + \frac{c^2}{2}(\tilde{\phi}(t) * \tilde{\phi}(t))_k, \quad \tilde{\phi}_k(t) = (k+1)\phi_{k+1}(t), \\ \phi_k(0) &= v(T)q_k. \end{aligned} \tag{F.11}$$

In order to prove the Theorem, we first link the PDE (F.9) to the system of ODE (F.11). Suppose f does not vanish and set $g(t, x) = \log f(T-t, x)$, then the following PDE holds:

$$\begin{cases} g_t(t, x) = \sum_{i=1}^3 u_i(T-t)p_i(x) + g_x(t, x)(a+bx) + \frac{1}{2}c^2g_{xx}(t, x) + \frac{1}{2}c^2g_x^2(t, x), \\ g(0, x) = v(T)q(x). \end{cases}$$

Since by Theorem F.11, f is C^1 in t and C^∞ in x , by Lemma I.2, $\log f$ is also C^1 in t and C^∞ in x . Then so is g .

To prove this theorem, we will need the following lemma.

Lemma F.13. *If a function h is C^1 in (t, x) , and one of the partial derivatives $\frac{\partial^2 h}{\partial t \partial x}$ and $\frac{\partial^2 h}{\partial x \partial t}$ exists and is continuous, then they both exist and are equal.*

Now we can prove Theorem F.12.

Proof of Theorem F.12. Since $F(t, x)$ does not vanish, we can define $\log F$ as per Definition I.5, i.e.

$$\exp(\log F(t, x)) = F(t, x), \quad \log F(T, 0) = 0.$$

We can now define $\log f(t, x) = \log F(t, x) + v(t)q(x)$ such that $\exp(\log f(t, x)) = f(t, x)$. By Theorem F.11, f is C^∞ in x and C^1 in t , then since $\log f$ is continuous, by Lemma I.2, $\log f$ is also C^∞ in x and C^1 in t . In addition, the PDE below can also be deduced:

$$\begin{aligned} -(\log f)_t(t, x) &= \sum_{i=1}^3 u_i(t)p_i(x) + (\log f)_x(t, x)(a+bx) + \frac{1}{2}c^2(\log f)_{xx}(t, x) + \frac{1}{2}c^2(\log f)_x^2(t, x) \\ \log f(T, x) &= v(T)q(x). \end{aligned}$$

The boundary condition comes from the fact $\log f(T, 0) = v(T)q(0)$, with the continuity of $\log f$ and the fact that $\exp(\log f)(T, x) = f(T, x) = \exp(v(T)q(x))$.

Now substitute $g(t, x) = \log f(T-t, x)$:

$$\begin{aligned} g_t(t, x) &= \sum_{i=1}^3 u_i(T-t)p_i(x) + g_x(t, x)(a+bx) + \frac{1}{2}c^2g_{xx}(t, x) + \frac{1}{2}c^2g_x^2(t, x) \\ g(0, x) &= v(T)q(x). \end{aligned} \tag{F.12}$$

Since the right side of (F.12) is C^∞ in x , so is g_t . Notice that by definition $\frac{\partial^k}{\partial x^k} g(t, x)|_{x=0} = k! \phi_k(t)$. We aim to take the k^{th} partial derivative with respect to x on both sides of (F.12) at $x = 0$ to deduce the equation in (F.11). For the left side of (F.11), Lemma F.13 allows the interchangeability of $\partial t, \partial x$ and the following equality can be proven by induction:

$$\frac{\partial^k}{\partial x^k} \frac{\partial}{\partial t} g(t, x)|_{x=0} = \frac{\partial}{\partial t} \frac{\partial^k}{\partial x^k} g(t, x)|_{x=0} = k! \phi'_k(t).$$

The right side of (F.11) can be deduced by noticing that $\frac{\partial^k}{\partial x^k} g(t, x)|_{x=0} = k! \phi_k(t)$ for all k . To obtain that

$$\frac{\partial^k}{\partial x^k} g_x^2(t, x)|_{x=0} = k! (\tilde{\phi}(t) * \tilde{\phi}(t))_k, \quad \tilde{\phi}_k(t) = (k+1)\phi_{k+1}(t),$$

first notice that $\frac{\partial^k}{\partial x^k} g(t, x)|_{x=0} = k! \phi_k(t)$ implies $\frac{\partial^k}{\partial x^k} g_x(t, x)|_{x=0} = k! \tilde{\phi}_k(t)$. The convolution follows naturally by applying the general Leibniz rule.

Lastly, the initial condition of (F.11) can be easily deduced from the fact that $g(0, x) = v(T)q(x)$. □

F.5 Putting everything together

Proof of Theorem D.4. Since p is negligible to double factorial by assumption, and $g_0 : [0, T] \rightarrow \mathbb{R}$, $g_1, g_2 : [0, T] \rightarrow \mathbb{C}$ are continuously differentiable such that $\Re(g_1) = 0, \Re(g_2) \leq 0$, then $F(t, x)$ in (4.1) is well-defined. We recall from Lemma D.8 that

$$F(t, x) = \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s) p_i(X_s) ds + v(T)q(X_T) - v(t)q(X_t) \right) \mid X_t = x \right],$$

with u_i, v are continuous in $[0, T]$, $\Re(u_1) \leq 0, \Re(u_2), \Re(u_3), \Re(v) = 0$, $p_1 = p^2$, and p_i, q are negligible to double factorial. Notice that $F(t, x) = f(t, x) \exp(-v(t)q(x))$. Theorem F.4 yields that F is continuous.

If in addition, F does not vanish, then so does f . By Lemma I.1, we can define $\log f, \log F$ such that $\log F = \log f - v(t)q(x)$. By Theorems F.4 and F.11, f is C^∞ in x and C^1 in t , and so is F . By Lemma I.2, $\log f, \log F$ are also C^∞ in x and C^1 in t . Then, we can take the k^{th} partial derivative with respect to x for $\log F = \log f - v(t)q(x)$ around $x = 0$. Then, the system of ODE (F.11) in Theorem F.12 induces the system of ODE (4.4) by setting $\psi_k := \phi_k - v(T-t)q_k$ with ϕ_k defined in (F.11), with the coefficients of the ODE (4.4) coming from the precise definition of $(u_i, p_i)_{i \in \{1, 2, 3\}}$ as defined in Appendix H. □

G Proof of Theorem D.12

Before proving Theorem D.12, we introduce the following definition of the function f_n :

Definition G.1. For $\mu_n = \frac{T}{n} \sum_{i=0}^{n-1} \delta_{iT/n}$, with δ_t the Dirac measure at point t , we define

$$\begin{aligned} f_n(t, x) &= \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s) p_i(X_s) \mu_n(ds) + v(T)q(X_T) \right) \mid X_t = x \right] \\ &= F_n(t, x) \exp(v(t)q(x)), \quad t \leq T, x \in \mathbb{R}, \end{aligned}$$

with F_n as defined in Definition D.9.

By the Markov property of X , we can remove the conditioning and write $f_n(t, x)$ equivalently as

$$f_n(t, x) = \mathbb{E} \left[\exp \left(\int_0^{T-t} \sum_{i=1}^3 u_i(s+t) p_i(e^{bs}x + w(s) + \widetilde{W}_s) \mu_n(ds+t) + v(T)q(e^{T-t}x + w(T-t) + \widetilde{W}_{T-t}) \right) \right]. \quad (\text{G.1})$$

Notice that $X_t = e^{bt}x + w(t) + \widetilde{W}_t$ for $X_0 = x$, $w(t) := a \int_0^t e^{b(t-u)} du$ and $\widetilde{W}_t := c \int_0^t e^{b(t-u)} dW_u$.

The function f_n is well-defined, since $\int_0^{T-t} \Re \left(u_1(s) p^2(e^{bs}x + w(s) + \widetilde{W}_s) \right) \mu(ds+t) \leq 0$ given the condition of u_i, p_i, v and q from Lemma D.8.

The main advantage of considering functionals (f_n, F_n) over (f, F) is their entire property in x shown in Theorem G.3 below. As we shall see, the results of (f, F) in Section F can be easily extended to (f_n, F_n) . We now layout the following steps to prove Theorem D.12:

- In Subsection G.1, we prove that f_n is entire in $x \in \mathbb{R}$, without any additional conditions imposed on f_n or F_n .
- In Subsection G.2, we prove discretized versions of the results in Section F, in the case when $f_n(t, \cdot)$ (or equivalently $F_n(t, \cdot)$) does not vanish on $\mathbb{R}$.
- In Subsection G.3, we prove Theorem D.12 under the assumption that the extension of $(F_n(t, \cdot))$ to the complex plane $(F_n(t, \cdot))_c$ does not vanish.

Lemma G.2. $f_n(t, x) \rightarrow f(t, x)$, when $n \rightarrow \infty$, for all $t \leq T$ and $x \in \mathbb{R}$.

Proof. This follows from a direct application of the bounded convergence theorem: the Riemann sum converges to the Riemann integral, and the exponential has a module at most 1. $\square$

G.1 Entire property

In this subsection, we do not need any additional condition imposed on f_n or F_n about their zeros.

Theorem G.3. For fixed t , $f_n(t, \cdot)$ is entire in $x \in \mathbb{R}$.

We first introduce a lemma that will be useful in proving Theorem G.3. The idea of the proof is standard and comes from exploiting the fact that the Gaussian density is entire. A similar argument has been used for single marginals in [12, Lemma 6.1].

Lemma G.4. Suppose that $(W_t)_{t \geq 0}$ is a centered Gaussian Process with independent increments such that $W_t - W_s$ has non-zero variance when $t > s$. Let $0 = t_0 < t_1 < \dots < t_n = T$. Take g a bounded measurable function from $\mathbb{R}^n$ to $\mathbb{C}$, then $\mathbb{E}[g(x + W_{t_1}, x + W_{t_2}, \dots, x + W_{t_n})]$ is entire in x .

Proof. Denote the variance of $W_{t_i} - W_{t_{i-1}}$ as $v_i > 0$, we have

$$\begin{aligned} & \mathbb{E}[g(x + W_{t_1}, x + W_{t_2}, \dots, x + W_{t_n})] \\ &= \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi v_i}} \right) \int_{\mathbb{R}^n} g(y_1, y_1 + y_2, \dots, y_1 + \dots + y_n) \exp \left(-\frac{(y_1 - x)^2}{2v_1} - \sum_{i=2}^n \frac{y_i^2}{2v_i} \right) dy_1 dy_2 \dots dy_n. \end{aligned}$$

Denote $g(y) = g(y_1, y_1 + y_2, \dots, y_1 + \dots + y_n)$, we have

$$\begin{aligned}
& \mathbb{E}[g(x + W_{t_1}, x + W_{t_2}, \dots, x + W_{t_n})] \\
&= \exp\left(\frac{-x^2}{2v_1}\right) \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi v_i}}\right) \int_{\mathbb{R}^n} g(y) \exp\left(-\sum_{i=1}^n \frac{y_i^2}{2v_i}\right) \exp\left(\frac{xy_1}{v_1}\right) dy_1 dy_2 \dots dy_n \\
&= \exp\left(\frac{-x^2}{2v_1}\right) \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi v_i}}\right) \int_{\mathbb{R}^n} g(y) \exp\left(-\sum_{i=1}^n \frac{y_i^2}{2v_i}\right) \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{xy_1}{v_1}\right)^k dy_1 dy_2 \dots dy_n \\
&= \exp\left(\frac{-x^2}{2v_1}\right) \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi v_i}}\right) \sum_{k=0}^{\infty} \frac{1}{k!} \left(\int_{\mathbb{R}^n} g(y) \exp\left(-\sum_{i=1}^n \frac{y_i^2}{2v_i}\right) \left(\frac{y_1}{v_1}\right)^k dy_1 dy_2 \dots dy_n\right) x^k,
\end{aligned}$$

where we applied Fubini in the last step since

$$\begin{aligned}
& \int_{\mathbb{R}^n} \sum_{k=0}^{\infty} |g(y)| \exp\left(-\sum_{i=1}^n \frac{y_i^2}{2v_i}\right) \frac{1}{k!} \left(\frac{xy_1}{v_1}\right)^k dy_1 dy_2 \dots dy_n \\
& \leq \|g\|_{\infty} \int_{\mathbb{R}^n} \exp\left(-\sum_{i=1}^n \frac{y_i^2}{2v_i} + \frac{|xy_1|}{v_1}\right) dy_1 dy_2 \dots dy_n < \infty.
\end{aligned}$$

This implies an infinite radius of convergence for

$$\exp\left(\frac{-x^2}{2v_1}\right) \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi v_i}}\right) \sum_{k=0}^{\infty} \frac{1}{k!} \left(\int_{\mathbb{R}^n} g(y) \exp\left(-\sum_{i=1}^n \frac{y_i^2}{2v_i}\right) \left(\frac{y_1}{v_1}\right)^k dy_1 dy_2 \dots dy_n\right) x^k$$

i.e. $\mathbb{E}[g(x + W_{t_1}, x + W_{t_2}, \dots, x + W_{t_n})]$ is equal to a power series with an infinite radius of convergence, so it is entire in $x \in \mathbb{R}$. □

Proof of Theorem G.3. We define $t_j = \frac{jT}{n}, j = 0, 1, \dots, n-1$, then

$$\begin{aligned}
f_n(t, x) &= \mathbb{E} \left[\exp \left(\sum_{i=1}^3 \sum_{t \leq t_j \leq T} u_i(t_j) p_i(e^{b(t_j-t)} x + w(t_j - t) + \widetilde{W}_{t_j-t}) \right. \right. \\
&\quad \left. \left. + v(T) q(e^{b(T-t)} x + w(T - t) + \widetilde{W}_{T-t}) \right) \right].
\end{aligned}$$

The exponential is bounded by 1. And notice that $e^{b(t_j-t)} x + \widetilde{W}_{t_j-t} = e^{b(t_j-t)} (x + e^{-b(t_j-t)} \widetilde{W}_{t_j-t})$. Recall that $\widetilde{W}_t = \int_0^t e^{b(t-u)} c dW_u$. Therefore, $e^{-bt} \widetilde{W}_t = c \int_0^t e^{-bu} dW_u$ is a Gaussian Process satisfying the condition of Lemma G.4, since $c \neq 0$. Then, an application of Lemma G.4 yields the result. □

G.2 Discretized versions of the theorems in Section F

In this subsection, we prove discretized versions of the results in Section F, in the case when $f_n(t, \cdot)$ (or equivalently $F_n(t, \cdot)$) does not vanish on $\mathbb{R}$.

Remark G.5. Recalling Corollary D.5, the conditions that $g_1 = 0$ and g_2 $\mathbb{R}$ -valued with $g_2 \leq 0$ guarantee that $f_n(t, \cdot)$ and $F_n(t, \cdot)$ do not vanish on $\mathbb{R}$.

Theorem G.6. (Discretized version of Theorem F.4) The function f_n defined in (G.1) is C^∞ in x with $\frac{\partial^k f_n}{\partial x^k}$ continuous in (t, x) in the region $(\frac{jT}{n}, \frac{(j+1)T}{n}) \times \mathbb{R}$ for $j = 0, 1, \dots, n-1$. In addition, there exists a power series q_k negligible to double factorial and constant C_k independent of (t, x) , such that $|\frac{\partial^k f_n}{\partial x^k}| \leq |q_k|(|x|) + C_k$.

Proof. Notice that all the lemmas and proofs used in Section F dealing with f being C^∞ in x and the bound of $\frac{\partial^k f_n}{\partial x^k}$ can be easily adopted to f_n by replacing the integral with a finite sum. $\square$

Theorem G.7. (Discretized version of Theorem F.11) For $j = 0, 1, \dots, n-1$, the function f_n

(i) is continuous in the region $(\frac{jT}{n}, \frac{(j+1)T}{n}] \times \mathbb{R}$. In addition, f_n is C^1 in t and C^∞ in x in the region $(\frac{jT}{n}, \frac{(j+1)T}{n}) \times \mathbb{R}$, $j = 0, 1, \dots, n-1$,

(ii) solves the following PDE

$$\begin{aligned} f_t + f_x(a + bx) + \frac{1}{2}c^2 f_{xx} &= 0, \quad t \notin \text{supp}(\mu_n) \cup \{T\}, \\ \lim_{s \rightarrow t^+} f(s, x) &= f(t, x) \exp\left(-\frac{T}{n} \sum_{i=1}^3 u_i(t) p_i(x)\right), \quad t \in \text{supp}(\mu_n), \\ f(T, x) &= \exp(v(T)q(x)), \end{aligned}$$

$$\text{with } \text{supp}(\mu_n) = \{0, \frac{T}{n}, \frac{2T}{n}, \dots, \frac{(n-1)T}{n}\}.$$

Proof. The proof follows along the same lines as the proof of Theorem F.11. By Theorem G.6 and Lemma F.3, $(f_n)_x(t, x)$ and $(f_n)_x^2(t, x)$ are bounded by $|q|(|x|) + C$, where q is negligible to double factorial. Therefore by Proposition F.7, the stochastic integral $\int_0^\cdot (f_n)_x(s, x) dW_s \in L^2$ and is a true martingale. The Riemann integral is still dominated, allowing the interchange between limit and the expectation and thus computing the the infinitesimal generator $\lim_{r \rightarrow 0^+} \frac{1}{r} (\mathbb{E}^x [f_n(t_0, X_r)] - f_n(t_0, x))$ as per the first way in Theorem F.11.

The second way of computing the generator in Theorem F.11 can also be adopted to compute $\lim_{r \rightarrow 0^+} \frac{1}{r} (\mathbb{E}^x [f_n(t_0, X_r)] - f_n(t_0, x))$ thanks to the Markov property of X_t . Indeed, define

$$Z_t^n = \exp\left(\int_0^t \sum_{i=1}^3 u_i(T-t+s) p_i(X_s) \mu_n(T-t+ds)\right), \quad Y_t = v(T)q(X_t).$$

we have

$$\begin{aligned} & \frac{1}{r} (\mathbb{E}^x [f_n(t_0, X_r)] - f_n(t_0, x)) \\ &= \frac{1}{r} \mathbb{E}^x \left[Z_{T-t_0+r}^n \cdot \exp\left(-\int_0^r \sum_{i=1}^3 u_i(t_0-r+s) p_i(X_s) \mu_n(t_0-r+ds)\right) \exp(Y_{T-t_0+r}) - Z_{T-t_0}^n \exp(Y_{T-t_0}) \right] \\ &= \frac{1}{r} \mathbb{E}^x [\exp(Y_{T-t_0+r}) Z_{T-t_0+r}^n - \exp(Y_{T-t_0}) Z_{T-t_0}^n] \\ &+ \frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r}^n \cdot \left(\exp\left(-\int_0^r \sum_{i=1}^3 u_i(t_0-r+s) p_i(X_s) \mu_n(t_0-r+ds)\right) - 1 \right) \right] \\ &= \frac{1}{r} (f(t_0-r, x) - f(t_0, x)) \\ &+ \frac{1}{r} \mathbb{E}^x \left[\exp(Y_{T-t_0+r}) Z_{T-t_0+r}^n \cdot \left(\exp\left(-\int_0^r \sum_{i=1}^3 u_i(t_0-r+s) p_i(X_s) \mu_n(t_0-r+ds)\right) - 1 \right) \right]. \end{aligned}$$

Notice that, if $t_0 \notin \text{supp}(\mu_n) \cup \{T\}$, where $\text{supp}(\mu_n) \cup \{T\}$ is a finite set, then there exists $r_0 > 0$ such that $r_0 < |t_0 - t|$ for all $t \in \text{supp}(\mu_n) \cup \{T\}$, thus for $0 < r < r_0$,

$$\mathbb{E}^x \left[\frac{1}{r} \exp(Y_{T-t_0+r}) Z_{T-t_0+r}^n \cdot \left(\exp\left(-\int_0^r \sum_{i=1}^3 u_i(t_0-r+s) p_i(X_s) \mu_n(t_0-r+ds)\right) - 1 \right) \right] = 0.$$

So similar the proof of Theorem F.11, $(f_n)_t$ exists and $0 = (f_n)_t + (f_n)_x(a + bx) + \frac{1}{2}c^2(f_n)_{xx}$. The boundary condition when $t \notin \text{supp}(\mu_n) \cup \{T\}$ can be deduced by applying the definition of μ_n .

□

Remark G.8. Theorem G.7 shows that for fixed x , $f_n(t, x)$ is left continuous for $t \in [0, T]$ and is discontinuous at $t \in \text{supp}(\mu_n)$.

We now define $\log f_n$ in the following Lemma G.9.

Lemma G.9. If $f_n(t, \cdot)$ does not vanish for $x \in \mathbb{R}$, then there exists $\log f_n$ such that $\exp(\log f_n) = f_n$, and in the regions $(\frac{jT}{n}, \frac{(j+1)T}{n}] \times \mathbb{R}$, $j = 0, 1, \dots, n-1$, $\log f_n(t, x)$ is continuous in (t, x) , and

$$\lim_{s \rightarrow t^+} \log f_n(s, x) = \log f_n(t, x) - \frac{T}{n} \sum_{i=1}^3 u_i(t) p_i(x), \quad t \in \text{supp}(\mu_n),$$

$$\log f_n(T, x) = v(T)q(x).$$

Proof. We want to use Lemma I.1 to define $\log f_n$ from f_n . However, Theorem G.7 shows that f_n is discontinuous when $t \in \text{supp}(\mu_n)$, with

$$\lim_{s \rightarrow t^+} f_n(s, x) = f_n(t, x) \exp \left(-\frac{T}{n} \sum_{i=1}^3 u_i(t) p_i(x) \right), \quad t \in \text{supp}(\mu_n),$$

we need to first define

$$\tilde{f}_n(t, x) = f_n(t, x) \exp \left(-\frac{T}{n} \sum_{\substack{s \in \text{supp}(\mu_n) \\ s \geq t}} \sum_{i=1}^3 u_i(s) p_i(x) \right),$$

where $\tilde{f}_n$ is continuous on $[0, T] \times \mathbb{R}$ and $\tilde{f}_n(T, x) = f_n(T, x) = \exp(v(T)q(x))$. And since f_n is non zero, so is $\tilde{f}_n$. We can now apply Lemma I.1 to yield the existence of $\tilde{g}$ continuous on $[0, T] \times \mathbb{R}$, such that $\exp(\tilde{g}) = \tilde{f}_n$, and to specify $\tilde{g}$, we choose that $\tilde{g}(T, 0) = v(T)q(0)$. We can now define:

$$\log f_n(t, x) = \tilde{g}(t, x) + \frac{T}{n} \sum_{\substack{s \in \text{supp}(\mu_n) \\ s \geq t}} \sum_{i=1}^3 u_i(s) p_i(x). \quad (\text{G.2})$$

Since $\sum_{s \in \text{supp}(\mu_n), s \geq t} \frac{T}{n} \sum_{i=1}^3 u_i(s) p_i(x)$ is continuous in the regions $(\frac{jT}{n}, \frac{(j+1)T}{n}] \times \mathbb{R}$, for $j = 0, 1, \dots, n-1$, the definition of $\log f_n$ in (G.2) yields that

1. $\exp(\log f_n) = f_n$.
2. In the regions $(\frac{jT}{n}, \frac{(j+1)T}{n}] \times \mathbb{R}$, $j = 0, 1, \dots, n-1$, $\log f_n(t, x)$ is continuous in (t, x) .
3. $\lim_{s \rightarrow t^+} \log f_n(s, x) = \log f_n(t, x) - \frac{T}{n} \sum_{i=1}^3 u_i(t) p_i(x)$, $t \in \text{supp}(\mu_n)$.

Furhtermore, since $\exp(\log f_n(T, x)) = \exp(\tilde{g}(T, x)) = \tilde{f}_n(T, x) = f_n(T, x) = \exp(v(T)q(x))$, $\log f_n(T, x) - v(T)q(x) \in \{2k\pi i, k \in \mathbb{Z}\}$. Noticing that $\log f_n(T, x) = \tilde{g}(T, x)$ is continuous in x , and $\log f_n(T, 0) - v(T)q(0) = \tilde{g}(T, 0) - v(T)q(0) = 0$, we deduce that $\log f_n(T, x) = v(T)q(x)$. □

We now introduce the discretized version of Theorem F.12.

Theorem G.10. Suppose that $f_n(t, \cdot)$ does not vanish on $\mathbb{R}$, then

$$\phi_{n,k}(t) = \frac{1}{k!} \partial_x^k \log f_n(T-t, x) \Big|_{x=0}, \quad t \geq 0$$

solves the following system of ODE:

$$\begin{aligned} \phi'_{n,k}(t) &= bk\phi_{n,k}(t) + a(k+1)\phi_{n,k+1}(t) + \frac{c^2(k+2)(k+1)}{2}\phi_{n,k+2}(t) \\ &\quad + \frac{c^2}{2}(\tilde{\phi}_n(t) * \tilde{\phi}_n(t))_k, \quad \tilde{\phi}_{n,k} = \phi_{n,k+1}(k+1), \quad t \in \{T-u : u \notin \text{supp}(\mu_n) \cup \{T\}\} \\ \phi_{n,k}(t) &= \lim_{s \rightarrow t^-} \phi_{n,k}(s) + \frac{T}{n} \sum_{i=1}^3 u_i(T-t)(p_i)_k, \quad t \in \{T-u : u \in \text{supp}(\mu_n)\} \\ \phi_{n,k}(0) &= v(T)q_k. \end{aligned} \tag{G.3}$$

Proof. By Theorem G.7, in the regions $(\frac{jT}{n}, \frac{(j+1)T}{n}) \times \mathbb{R}$, for $j = 0, 1, \dots, n-1$, f_n is C^1 in t and C^∞ in x . Since $f_n \neq 0$, thus by Lemma I.2, $\log f_n$ in G.9 is C^1 in t and C^∞ in x in these regions. Set $g_n(t, x) = \log f_n(T-t, x)$, g_n then satisfies the following PDE:

$$\begin{aligned} g_t(t, x) &= g_x(t, x)(a + bx) + \frac{1}{2}c^2 g_{xx}(t, x) + \frac{1}{2}c^2 g_x^2, \quad t \in \{T-u : u \notin \text{supp}(\mu_n) \cup \{T\}\} \\ \lim_{s \rightarrow t^-} g(s, x) &= g(t, x) - \frac{T}{n} \sum_{i=1}^3 u_i(T-t)p_i(x), \quad t \in \{T-u : u \in \text{supp}(\mu_n)\} \\ g(0, x) &= v(T)q(x). \end{aligned}$$

Next, we apply Lemma F.13 as in the proof of Theorem F.12, and then comparing the derivatives of both sides in x yields the system of ODE. Lastly, the initial condition of (G.3) can be easily deduced from the fact that $g_n(0, x) = v(T)q(x)$. $\square$

G.3 Putting everything together

In this subsection, we will use the condition that the extension of f_n in the complex plane $(f_n(t, \cdot))_c$, defined in D.10 does not vanish for $x \in \mathbb{C}$ as opposed to $\in \mathbb{R}$. This condition is both necessary and sufficient to guarantee that $\log f_n$ is entire in $x \in \mathbb{R}$, see Lemma I.3 and Remark I.4.

Lemma G.11. If $(f_n)_c(t, \cdot)$ does not vanish on $\mathbb{C}$, then $f_n(t, x) = \exp\left(\sum_{k \geq 0} \phi_{n,k}(T-t)x^k\right)$, where $\phi_{n,k}$ are defined as in Theorem G.10.

Proof. By Lemma I.3, $\log f_n$ is entire on $\mathbb{R}$, and notice that $\phi_{n,k}(t) = \frac{1}{k!} \partial_x^k \log f_n(T-t, x) \Big|_{x=0}$. $\square$

Proof of Theorem D.12. By Lemma G.11, $f_n(t, x) = \exp\left(\sum_{k \geq 0} \phi_{n,k}(T-t)x^k\right)$, where $\phi_{n,k}$ are defined as in Theorem G.10.

We set $\psi_{n,k}(t) = \phi_{n,k}(t) - v(T-t)q_k$. By Lemma D.8 and Lemma H.1, this is equivalent to

$$\begin{aligned} \psi_{n,k}(t) &= \phi_{n,k}(t) - \rho g_1(t)g_0(T-t)\frac{p_{k-1}}{ck}, \quad k \geq 1, \\ \psi_{n,0}(t) &= \phi_{n,0}(t). \end{aligned}$$

Therefore, by Lemma G.2,

$$\begin{aligned} F(t, x) &= f(t, x) \exp(-v(t)q(x)) \\ &= \lim_{n \rightarrow \infty} f_n(t, x) \exp(-v(t)q(x)) \end{aligned}$$

and, by Lemma G.11,

$$\begin{aligned}
f_n(t, x) \exp(-v(t)q(x)) &= \exp\left(\sum_{k \geq 0} \phi_{n,k}(T-t)x^k\right) \exp(-v(t)q(x)) \\
&= \exp\left(\sum_{k \geq 0} (\phi_{n,k}(T-t) - v(t)q_k)x^k\right) \\
&= \exp\left(\sum_{k \geq 0} \psi_{n,k}(T-t)x^k\right).
\end{aligned}$$

Recalling that $F_n(t, x) = f_n(t, x) \exp(-v(t)q(x))$ by Definition G.1 ends the proof. $\square$

H Proof of Lemma D.8

Lemma H.1. *Let p be a power series with an infinite radius of convergence. Define the power series r_1, r_2 by*

$$r_1(x) = -\frac{1}{c}((a+bx)p(x) + \frac{1}{2}c^2p'(x)), \quad r_2(x) = \frac{1}{c} \int_0^x p(y)dy, \quad x \in \mathbb{R},$$

which again have an infinite radius of convergence. Then, for any continuously differentiable function $h : [0, T] \rightarrow \mathbb{C}$, it holds that

$$\int_t^T h(s)p(X_s)dW_s = \int_t^T (h(s)r_1(X_s) - h'(s)r_2(X_s))ds + h(T)r_2(X_T) - h(t)r_2(X_t).$$

Proof. This follows from a straightforward application of Itô's Lemma on the process $(h(s)r_2(X_s))_{s \leq T}$ between t and T . Recall the dynamics of X in (2.1). $\square$

Proof of Lemma D.8. Using $d \log S_t = -\frac{\sigma^2}{2}dt + \sigma_t dB_t$, we have:

$$\begin{aligned}
\int_t^T g_1(T-s)d \log S_s &= \int_t^T -\frac{1}{2}g_1(T-s)g_0^2(s)p^2(X_s)ds + \rho g_1(T-s)g_0(s)p(X_s)dW_s \\
&\quad + \sqrt{1-\rho^2}g_1(T-s)g_0(s)p(X_s)dW_s^\perp.
\end{aligned}$$

Conditioning on $\mathcal{F}_t \vee \mathcal{F}_T^W$, the integral $\int_t^T \sqrt{1-\rho^2}g_1(T-s)g_0(s)p(X_s)dW_s^\perp$ is Gaussian with conditional variance $\int_t^T (1-\rho^2)g_1^2(T-s)g_0^2(s)p^2(X_s)ds$. Therefore,

$$\begin{aligned}
&\mathbb{E} \left[\exp \left(\int_t^T \sqrt{1-\rho^2}g_1(T-s)g_0(s)p(X_s)dW_s^\perp \right) \mid \mathcal{F}_t \vee \mathcal{F}_T^W \right] \\
&= \exp \left(\frac{1}{2} \int_t^T (1-\rho^2)g_1^2(T-s)g_0^2(s)p^2(X_s)ds \right),
\end{aligned}$$

so that

$$\begin{aligned}
F(t, X_t) &= \mathbb{E} \left[\exp \left(\int_t^T g_1(T-s)d \log S_s + \int_t^T g_2(T-s)\sigma_s^2 ds \right) \mid \mathcal{F}_t \right] \\
&= \mathbb{E} \left[\exp \left(\int_t^T \left(\frac{1}{2}(1-\rho^2)g_1^2(T-s) - \frac{1}{2}g_1(T-s) + g_2(T-s) \right) g_0^2(s)p^2(X_s)ds \right. \right. \\
&\quad \left. \left. + \int_t^T \rho g_1(T-s)g_0(s)p(X_s)dW_s \right) \mid \mathcal{F}_t \right].
\end{aligned}$$

By applying Lemma H.1 and choosing $h(t) = \rho g_1(T-t)g_0(t)$, we can rewrite the following:

$$\int_t^T \rho g_1(T-s)g_0(s)p(X_s)dW_s = \int_t^T h(s)r_1(X_s) - h'(s)r_2(X_s)ds + h(T)r_2(X_T) - h(t)r_2(X_t).$$

Next, define $u_1(s) = (\frac{1}{2}(1-\rho^2)g_1^2(T-s) - \frac{1}{2}g_1(T-s) + g_2(T-s))g_0^2(s)$, $p_1 = p^2$, $u_2(s) = h(s)$, $p_2 = r_1$, $u_3(s) = h'(s)$, $p_3 = -r_2$, $v(s) = h(s)$, $q = r_2$. Since p is negligible to double factorial, g_0, g_1, g_2 are continuously differentiable, and $\Re(g_1) = \Im(g_0) = 0$, $\Re(g_2) \leq 0$, we deduce that u_i, v are continuous in $[0, T]$, $\Re(u_1) \leq 0$, $\Re(u_2), \Re(u_3), \Re(v) = 0$, $p_1 = p^2$, p_i, q are negligible to double factorial by Lemma F.3, and

$$F(t, x) = \mathbb{E} \left[\exp \left(\int_t^T \sum_{i=1}^3 u_i(s)p_i(X_s)ds + v(T)q(X_T) - v(t)q(X_t) \right) \mid X_t = x \right].$$

□

I A small remark on the complex logarithm

For complex-valued functions, such as F in (4.1), the definition of $\log F$ is not trivial, especially if the range of F is not simply connected. We use the lifting property in Algebraic Topology.

Lemma I.1. *If $f : I \times \mathbb{R} \rightarrow \mathbb{C} \setminus \{0\}$ is a continuous function where I is an interval of $\mathbb{R}$. Then, there exists a continuous function $g : I \times \mathbb{R} \rightarrow \mathbb{C}$ such that $\exp(g) = f$. And if the value of g at one point is specified, then g is entirely specified, i.e. there exists only one g that satisfies these conditions in this case.*

Proof. Notice that $p = \exp(z)$ is a covering projection from $\mathbb{C}$ to $\mathbb{C} \setminus \{0\}$, i.e. a local homeomorphism. And $I \times \mathbb{R}$ is path-connected, locally path-connected, and with a trivial fundamental group. Then by Proposition 1.33 of [22], the lifting criterion, f as a continuous function from $I \times \mathbb{R}$ to $\mathbb{C} \setminus \{0\}$, can be lifted to a continuous function g from $I \times \mathbb{R}$ to $\mathbb{C}$, i.e. $f = p \circ g = \exp(g)$. And by Proposition of [22, 1.34], the unique lifting property, and the fact that $I \times \mathbb{R}$ is connected, if the value of g at one point is given, then g is specified. □

Lemma I.2. *Let $f : I \times \mathbb{R} \rightarrow \mathbb{C} \setminus \{0\}$ be a continuous function, where I is an interval of $\mathbb{R}$. Assume that f is C^1 in the first variable t and C^∞ in the second variable x . Then, the function g as defined in Lemma I.1 is also C^1 in t and C^∞ in x .*

Proof. Suppose that we define g as in the Lemma I.1. We need only to prove that for all fixed point $(t_0, x_0) \in I \times \mathbb{R}$, g is C^1 in t , C^∞ in x in neighborhood of (t_0, x_0) . Since $f(t_0, x_0) \neq 0$, there exists a $\delta > 0$, for all $(t, x) \in I \times \mathbb{R}$ such that $|t-t_0|+|x-x_0| < \delta$, we have that $|f(t, x) - f(t_0, x_0)| < |f(t_0, x_0)|$. Therefore, for all $(t, x) \in I \times \mathbb{R}$ such that $|t-t_0|+|x-x_0| < \delta$, $f(t, x)$ is contained in an open ball B , with center $f(t_0, x_0)$ and radius $|f(t_0, x_0)|$. Particularly, B is simply connected and does not contain 0.

We define h from B to $\mathbb{C}$ as $h(z) = \int_{l_{f(t_0, x_0), z}} \frac{1}{w} dw + g(t_0, x_0)$, where $l_{f(t_0, x_0), z}$ is the line segment between 0 and z . We aim to prove that $g(t, x) = h(f(t, x))$ for all $(t, x) \in I \times \mathbb{R}$ such that $|t-t_0|+|x-x_0| < \delta$.

By the definition of h , $h'(z) = \frac{1}{z}$, thus $(\frac{\exp(h(z))}{z})' = 0$, so $\frac{\exp(h(z))}{z}$ is a constant on B . What is more, $\frac{\exp(h(f(t_0, x_0)))}{f(t_0, x_0)} = \frac{\exp(g(t_0, x_0))}{f(t_0, x_0)} = 1$. So $\exp(h(z)) = z$, and thus $\exp(h(f(t, x))) = f(t, x) = \exp(g(t, x))$ for all $(t, x) \in I \times \mathbb{R}$ such that $|t-t_0|+|x-x_0| < \delta$. Thus $h(f(t, x)) - g(t, x) \in \{2k\pi i, k \in \mathbb{Z}\}$. Notice again that $h(f(t_0, x_0)) - g(t_0, x_0) = 0$, and by continuity, we have that $g(t, x) = h(f(t, x))$ for all $(t, x) \in I \times \mathbb{R}$ such that $|t-t_0|+|x-x_0| < \delta$. Notice that $h'(z) = \frac{1}{z}$, so h is holomorphic on B . Therefore h is C^∞ on B when regarding B as a subset of $\mathbb{R}^2$. Then since g is a composition of f and h , g is also C^1 in t , C^∞ in x . □

Lemma I.3. *Let $f : \mathbb{R} \rightarrow \mathbb{C} \setminus \{0\}$ be an entire function on $\mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{C}$ a continuous function such that $\exp(g(x)) = f(x)$. Then, g is entire on $\mathbb{R}$ if and only if the extension f_c of f to the complex plane, as defined in Definition D.10, does not vanish on $\mathbb{C}$.*

Proof. $\Leftarrow$ Assume that $f_c : \mathbb{C} \rightarrow \mathbb{C}$ does not vanish. Then, for $z \in \mathbb{C}$, we can define $h(z) = \int_{l_{0,z}} \frac{f'_c(w)}{f_c(w)} dw + g(0)$, where $l_{0,z}$ is the line segment between 0 and z . It follows that $h'(z) = \frac{f'_c(z)}{f_c(z)}$, showing that h is holomorphic on $\mathbb{C}$. What is more, consider the function $F(z) = f_c(z) \exp(-h(z))$, we know that $F(0) = f_c(0) \exp(-h(0)) = f(0) \exp(-g(0)) = 1$, $F'(z) = 0$, thus $F(z) = 1$, for all $z \in \mathbb{C}$. Thus $\exp(g(x) - h(x)) = \exp(g(x)) \exp(-h(x)) = f(x) \exp(-h(x)) = F(x) = 1$ for all $x \in \mathbb{R}$. Thus there exists a $k \in \mathbb{Z}$, such that $g(x) - h(x) = 2k\pi i$. And notice that $h(0) = g(0)$, we have that $g(x) = h(x)$ for $x \in \mathbb{R}$. Particularly, h is holomorphic on $\mathbb{C}$, so it is entire on $\mathbb{C}$ and thus on $\mathbb{R}$, then so is g .

$\Rightarrow$ Assume $g : \mathbb{R} \rightarrow \mathbb{C}$ is entire on $\mathbb{R}$. Then, it can be written in the form $g(x) = \sum_{n=0}^{\infty} a_n x^n$, which is a power series with infinite radius of convergence. Then $g_c(z) = \sum_{n=0}^{\infty} a_n z^n$ is entire on $\mathbb{C}$, and so is $\exp(g_c(z))$. Notice that $\exp(g_c(z))$ and $f_c(z)$ are both entire on $\mathbb{C}$, and $\exp(g_c(x)) = \exp(g(x)) = f(x) = f_c(x)$ for all $x \in \mathbb{R}$. Since the zeros of an entire function are isolated, except for the zero function, we have that $f_c(z) = \exp(g_c(z))$, and hence $f_c(z) \neq 0$ for all $z \in \mathbb{C}$. □

Remark I.4. *We point out that for a real entire function $f : \mathbb{R} \rightarrow \mathbb{C}$ that does not vanish on $\mathbb{R}$, there exists a continuous function $\log f : \mathbb{R} \rightarrow \mathbb{C}$ such that $\exp(\log f) = f$, (choosing $I = [0, 0] = \{0\}$ in Lemma I.1). However $\log f$ is not necessarily entire on $\mathbb{R}$. For instance the function $f(x) = 1 + ix$, is entire on $\mathbb{R}$ and does not vanish on $\mathbb{R}$. However, $\log f = \frac{1}{2} \log(1 + x^2) + \arctan(x)i + 2k\pi i, k \in \mathbb{Z}$ is no longer entire on $\mathbb{R}$. In fact, the Taylor series at $x = 0$ for both $\log(1 + x^2)$ and $\arctan(x)$ have a finite convergence radius of 1 (can easily be checked) and not ∞ .*

Definition I.5. *If the joint characteristic functional $F(t, x)$ is continuous on $[0, T] \times \mathbb{R}$ and $F(t, x) \neq 0$, for all $(t, x) \in [0, T] \times \mathbb{R}$, we define $\mathbf{log F}$ by Lemma I.1, as the function g such that $\exp(g) = F$ and $g(T, 0) = 0$.*

Remark I.6. *Notice that by the definition of F in 4.1, $F(T, x) = 1$, so the condition $g(T, 0) = 0$ is satisfied.*